

CHLORO WATCH: WEB SYSTEM FOR WATER QUALITY MONITORING OF CHLOROPHYLL LEVELS

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In partial fulfillment of the requirement for the degree
Bachelor of Science in Information Technology
*Specialized in Web and Mobile Application
Development*

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APPROVAL SHEET

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The Researchers

DEDICATION

The researchers dedicate this study to the All-Powerful God, whose guidance, strength, courage, motivation, and wisdom have been essential throughout this journey. This study is also dedicated to the parents of the researchers, whose unwavering love, encouragement, and support have fostered their resilience.

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ABSTRACT

This study presents the Chloro Watch, an innovative web-based system designed for monitoring chlorophyll-A levels, using Arduino-based sensors. The objective was to assess eutrophication risks indicated by chlorophyll-A level augmentation through tracking water quality parameters, including pH, dissolved oxygen, temperature, and turbidity. Data was gathered from Tadolac Lake, utilizing calibrated Arduino-based sensors validated through comparative laboratory analyses yielded a non-significant difference for pH ($p = 0.356$), temperature ($p = 0.619$), and dissolved oxygen ($p = 0.412$) while significant difference for turbidity ($p = <.001^*$) which signifies the need in enhancing calibration aligned with the cheap or generic sensor. The system was developed using an Agile Scrum methodology, the architecture was implemented, and all modules were developed in a timely manner, ESP8266 and Firebase were used for the real-time data management to ensure the sensor readings efficiency. The development emphasized iterative testing and user feedback. Unit, integration, and system testing collectively ensured the system's reliability, confirming reliable sensor readings and a stable UI, verifying smooth module interactions, and validating end-to-end performance. Evaluation was conducted the ISO 25010 software quality standards, focusing on functional completeness, performance efficiency, and usability. Results indicated significant robustness, with an overall score of 3.25. The outcomes emphasize the system's effectiveness in real-time monitoring and its potential for scaling to broader ecological applications. As a low-cost solution, Chloro Watch provides an accessible approach to water quality management, making advanced monitoring technologies available to resource-limited settings. It emerges as a promising tool for advancing sustainable water practices and awareness.

Keywords: *Eutrophication, Chlorophyll, Water Quality Monitoring, Arduino*

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Definition of Terms

Chlorophyll-a

A specific form of chlorophyll used in oxygenic photosynthesis. It is a key indicator of algal biomass and water quality in aquatic ecosystems.

Soft sensor

A virtual sensor that uses mathematical models and algorithms to estimate process variables that are difficult or impossible to measure directly.

Water Quality

A measure of the condition of water relative to the requirements of one or more biotic species and or to any human need or purpose.

Lakes

A lake is an often naturally occurring, relatively large and fixed body of water on the Earth's surface. It is localized in a basin or interconnected basins surrounded by dry land.

Data Collection

A collection of data used for training, validating, and testing machine learning models. It typically includes features (input variables) and labels (output variables) and is essential for model development.

Algal Blooms

Excessive nutrient levels usually cause rapid increases in algal populations in aquatic ecosystems. The blooms can cause harmful toxins, decrease oxygen levels, and negatively affect water quality as well as marine life.

Eutrophication

Eutrophication is a process in which a body of water becomes too fortified with necessary nutrients, often as a result of human activity. These nutrients, notably nitrogen and phosphorus, stimulation of algae overgrowth and other aquatic plants, hence creating a succession. of Environmental problems. Eutrophication can lead to enormous Changes in the hydrological dynamics of water ecosystems, often decreasing diversity Impairing water quality.

Ph	A scale used to specify the acidity or basicity of an aqueous solution. It ranges from 0 to 14, with 7 being neutral, lower values being more acidic, and higher values being more basic.
Temperature	A quantification of the thermal energy in a matter, often measured in degrees Celsius (°C) or Fahrenheit (°F) in scientific contexts. In aquatic environments, temperature affects the metabolism and health of organisms.
Dissolve Oxygen (DO)	The total of oxygen that is present in water, crucial for the survival of aquatic organisms. It is typically measured in milligrams per liter (mg/L) and indicates the health of an aquatic system.
Turbidity	The measure of the cloudiness or haziness of a liquid, typically water, caused by the presence of suspended particles such as sediment, algae, plankton, or other microscopic organisms.
Arduino	Arduino is an open-source electronics platform based on easy-to-use hardware and software. It's designed to make electronics more accessible to beginners and professionals alike, allowing users to create interactive projects that can sense and control objects in the physical world.
SMS	SMS, or Short Message Service, is a text messaging service component of most telephone, internet, and mobile device systems. It allows users to send short text messages, typically limited to 160 characters, between mobile phones and other systems capable of sending and receiving text.
Wi-Fi	Wi-Fi is a wireless networking technology that allows devices to connect to the internet and communicate with each other over a wireless signal. It uses radio waves to transmit data between devices like smartphones, laptops, smart home devices, and the internet router, making

it possible to connect without physical cables.

Parameters

Parameters are values or settings that define certain characteristics or behaviors within a system, function, or model. They are used across various fields, such as programming, mathematics, engineering, and research, to control or influence outcomes. Here are some different contexts for parameters:

Model

Model refers to an algorithm or mathematical model that learns patterns or structures from data in order to make predictions or decisions without being explicitly programmed for the specific task. The model "learns" from historical data by detecting patterns and relationships within that data and then generalizes to make predictions on new, unseen data.

Sensor

A sensor is a device that detects and responds to physical stimuli, such as light, heat, motion, pressure, or sound, and converts them into signals that can be interpreted or measured by humans or machines. Sensors are fundamental components in a wide variety of systems, including automated control systems, robotics, environmental monitoring, and Internet of Things (IoT) devices.

Calibration

Calibration is the process of adjusting and fine-tuning a measurement instrument or sensor to ensure that it provides accurate and reliable measurements. It involves comparing the measurements of the instrument or sensor against a known, traceable standard and then adjusting the instrument to minimize any discrepancies.

CHAPTER I

INTRODUCTION

PROJECT CONTEXT

Chlorophyll-A is a great indicator of eutrophication in water bodies (Zhang, 2023). Chlorophyll-A concentration represents as a water quality parameter used to evaluate the condition of eutrophication; due to its visibility, algae growth may be noticed (Yang X., 2020). Seasonal events such as rainfalls and storms, and agricultural activities are the key factors contributing to an increasing amount of nutrients in water bodies (Akinnowo, 2023). Indeed, aquaculture can also contribute to eutrophication in the lake (Villaruel, 2022). Eutrophication threatens water quality worldwide; reducing aquatic food supplies and fish production are concerns (Wurtsbaugh, 2019). Eutrophication can be a cause for the increasing number of algae. Harmful Algal Blooms can affect the ecosystem in aquatic life because of toxin production. Additionally, they cover large areas of freshwater bodies, indicating that sunlight is not reaching lower depths, which can inhibit the growth of other plants. When the algae die, it sinks to the bottom, and the decomposition of bacteria occurs; it produces toxic substances that can harm aquatic organisms, and these effects lead to a decrease in oxygen levels in the water. Concerning oxygen depletion, algae do not process photosynthesis at night; it is their respiration period. During the mentioned period, algae consume oxygen quickly, so oxygen levels drop rapidly due to their massive number. With these factors, some species may die due to suffocation, which may lead to ecosystem imbalances.

Inland municipal fisheries production in the Philippines was estimated at 205.54 metric tons in the year 2021; it was reduced by 14.7% in the year 2022, estimated at 175.35 thousand metric tons; suddenly, another reduction occurred in the year 2023, 174.58 thousand metric tons was recorded (PSA, 2024). There must be a cause for the considerable reduction. As Santiago (2022) stated, tons of bangus, tilapia, and other fish have died due to fishkill in Lake Tawiran, located at Obanda, Bulacan, in 2022, causing losses for fishers. Within the same year, another fish kill occurred at fishponds in Taal Lake in part of Agoncillo, Batangas, affecting at least 29 tons of tilapia, amounting to P3.2 million (BFAR, 2022). With the ongoing terrible incidents, a massive wave of fish kill hit South Cotabato's Lake Sebu, which began on Wednesday, January 25, 2023, and has affected millions of pesos in tilapia over three days.

The continuing fish kill, known locally as "kamahong," is the third to attack the lake since early January, bringing the total number of afflicted fish cages to 2,023 as of Friday, January 27 (Rebollido, 2023). Fish kills can occur due to several factors, including excessive nutrients leading to algae growth, low levels of dissolved oxygen, toxic compounds, and sustained increases in water temperature. Therefore, it is vital to maintain water quality, manage nutrient levels, promote sustainable practices, and prevent toxic releases to mitigate the occurrence of fish kills in freshwater ecosystems. With the said issues, people should come up with a solution, providing environmental insights to locals and an efficient way of monitoring water quality through modern technologies.

Remote sensing gives a huge perspective when visualizing an entire lake and tracking how it changes over time. The gathered data can be used to

understand how lakes work and determine their water quality (Dörnhöfer, 2016). Sentinel-2 high-resolution satellite images are one of the most efficient ways to detect harmful algal blooms, particularly in small areas (Khalili, 2019). Accessing quality remote sensing data is crucial, but it can be limited or expensive, and the accuracy and applicability of existing tools and algorithms can vary depending on the specific condition of the lake being monitored. Factors such as the depth and size of the lake and the type of water pollutants can influence how accurately these tools perform (Dörnhöfer, 2016). Additionally, there can be an issue regarding using Satellite-based sensors due to climate change; cloud contamination would be the factor, and the low transparency of processed images will be a concern regarding their performance and accuracy (Zhang, 2023).

The researchers came up with the analysis, as mentioned, that rainfalls and storms are causes of eutrophication due to floods or runoffs; it was clearly important that during rainy seasons, monitoring of chlorophyll levels in lakes is a must. However, the remote sensing method has a low accuracy during rainy seasons. To come up with the solution, the researchers executed an on-site deployment of sensors, employing Arduino technologies for data retrieval and transmission. As low accuracy was not a concern, the obtained values of the water quality parameters could be utilized to determine and forecast chlorophyll levels in the lakes. A web-based system, facilitated by access to a database, is implemented to display transmitted data that offers real-time monitoring capabilities.

This study is conducted to address the alarming issue of eutrophication in Lakes, recognizing its impact on water quality and ecosystem health.

Through monitoring the chlorophyll-A level, the study aims to develop a practical method to assess the eutrophication state, mainly focusing on the lakes' vulnerability to nutrient enrichment and algal blooms.

PROJECT OBJECTIVES

The general objective of this capstone project is to develop a web-based water quality monitoring system for the estimation of chlorophyll-A levels.

The specific objectives are as follows:

1. Gather data for a water quality parameter such as pH, dissolved oxygen, temperature, and turbidity through Arduino-based sensors.
2. Develop a web-based system that monitors water quality such as pH, DO, Temp, Turb, and Chl-A.
3. Test and perform an evaluation of the web-based system through a software review using ISO 25010.

PROJECT PURPOSE

The Chloro Watch Web-based System is a significant step toward attaining several Sustainable Development Goals. First, it is aligned with SDG 11: Sustainable Communities and Cities. The system provides continuous, reliable data, empowering local authorities and people to make knowledgeable decisions regarding water quality. This data accessibility promotes a proactive approach to a sustainable ecosystem, critical for developing resilient and sustainable cities.

In addition, the project also contributes directly towards SDG 14 which is Life Below Water. Aquatic ecosystems are the planet's lifeblood, sustaining biodiversity and delivering vital benefits to populations worldwide. This

project is essential for the protection and sustainable use of aquatic resources, ensuring that life below the water continues to thrive for future generations.

Understandably, this specific project relates to SDG2: Zero hunger. Chloro Watch protects the freshwater environment and contributes to Food security initiatives. Poor quality water can also lead to loss of food. This project can reduce starvation through improving ecosystems thus affecting communities' welfare assertively.

This project, therefore, will aim at developing real-time monitoring systems that will come in handy in preserving the weak equilibrium among aquatic organisms in freshwater bodies with regard to persistent water quality problems. The Chloro Watch will enhance surveillance techniques which include current technological developments, thereby aiding in facilitating life within freshwater bodies. The system has the capacity of promoting proactive actions in managing the resource sustainably because it has the ability to generate current accurate data in real-time. Finally, it will enhance perceptions toward aquatic environments and protection so that resilience and health are maintained.

The following are the beneficiaries of the project:

Fisher Folks. Fisher folk will be able to sustain their livelihood yet, raise more cultured fish that could be sold to the public markets. Better, an earlier warning will also benefit them against a HABs and other deteriorating water quality. Algal blooms affect the fish populations due to low levels of oxygen produced, causing toxicity, and resulting in hypoxic conditions, hence fish kill. With this knowledge, the Chloro Watch system may assist the fisher folks to monitor beforehand because it provides information relating to the real-time

changes on the quality of the water. The community can produce more and quality types of food products,

Communities. The community can produce a wider variety and better quality of food products, which in turn enhances their economic stability, with lower cost products as well. As with the previous aspect, this initiative is supportive of the well-being of society, through provision for livelihoods and sustained development.

Future Researchers. This allows future researchers to understand lakes' ecosystem dynamics and how these react to external factors; for example, climate change and nutrients pollution. It will be a ready-made reference for other researchers who take or have an interest in exploring some sort of similar concept.

SCOPE AND LIMITATIONS OF THE STUDY

The project was developed to ensure the sustainability of aquatic water life in the lake of Tadalac in Brgy. Tadalac, Los Baños, Laguna, by continuously monitoring its water quality through devices. The project was built with a database connected to an Arduino-based sensor network so that the web application could access real-time data and provide visualization for obtained water quality values. The following modules are separated into two categories: hardware and software.

Hardware

The project's hardware components are the Arduino-based Sensor Network Module, Wi-Fi Module, and SMS Module.

Arduino-based Sensor Network Module

The Arduino-based Sensor Network Module integrates various sensors to monitor the water quality parameters and chlorophyll concentration in Tadlac Lake.

The module utilized the following sensors:

PH-4502C pH Sensor: The PH-4502C pH Sensor is a reliable and affordable option for measuring water acidity or alkalinity by detecting hydrogen ion concentration. And it provides accurate readings.

Gravity Analog Dissolved Oxygen Sensor: The Gravity Analog Dissolved Oxygen Sensor is a cost-effective choice for monitoring dissolved oxygen levels in lake water. It offers reliable performance.

DS18B20 Waterproof Temperature Sensor: The DS18B20 Waterproof Temperature Sensor features an accurate temperature reading with suitability for any environmental application.

Generic Turbidity Sensor: The Generic Turbidity Sensor is valued for being inexpensive and durable and has effectiveness in the measurement of measuring cloudiness of water bodies over very extended periods.

The module will use an Arduino Uno GeekCreit board to read data off the sensors and transmit it to a web-based system. The Arduino Uno GeekCreit was selected due to its good price and reviews, and reliability. For the wireless data transmission, the module will incorporate a Wi-Fi module, specifically the ESP8266, enabling Chloro Watch to provide up-to-date information about the water quality in the monitored lake.

Wi-Fi Module

The ESP8266 is a low-cost, low-power Wi-Fi microchip that can add Wi-Fi connectivity to the Arduino Uno GeekCreit board; it establishes the connection to the designated database server using appropriate network protocols. The collected data will be transmitted wirelessly to the web-based system.

SMS Module

Providing a notification for agencies through SMS to further enhance the system's monitoring capabilities. The module will continuously monitor the sensors' data and water quality thresholds. When any of the water quality parameters or chlorophyll levels exceed pre-defined critical levels, the module will automatically generate and send SMS alerts to designated contacts, such as local staffs or environmental authorities using GSM-based SIM900A, and trigger an alert indicator to web application, for the users to be informed regarding in water quality condition of the lake. This holistic approach provides appropriate response and intervention toward such water quality issues and protects aquatic ecosystems in time.

Power Source Module

In order to ensure that the parameters of water quality in Tadlac Lake are continuously monitored, a reliable source of power must be secured. For this purpose, it uses 12V solar panel kits and a solar charge controller with a 12V Yuasa NPL24-12l battery. The solar panel will transform sunlight to electricity and distribute the generated electricity to the battery to be stored. The reliability, strength, and capabilities of the battery maintain good performance in the system regardless of cloudy days or at night. A solar charge controller

prevents overcharging or deep discharging of the battery while ensuring the stability of the power supply of the system.

Software

The following software components of the project are the Data Visualization Module, and Data Report Module.

Data Visualization Module

Users can access real-time data on the dashboard. From the data insights, pH, dissolved oxygen, temperature, and turbidity, users can visualize the chlorophyll-A levels through "AquaSense", a soft sensor model for predicting chlorophyll-A levels that has been integrated into the "Chloro Watch" web-based system.

Data Report Module

The module is responsible for processing and analyzing the data collected from the sensors. It will incorporate various data visualization tools, such as graphs that display sensor values, and prediction for chlorophyll-A, tasked with presenting the water quality information in a user-friendly and intuitive manner. Furthermore, users can visualize the data records are presented in table form.

CHAPTER II

THEORETICAL FRAMEWORK OF THE PROJECT

REVIEW OF RELATED LITERATURE, STUDIES, AND SYSTEMS

This chapter discusses the previous researches and existing technologies that are held to serve as a basis regarding water quality and focus on chlorophyll levels. The researchers explore different publications and journals. Additionally, it serves as the foundation for developing and implementing the Chloro Watch System, designed for the specific context and relevance to Lake of Tadolac.

Overview to Eutrophication and Chlorophyll Levels in Lakes

Eutrophication is a phenomenon in which a water body becomes enriched with minerals and nutrients that stimulate the excessive growth of algae and other aquatic plants. This may lead to low oxygen levels in the water, thus aquatic creatures will be harmed. The main sources of these additional nutrients are mostly agricultural runoff, sewage discharge, and other human activities that highly introduce nitrogen and phosphorus into the water (WRMA, 2019).

The subsequent rise in the concentration of chlorophyll in lakes and other water bodies is a prime indicator of eutrophication. Chlorophyll is the pigment that allows plants and algae to carry out photosynthesis and abundant chlorophyll may lead excessive plant life in water (Gökçe, 2016). Consequently, chlorophyll monitoring is one of the basic components in the lake health assessment and the mechanism of eutrophication (Chen, 2021). Increased chlorophyll can trigger algal blooms, contributing to the production of toxins that

can harm fish and humans. Also, excessive decomposition of vegetation can lower water's oxygen levels leading to "dead zones" where animals and other living organisms find it hard to survive (Ogueri, 2016).

Eutrophication and high chlorophyll levels in lakes can be averted in several ways. These are nutrient management for nitrogen and phosphorus reduction, wetland rehabilitation and setting up of natural buffers, and water aeration technology to increase the dissolved oxygen concentration in water (Sfriso, 2020). Hence, it is necessary to know the causes and consequences of eutrophication and methods that are right for measuring chlorophyll levels so that effective management schemes can be developed to protect freshwater ecosystems. (Hussein, 2019).

Identifying nutrient pollution sources and caring for the lake's long-term ecological balance contribute much to managing situations like eutrophication with high chlorophyll content (Smith, 1999). Ramião (2024) has asserted that several sustainable farming practices, including accurate nutrient control and proper soil retention, amongst others, could reduce nutrient runoff from agriculture into nearby waters. Also, cutting down the rates of nutrient flow into lakes and streams can be achieved by adopting wastewater treatment technologies and implementing best management practices in urban and industrial areas.

Significance of Chlorophyll Monitoring

Harmful Algal Blooms (HABs) have become increasingly common in recent decades. This dangerous phenomenon affects the lives of fishers and can harm the region's economy (Khalili, 2019). Detecting HABs is closely connected to chlorophyll monitoring. Chlorophyll is a green pigment in algae,

and its concentration in water is commonly used to measure algal biomass. Algal blooms, particularly harmful ones, are distinguished by their fast development and algae accumulation, which can substantially influence water quality and aquatic ecosystems.

Added to the statement presented, Ramesh (2023) mentioned that HABs are already causing serious issues for fishermen and other people who rely on aquaculture life and fish production. HABs have a wide range of negative consequences for climate change, fishermen, and the Indian economy. For sustainable economic growth and aquatic ecosystem concerns, detecting and monitoring harmful algal blooms in various aquatic areas is important. By recognizing the presence of harmful algae blooms and comprehending them well, communities can forestall possible damage to fishery stocks, tourism or other fisheries-related industries.

HABs are one of the most serious threats to lake water quality because they produce toxins. Remote sensing using Landsat-8 and Sentinel-2 data might be an alternate method of monitoring algae levels in the water (Azarderakhsh, 2021). This meant remote sensing data could be utilized to monitor lake water quality.

Turbid waters, dissolved oxygen (DO) depletion, and animal poisoning can impact public health, fisheries, tourism, monitoring, and management plans. Providing stakeholders and policymakers with quick inexpensive tools to understand, predict and deal with HABs are the solution (Mozo, 2022). Besides, proactive monitoring helps prevent the impacts of algal toxins and oxygen depletion on various aquatic animal life forms. This way, biodiversity and the overall health of aquatic ecosystems may be retained within a framework that

maintains the long-term sustainability of both the environment and economies that depend on the coast for resources.

Water Quality Parameters for Assessing Elevated Chlorophyll Levels

The management of high chlorophyll content in aquatic ecosystems depends on water quality parameters since they give general knowledge about water quality that people have to know about (Bhardwaj et al., 2015). Similarly, advanced ways of determining water quality such as using particular low-cost sensors combined with high-resolution spectrometers for accurate determination of chlorophyll-A and other variables were developed lately (Bhardwaj et al., 2015). Dissolved oxygen, turbidity, pH, E-Coli levels, effective chlorination, biochemical oxygen demand and fluoride - all these parameters to be taken into factors (Bhardwaj 2023). Knowledge about the interconnection of water quality parameters and their correlation with elevated levels of chlorophyll is necessary to properly administer aquatic ecosystems (Pérez- Ruzafa, 2019).

The presence of high levels of chlorophyll may indicate reduced dissolved oxygen due to algal blooms in the water. The measurement of turbidity, also known as clarity, can help understand how much sunlight gets through it because it affects photosynthesis. It can vary depending on suspended particles in the water, such as algae (Kazbar, 2019).

pH level has an important effect on the general health status of aquatic environments since extreme values tend to cause stress or negatively impact aquatic organisms, as well as regulate nutrient availability and chemical processes (Carstensen, 2019). E. Coli levels indicate fecal contamination, thereby putting human health and that of aquatic life at risk. Chlorination efficiency, biochemical oxygen demand, and fluoride values are among other

factors determining the overall status of water quality and can indicate potential sources of pollution and their impact on aquatic ecosystems (Gu, 2021).

Therefore, monitoring and evaluating these water quality indicators can identify concerns and raise the alarm when there are elevated chlorophyll levels. It is important to undertake all-inclusive monitoring and assessing for informed decision-making as well as preventive measures concerning probable issues on water quality. The preservation and sustainable management of aquatic systems can be enhanced through understanding the complex relationships between water quality parameters and elevated chlorophyll levels.

Chlorophyll Monitoring Techniques

Remote sensing technology holds significant promise for monitoring chlorophyll a (Chl-a), a critical indicator of eutrophication in aquatic environments. Nevertheless, the consistent spatial and temporal coverage of remote sensing data is closely tied to limitations in sensor resolution and interference from cloud cover, which obstruct the effective real-time monitoring of water quality in small water bodies (Zhang, 2023). As Seegers (2021) also stated, readily accessible and standardized satellite data sets would greatly ease and enable future endeavours in this field. Furthermore, verified satellite algorithms can establish a historical dataset of inland waters, enabling the assessment of changes in water bodies over time.

Landsat-based spectrals can estimate chlorophyll-a (Chl-a) levels in the water. Through cross-validation of linear machine learning (ML) algorithms, the study assesses the uncertainties in Chl-a estimation. The findings from comparing Chl-a estimates from four indices—normalized difference vegetation index (NDVI), maximum chlorophyll index (MCI), normalized difference

chlorophyll index (NDCI), and multi-linear regression (MLR)—reveal that MLR, a composite of the three spectral indices (NDVI, MCI, and NDCI), which evaluates the combined impact of these variables on Chl-a outcomes, outperformed the other indices. Consequently, the indices' performances, ranked in ascending order based on the coefficient of determination (R^2), are as follows: $MLR > NDCI > MCI > NDVI$ (Mukonza, 2022). It just shows an applicable MLR composite index utilizing satellite remote sensing to monitor water quality in a small reservoir located in southern Taiwan. The findings show the importance of readily available satellite data for assessing water quality in near-real-time.

There is a developing case for using sensor networks to monitor and control the effects of non-point-source pollution, like nutrients seeping from watersheds with intensive agricultural practices, which is increasing. To minimize the irregular distribution of events such as algal blooms, it is important to keep monitoring and enhancing. This can be made possible through sensor-based measuring of parameters like temperature, pH, conductivity, dissolved oxygen and turbidity. (Saleh, 2023). To track the variations in essential parameters, such as evaluating fluctuations in chlorophyll levels and cyanobacterial biomass.

Another use of IoT in water quality monitoring is to allow data transmission over the internet. pH, turbidity, water level, temperature and humidity sensors should all be linked to an Arduino Mega microcontroller unit. This will ensure data is forwarded to an online storage or cloud through the Thing Speak platform. It would be possible to develop a watershed algorithm for water quality detection that works in real-time using these tools. Water

quality detection can also include more factors into its formulas like a dissolved oxygen sensor or an oxygen reduction potential sensor (Zainurin,2022).

Firestore Backend and Real-Time Data Management

To simplify monitoring tasks, IoT-based sensors are utilized to send real-time monitoring data to users through a progressive web application integrated with the Firestore Real-time Database. The data is then stored in cloud storage (Prasetyo, 2020). A progressive web application can monitor the condition of all sensors used. This application gets all sensor data from the Firestore Real-time Database, which is sent using push messaging techniques and microcontrollers' help.

Through Firestore, a real-time database within an IoT setting, particularly with integrated smartphone sensors, presents a practical solution for real-time monitoring applications (Younis, 2023). The capability to evaluate and analyze performance metrics via the Firestore console and Google Cloud monitoring dashboard offers valuable insights for enhancing the database's effectiveness and responsiveness in practical situations. Oo (2019) stated that the Firestore database successfully managed real-time data from numerous sensors. It served as a dependable platform for storing and displaying the data in a user-friendly manner through the Android app and web applications, demonstrating the practicality of employing Firestore for real-time monitoring applications.

Predictive Algorithms for Water Quality Assessment

Exploring new methods for analyzing and predicting water quality (WQ) is crucial. It is advised to incorporate the temporal aspect for predicting WQ patterns, ensuring monitoring of seasonal variations in WQ. Developing highly effective advanced artificial intelligence models for predicting the Water Quality

Index (WQI) using artificial neural networks and deep learning algorithms is recommended (Aldhyani, 2020). Creating a model through advanced artificial intelligence algorithms is crucial for predicting and monitoring water quality, which is essential for safeguarding the environment. This model can be instrumental in forecasting future water quality conditions.

In chlorophyll level monitoring, online sequential extreme learning machine, M5 Prime tree, multilayer perceptron artificial neural network, response surface methodology, and multivariate adaptive regression spline models can be a factor used for daily chlorophyll-A concentration prediction, examining the correlations between Chl-a and various water quality parameters such as water temperature, pH, specific conductance, and turbidity. (Alizamir, 2021). Extreme Learning Machine (ELM) models offer a dependable method for predicting Chl-a values in aquatic settings, providing advantages for ecological and environmental management and the control of algal blooms.

Predicting chlorophyll-a concentrations in regions with a monsoon-like climate found that SVM (Support Vector Machine) had the best prediction performance (Mamun, 2019). Nonetheless, RNN (Recurrent Neural Network) was used to grasp temporal patterns of data for prediction purposes hence it demonstrated better prediction performance compared to other machine learning models like SVM (Shin, 2020). This is a way of showing how predictive modeling in environmental studies is morphing and how non- conventional machine learning methods can be better than those of conventional ones in some cases with different data characteristics that are useful for predicting chlorophyll-a content in places with similar climates. As environmental data becomes increasingly dynamic and multifaceted, embracing such innovative

machine learning techniques will be essential for more accurate and reliable predictions.

SYNTHESIS AND RELEVANCE OF THE STUDY

Agricultural runoff, sewage discharge, and other human-related activities are the essential sources of nutrients which are phosphorus and nitrogen. Elevated chlorophyll levels serve as an indicator of eutrophication and play a crucial role in evaluating the overall condition of a lake. This means that chlorophyll is a measure of the amount of algae growing in a water body. High chlorophyll concentrations can result in harmful algal blooms and oxygen depletion, adversely impacting aquatic life and human well-being. Monitoring and comprehending the parameters for chlorophyll is a must because it facilitates the identification of areas of concern and formulating appropriate management strategies.

Different management controls are implemented for eutrophication and elevated chlorophyll levels, including nutrient management practices, wetland restoration initiatives, and aeration technologies. The researches show the different tools and methodologies available for real-time water quality monitoring and prediction. The integration of satellite-based spectral estimation and IoT-based sensors, coupled with advanced artificial intelligence models and usage of water quality parameters, presents various approaches to understanding and managing chlorophyll-A levels and algal blooms in water bodies. The emphasis on the temporal aspect of water quality monitoring and advanced predictive algorithms further accentuates the nature of environmental research and the potential for more accurate and context-specific predictions.

RELATED LITERATURE MATRIX

Table 1. Matrix of Review of Related Literature

Author/Year	Title	Findings
José Pedro Ramião 2024	Mitigating water pollution in a Portuguese river basin under climate change through agricultural sustainable practices	Future climatic circumstances can affect how well sustainable agriculture techniques reduce nutrients and sediments. This is due to changes in temperature and precipitation.
Lei Zhang 2023	The Impact of Atmospheric Correction Processors in Spatio-Temporal Fusion for Monitoring Chlorophyll-A Concentration in Inland Lakes	Remote sensing technology offers enormous promise for monitoring chlorophyll-A (Chl-a), a key indication of eutrophication in aquatic bodies. However, remote sensing data's spatial and temporal continuity is inextricably linked to sensor resolution limitations and cloud pollution, which hinder highly dynamic monitoring of water quality in inland medium and small water bodies.
Solomon Oluwaseun Akinawo 2023	Eutrophication: Causes, consequences, physical, chemical and biological techniques for mitigation strategies	Using physical, chemical, and biological procedures as mitigation solutions for eutrophication is beneficial, yielding the desired effects.
Solomon Oluwaseun Akinawo 2023	Tilapia losses soar as new fish kill hits South Cotabato lake	Another round of fish kills has hit South Cotabato's Lake Sebu, beginning on Wednesday, January 25, and affecting millions of pesos' worth of tilapia over three days.
Shweta Sharma 2023	Multidisciplinary Approach: Enhanced Agriculture Production in a Sustainable Way	Tools for determining water's physical, chemical, and biological characteristics are parameters used to measure the quality of the water. These elements help assess the overall quality of the water and determine if it is suitable for

		industrial, agricultural, swimming, or drinking purposes.
Mc Jervis Soltura Villaruel 2022	Phytoplankton Community Dynamics in Tadalac Lake, Los Baños, Laguna, Philippines	Poultry farms and construction have witnessed garbage being thrown into the lake, which might substantially contribute to the loading of organic matter. This might explain why the lake remains eutrophic despite the absence of aquaculture operations.
Jay-r Santiago 2022	Tons of Bangus, Tilapia & Fish Dies of Fishkill in Obando, Bulacan	Since Monday, May 16, 2022, numerous fish farms in Barangay Tawiran, Obando, Bulacan, have had significant fish losses due to fish kill. In Tawiran Lake, dead fish have been discovered floating.
BFAR 2022	BFAR reports fish kill in Batangas	At least 29 tons of tilapia, valued at P3.2 million, were affected by fish kill at fishponds in Agoncillo, Batangas, according to a report released Wednesday by the Bureau of Fisheries and Aquatic Resources (BFAR). Low dissolved oxygen levels were the cause of the fish mortality, according to BFAR 4-A Director Sammy Malvas.
Siti Nadhirah Zainurin 2022	Advancements in Monitoring Water Quality Based on Various Sensing Methods	The most recent approaches for monitoring water quality are also examined to identify any shortcomings in the current approaches. The paper discovered that Cyber Physical System's effective integration of computational and physical techniques makes it appropriate for water quality monitoring.
Xi Chen 2021	Remote Sensing Retrieval and Evaluation of Chlorophyll-a	Based on the ENVI software platform, the quantitative remote sensing statistical model was utilized to assess

	Concentration in East Dongting Lake, China	East Dongting Lake's Chl-a content across several seasons and years between 2016 and 2017. According to the findings, the highest Chl-a concentration was 28 mg/m ³ . with sixteen mg/m ³ on average. Throughout the analysis time.
Xi Chen 2021	Benthic microbial assembly and biogeochemical potentials in E. coli - Stressed aquatic ecosystems during plant decomposition	Benthic bacteria are essential to the aquatic ecosystem's biogeochemical balance, particularly for the material cycling during plant breakdown. However, agricultural runoff carrying many typical pathogenic invaders, such as Escherichia coli, constantly threatens those systems in agricultural areas.
Mezzy Rynee Romin 2021	Variations in the Physicochemical Water Parameters and Phytoplankton Community in Coastal Water of Kudat, Sabah, Malaysia	Benthic bacteria are essential to the aquatic ecosystem's biogeochemical balance, particularly for the material cycling during plant breakdown. However, agricultural runoff carrying many typical pathogenic invaders, such as Escherichia coli, constantly threatens those systems in agricultural areas.
Mezzy Rynee Romin 2021	Variations in the Physicochemical Water Parameters and Phytoplankton Community in Coastal Water of Kudat, Sabah, Malaysia	The physicochemical water conditions did not significantly impact the diversity and quantity of phytoplankton. In addition to these factors, the nutrients in the water may also play significant roles in phytoplankton blooming, so this kind of research must address these factors.
Libu Manjakkal 2021	Connected Innovative Sensors, Sensor Deployment, and Intelligent Data Analysis	Artificial intelligence (AI) and other intelligent algorithms might be used for forecasting and standardized data

	for Online Water Quality Monitoring	analysis. An extensive overview of the water quality monitoring sensor, deployment, and analysis technologies is provided.
Adriano Sfriso 2020	Management and Exploitation of Macroalgal Biomass as a Tool for the Recovery of Transitional Water Systems	The welfare of people, marine life, and ecological services are all favored by aquatic angiosperms.
Xujie Yang 2020	Temporal and Spatial Variations of Chlorophyll-A Concentration and Eutrophication Assessment (1987–2018) of Donghu Lake in Wuhan Using Landsat Images	The eutrophication state of Donghu Lake was monitored and assessed using the complete trophic level index (TLI). Donghu Lake had a TLI (Σ) of 63.49 in April 2016, indicating eutrophic conditions based on previous Chl-a concentrations.
Rabar Mohammed Hussein 2020	Eutrophication Process and Water Quality Indices	Many human activities, from large-scale agriculture and industry to commonplace actions like driving vehicles and fertilizing gardens, damage freshwater environments.
Marta Umberto 2020	Remote Sensing Synergy between Ocean Variables: Remotely Sensed Surface Temperature and Chlorophyll Concentration Coherence	Chlorophyll concentration-measuring satellite missions have restricted coverage because of orbital gaps, clouds and aerosols, sun glare, and inadequate retrieval of the geophysical variables.
Vasit Sagan 2020	Monitoring inland water quality using remote sensing: potential and limitations of spectral indices, bio-optical simulations, machine learning, and cloud computing	The paper concludes that anomaly detection using multi-sensor data fusion and virtual constellation in cloud computing is the most promising way to predict impending water pollution outbreaks like algal blooms. This is the first review of deep neural networks for water quality estimation.

Gonzalo Cucho-Padin 2020	Development of low-cost remote sensing tools and methods for supporting smallholder agriculture	Tools for agricultural remote sensing using unmanned aerial vehicles (UAVs) were created and tested to help with decision-making and boost production in underdeveloped nations.
Ramón Martínez 2020	On the Use of an IoT Integrated System for Water Quality Monitoring and Management in Wastewater Treatment Plants	A decentralized smart water quality monitoring system has been designed. It enables instantaneous online communication with many devices spread throughout water treatment facilities. It also offers data analytics and preventive measures to aid in decision-making.
Fan-Lun Chen 2020	Applying a deployment strategy and data analysis model for water quality continuous monitoring and management	The proposed project was Internet of Things water quality monitoring system that incorporates data analysis models, deployment strategies, and water quality sensors. The aim is to provide a standard operating manual for automatic continuous water quality monitoring devices.
Mohammad Hossein Khalili 2019	HARMFUL ALGAL BLOOMS MONITORING USING SENTINEL-2 SATELLITE IMAGES	We can more accurately detect and monitor HABs in tiny areas using sentinel-2 high spatial and temporal resolution satellite photos, the suitable indices, and their related threshold.
Wayne A. Wurtsbaugh 2019	Nutrients, eutrophication, and harmful algal blooms along the freshwater to marine continuum	Scientists have questioned whether phosphorus, nitrogen, or both should be decreased to regulate eutrophication from freshwater to marine, but many management organizations worldwide are increasingly embracing dual control.

Water Resources Mission Area 2019	Nutrients Eutrophication	and Dead algal mats that decompose can give the water an unpleasant taste and smell. The bacteria that cause the mats' disintegration also take up dissolved oxygen, which can occasionally kill fish.
Angel Pérez-Ruzafa 2019	Long-Term Dynamic in Nutrients, Chlorophyll-A, and Water Quality Parameters in a Coastal Lagoon During a Process of Eutrophication for Decades, a Sudden Break and a Relatively Rapid Recovery	Homeostatic characteristics in the Mar Menor lagoon have prevented eutrophication through several mechanisms, including exporting excess production outside the system, directing its production toward the benthic system (maintaining high biomasses of primary producers, filter feeders, and detritivores), and top-down control of the pelagic trophic web exercised by ichthyoplankton and jellyfish.
A. Kazbar 2019	Effect of dissolved oxygen concentration on microalgal culture in photobioreactors	In a torus-shaped photobioreactor operating in chemostat mode, <i>Chlorella vulgaris</i> , a eukaryotic microalgae, was cultured under variable CO ₂ and constant light conditions. The biomass productivity decreased at CO ₂ ≥ 31 g·m ⁻³ , according to the results. The kinetic growth model predicted the effect of CO ₂ by recalculating the specific rate of cofactor renewal on the respiratory chain (JNADH ₂).
Jacob Carstensen 2019	Drivers of pH Variability in Coastal Ecosystems	Land-based inputs are the primary source of the coastal ecosystems' high pH fluctuation. These include nutrients that raise pH and productivity, organic matter that supports excessive respiration that drives acidification, and freshwater imports that generally dilute the alkalinity of saltwater, reducing balancing.

Max Gerhards 2019	Challenges and Future Perspectives of Multi-/Hyperspectral Thermal Infrared Remote Sensing for Crop Water-Stress Detection	In a multi-sensor approach, data from multi- and hyperspectral TIR sensors combined with those from VNIR/SWIR and SIF sensors can offer deep insights into the actual state of plants (water) and the causes of physiological and biochemical changes.
Alexander T. Demetillo 2019	A system for monitoring water quality in a large aquatic area using wireless sensor network technology	The system's basic hardware comprises the microprocessor, wireless communication system, off-the-shelf electrochemical sensors, and customized buoy. It measures the pH, dissolved oxygen, and water temperature within a certain time.
Katja Dörnhöfer 2016	Remote sensing for lake research and monitoring – Recent advances	Lake water quality and ecosystem are impacted by climate change, which is becoming a more obvious stressor for lakes. A thorough analysis summarizing the important variables influencing lakes' responses to climate change.
Didem Gökçe 2016	Algae as an Indicator of Water Quality	Qualitative samples and community assessments of algae are useful for monitoring aquatic ecosystems. Most of the interannual variations in the temporal dynamics of algae in lakes that range from oligotrophic to hyper-eutrophic can be attributed to morphometry, climate, and ecological variables.
Doris Ifeoma Ogueri 2016	Water Quality Characterization and Restoration Measures of University of Nottingham Malaysia Campus (UNMC) Lake	The lake's water quality in Malaysia was acidic at night and alkaline during the day, based on dissolved oxygen and pH values taken in-situ. These data indicated that

oxygen depletion occurred in the lake at night and increased during the day.

Mohammad Haji Gholizadeh 2016	A Comprehensive Review on Water Quality Parameters Estimation Using Remote Sensing Techniques	The qualitative characteristics of waterbodies have been measured extensively using remote sensing methods.
V.H. Smith 1999	Eutrophication: impacts of excess nutrient inputs on freshwater, marine, and terrestrial ecosystems.	The structure and function of the environment have undergone significant changes due to human activity in the centuries following the industrial revolution. The growing human population has increased the pressures on land and marine ecosystems.

CONCEPTUAL FRAMEWORK

The conceptual framework for the Chloro Watch system is centered around a dynamic and responsive design that integrates cutting-edge web technologies with real-time sensor data to monitor and communicate water quality. By leveraging Agile Scrum methodologies, the development process prioritizes flexibility and continuous improvement, ensuring that the system evolves to meet changing user needs and environmental conditions. Central to this framework is the use of Chlorophyll-A soft sensor, which provide essential readings for assessing water health, and a web application that facilitates easy access to this data, empowering users with timely and accurate information. This iterative approach fosters a user-centric, adaptive system that is both resilient and scalable, making it a critical tool for effective water quality management.

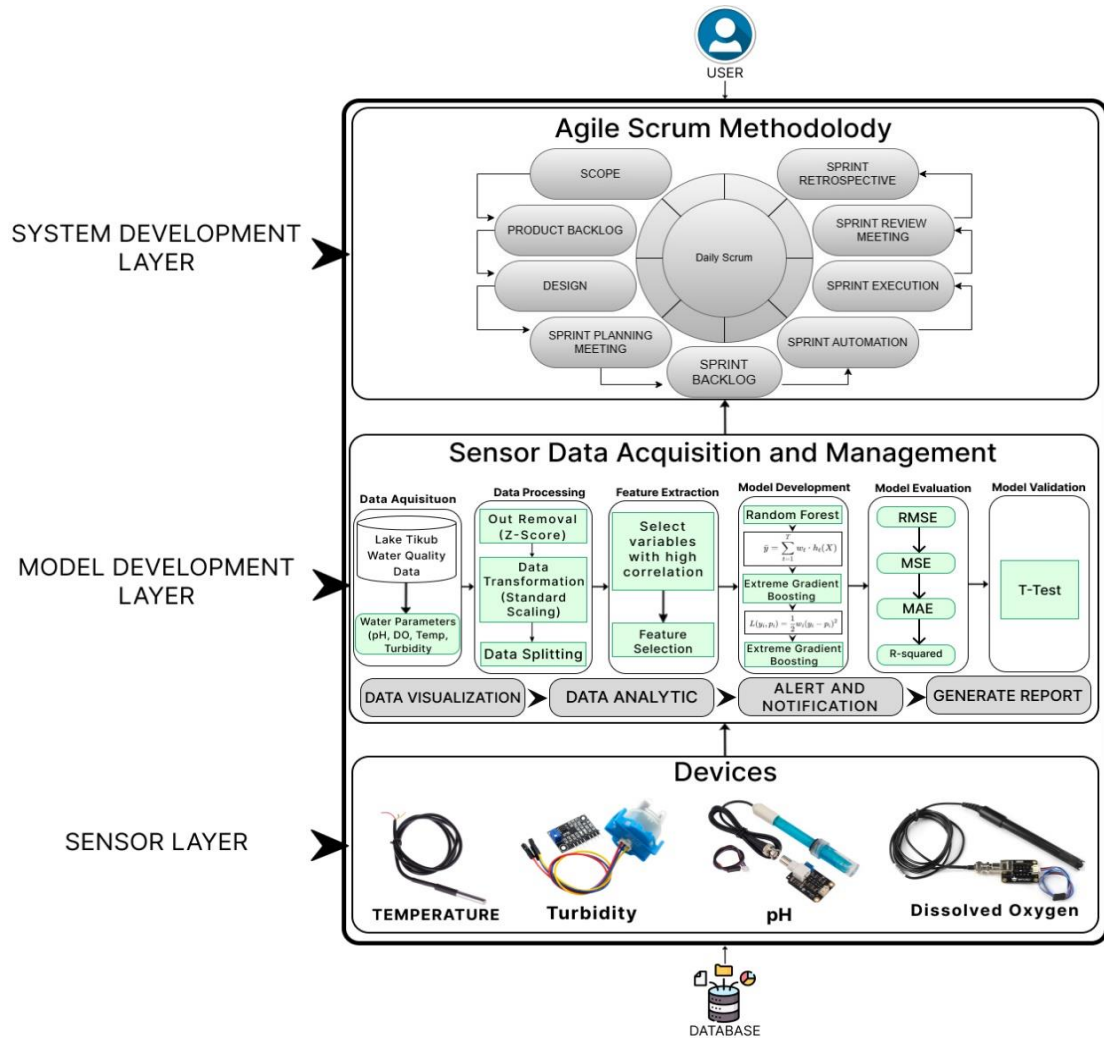


Figure 1. Conceptual Framework

Figure 1 shows the system components. The web application established online accessibility by adopting essential web developments, allowing users to gain specific information regarding water quality. The readings of sensors are needed for AquaSense (Chlorophyll-A soft sensor) to fulfill its function. Incorporating Agile Scrum methodology flows, the development process emphasizes iterative progress and responsiveness to evolving requirements. This iterative approach ensures continuous refinement and adaptation throughout the development lifecycle, creating a resilient and user-centric system.

CHAPTER III

METHODOLOGY

This chapter demonstrated the methodical evaluation of the study's methodology. A proper approach was conducted when developing the web system for water quality monitoring as a guide. This chapter included project development methods for web system creation and device setup, testing procedures, and assessment techniques.

Project Design

The system architecture efficiently connects user requirements with a technical plan for water quality monitoring. It ensures efficient data collection, processing, and presentation, aligning with monitoring and management objectives.

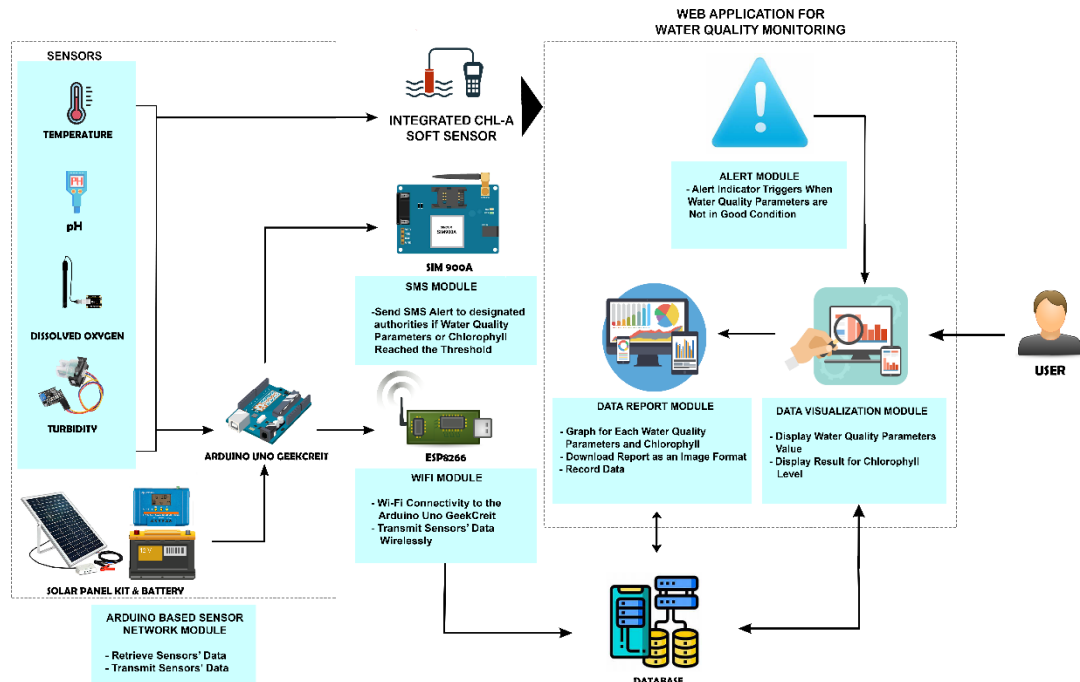


Figure 2. System Architecture

As shown in Figure 2, the web application acts as the interface facilitating user interaction with the monitoring and data retrieval process. When users

interact with the web application, an established connection to sensors is implemented, and they are tasked with measuring water quality parameters, such as pH, dissolved oxygen, temperature, and turbidity in the designated environment. These parameters form the basis for fulfilling the Chlorophyll-A soft sensor model requirements. The parameters' sensors are instrumental in gathering real-time data, subsequently used to evaluate, determine, and predict the chlorophyll levels within the monitored area. The data transmission occurs from the microcontroller device, powered by Solar Panel Kit and Battery. Data is transmitted wirelessly to the database, enabling users to access timely and accurate information regarding water quality parameters and chlorophyll concentration values, especially data reports to determine the changes and behaviour of the data over time and when the threshold value of water quality parameters, and chlorophyll levels exceed, there will be an alert indicator features for users to be notified due to alarming critical levels occurred, and an SMS will be sent to the designated authorities responsible for monitoring the area.

Materials

This section shows the software and hardware equipment used in developing the project.

Software - Listed below is the software developers use to develop the web-based system, including the reasons and features.

Database set up - Firebase Real-time Database is a NoSQL cloud-hosted database that stores data as JSON and synchronizes it across all connected clients in real-time. Any changes made to the database are immediately reflected on all connected devices without the need for manual updates or

refreshments. This feature is handy for collaborative applications where multiple users may access and modify the same data simultaneously.

React - React is a JavaScript framework aligned with creating creative and responsive user interfaces. React is component-based. An example of a component could be a form, form field, or button on a website.

JavaScript - It is mainly utilized to make websites more dynamic and interactive. It can enable the developers to create drop-down menus, image slideshows, forms among other features that promote user's engagement in the website.

HTML – HTML provides the framework for web pages, enabling the creation of headlines, paragraphs, lists, tables, and other elements for organizing content. Using semantic tags like ``<header>``, ``<nav>``, and ``<section>`` improves accessibility and search engine rankings.

Hardware

Table 2. Minimum Requirements for PC or Laptop

MINIMUM REQUIREMENTS FOR PC OR LAPTOP	SPECIFICATION
OS	WINDOWS 10
RAM	4GB
HARD DISK	At least 50GB
PROCESSOR	Intel Intel Core i5

Table 2 briefly shows all the important requirements necessary for smooth and error-free work on a computer or laptop. The Windows 10 operating system is recommended because some components may not be compatible with lower versions of the OS. Additionally, this system needs an internet browser and connection to fully exploit capabilities provided to it by online

service providers. If these basic requirements are met, then efficiency in performance will be guaranteed, and users will be able to maximize system resources according to their individual needs.

Data Analysis

The developer's collaborator conducted descriptive analysis on each variable, which involves calculating key descriptive statistics, including mean and standard deviation. This process provides a detailed understanding of the dispersion of all variables.

Table 3. Descriptive Statistics of Water Quality Parameters in Lake Tikub from September 1, 2022 to September 5, 2023.

	count	Mean	std	min	max
pH	390	7.89	0.6544	6.68	8.98
DO (mg/L)	390	4.72	3.44	0.31	11.83
TDS (mg/L)	390	114.94	27.92	21.5	179.9
Salinity (ppt)	390	0.168	0.147	0.04	0.67
Cond (uS/cm)	390	233.75	45.3	131.1	372
Temp (°C)	390	28.71	2.31	26.4	34.0
BOD (mg/L)	390	1.93	1.02	0.04	5.66
TSS (mg/L)	390	0.018	0.027	0.00	0.30
NO ₂ (ppm)	390	0.045	0.025	0.022	0.37
NO ₃ (ppm)	390	0.057	0.047	0.015	0.40
PO ₄ (ppm)	390	0.541	0.590	0.033	2.35
NH ₄ (ppm)	390	0.897	0.924	0.010	3.93
TN (ppm)	390	0.996	0.933	0.032	4.03
TP (ppm)	390	0.561	0.590	0.053	2.37
BGA-PC (ug/L)	390	0.178	0.210	-0.020	1.03
Chlorophyll (ug/L)	390	3.36	5.79	0.120	57.86
Turbidity (FNU)	390	1.09	1.21	-0.10	7.65
Coliform (CFU/100ml)	390	1041.01	485.27	43.0	1400.0
Cu (ppm)	390	13.63	2.08	8.9	22.4
Fe (ppm)	390	45.42	7.71	25.2	65.9
Mn (ppm)	390	63.28	7.13	46.6	82.7
Zn (ppm)	390	8.65	3.55	4.60	51.9
Cr (ppm)	390	6.53	1.66	2.7	11.8
Cd (ppm)	390	12.42	1.46	9.10	17.7
Hg (ppm)	390	1.01	0.033	1.0	1.2
As (ppm)	390	3.31	0.405	2.0	4.7
Pb (ppm)	390	4.51	0.826	1.90	7.10

The developers' collaborator performs an analysis that shows readings within the normal ranges for a freshwater lake ecosystem. A few examples of these key parameters are pH, dissolved oxygen, conductivity, and temperature, which are the usual values in healthy aquatic environments. Nutrient levels such as nitrogen and phosphorus fall within acceptable limits, indicating a balanced nutrient composition. Also, there is no significant contamination as its concentration remains within the typical ranges of heavy metals like copper, iron, and zinc. Afterwards, by examining each parameter deeply and correlating them with one another, the developers' collaborator sought to find out the parameters that had high correlations among themselves for use in predicting chlorophyll-a levels by developing a model based on such parameters. Acidity (pH), Dissolved Oxygen (DO), Temperature (Temp), and Total Phosphorus (TP) are chosen.

Due to the high cost of total phosphorus sensors, the developers' collaborators identified an alternative parameter. Turbidity was selected as a suitable replacement. Turbidity is closely linked to the presence of suspended algae in the water, and higher turbidity values often correspond to increased algal biomass, which in turn leads to higher chlorophyll-a levels. As a result, turbidity serves as a reliable indicator of potential changes in chlorophyll-a concentrations. Furthermore, the use of turbidity for chlorophyll-a prediction ensures a cost-effective and practical approach to real-time water quality monitoring. Moreover, turbidity sensors are relatively more affordable, easier to deploy, and capable of providing continuous real-time data. This makes turbidity a valuable alternative for assessing water quality in freshwater lakes, especially when resources for more expensive sensors are limited.

Model Development

As the developers' collaborator role in developing the AquaSense (Integrated Chlorophyll Soft Sensor), they recognize that each parameter employed possesses distinct characteristics that influence the model.

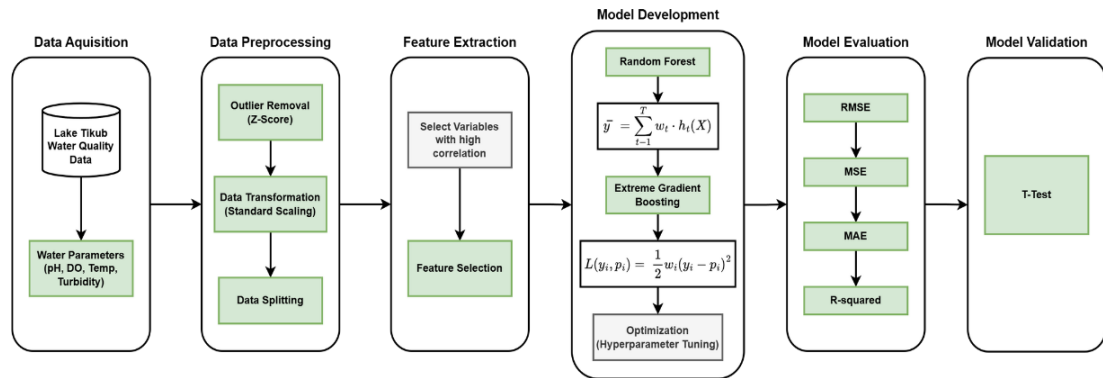


Figure 8. Model Development

To ensure the integrity of the data groups and preserve the original data trends, the developers' collaborator prioritizes the normalization of the parameters. Subsequently, a symmetrical transformation is applied to these normalized parameters, priming them to apply advanced machine-learning techniques. This method marks the initiation of the development phase of the AquaSense model.

Data

The highly correlated parameters necessitate the provision of sensors. The sensors will be responsible for gathering data originating from the lake. To validate the sensors' readings, a laboratory examination was done, which involved the collection of water samples from the same location where the sensors are deployed and comparing the results obtained from laboratory analysis with the readings recorded by the sensors at the time of sample

collection. The process facilitates precise comprehension and visualization for users, enhancing the accessibility and usability of the information provided.

Project Development

The developers skillfully crafted and implemented the web-based solution using the agile scrum software development life cycle approach. Utilizing the agile scrum method, the project's progression was effectively overseen by breaking it down into distinct phases. This is necessary to continuously improve in every phase and proceed with end user participation through the evolving needs-driven development journey.

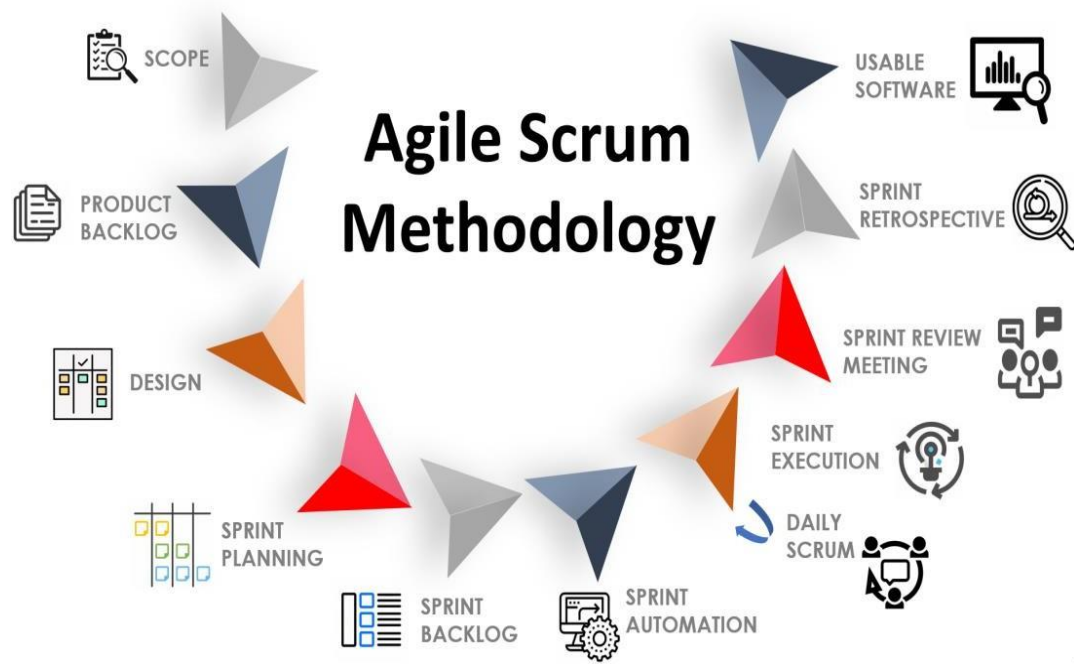


Figure 4. Agile Scrum Methodology

Figure 4 presents a project management methodology focusing on incremental development. The approach calls for dividing a project into pieces that can be accomplished in sprints of two to four weeks. These sprints are meant to prioritize and complete features quickly, which could produce a working system in stages. Projects may be continuously refined and adapted

through this iterative process to move along with changing requirements and feedback. With Agile Scrum, developers use a flexible approach, enabling them to deliver projects wisely, and satisfy more users with their work.

Scope - The water quality monitoring system's main users include agencies, and water management authorities. It was designed to gather and display information about various water quality parameters from monitoring stations on different water bodies like rivers or lakes. Hence, data collectors can use this technology to view all the relevant water quality data they needed for collection. System users such as agencies, researchers, or even standard residents can visit and search through this system to get insights on water conditions for further planning and research objectives.

Design – A strategy was also designed that upgrades the current environmental data collection methodology from a more labor-demanding method of a manual version to the more efficient digital version. Thus, this case's proposed framework optimizes the whole process with better user experience. This method will greatly benefit from idea to eventual implementation of the web application. The storyboard acts as a guide that is sure to ensure each module and feature is well thought out to impact optimally. The plan will also make sure that the developers come up with an intuitive and user-friendly interface based on principles that center around user experience. The core part of this methodology is the iterative development process. By moving from preliminary concepts to the final implementation in separate phases, the developers ensure that every phase allows testing, refinement, and enhancement with feedback and practical application as guides. This gives an

adaptive response to the changing need for the users, a characteristic that no doubt is paramount for this ever-changing world of environmental science.

Product Backlog

Table 4. Product Backlog (for users)

Priority	USER STORY	Est.hr./day
1	As a user, I want to view real-time water quality parameters from Tadalac Lake	3hrs
2	As a user, I want to know real-time chlorophyll-A levels from Tadalac Lake	4hrs
3	As a user, I want to see alert indicator for high Chlorophyll-A level	2hrs

The Product Backlog, presented in Table 4, describes the user stories whose characteristics have been seen to be significant for the project. Such functionalities of user stories provide the guideline for the developers. The hours spent in any activity per day serve as a basis of what's needed to complete properly. This is because the set of activities lies in the beginning stage of a project, thus setting up grounds for further development. It was compiled based on users' requirements and serves as a roadmap for system creation. These user stories are related to workflows and system performance, aiming to ensure the project's success and serve as a comprehensive guide for development tasks.

Sprint Planning

An overview of the plan comprises the general system flowchart, use case diagram, sensors' plan diagram, SMS module diagram, and Arduino circuit diagram. Each diagram contributes to separate areas, starting from user interfaces to sensor integration and data transmittance. They provide an overview of system flows and functional relationships that help in better

understanding its various components. As part of sprint planning, these diagrams help break the project into manageable tasks and allow for better task prioritization. They also enable the team to identify risks and areas that require further refinement.

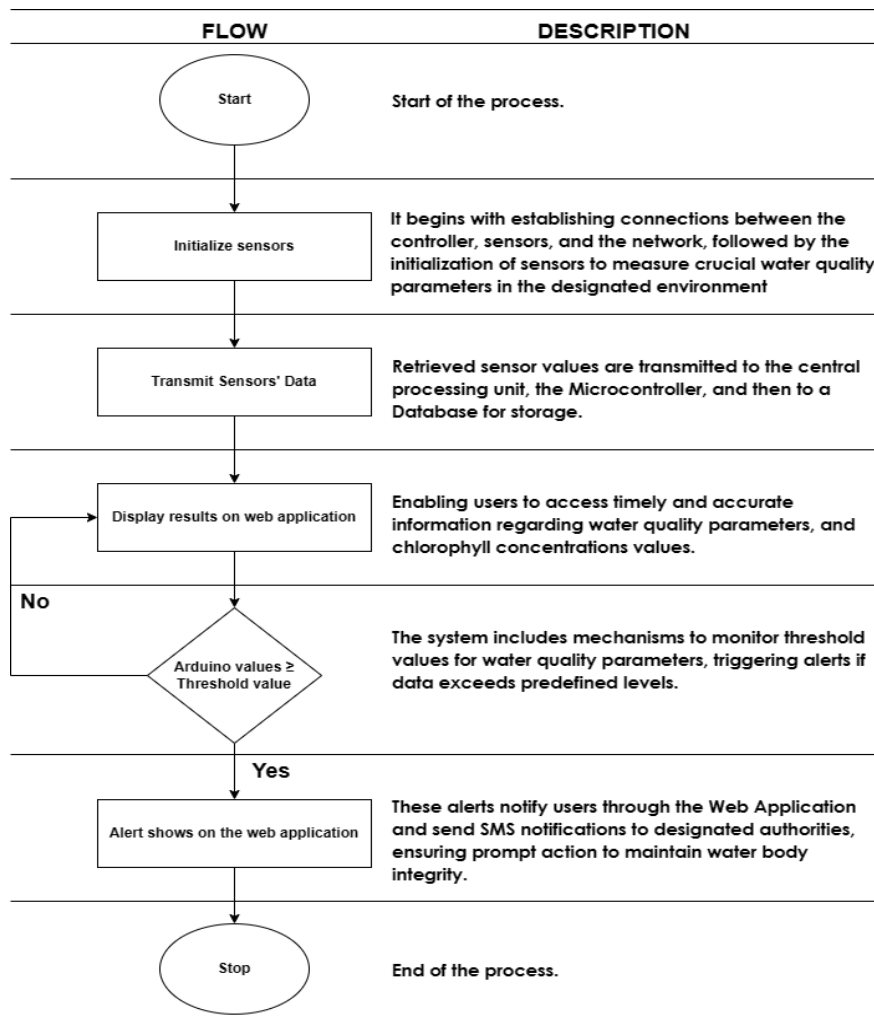


Figure 5. General System Flowchart

The flowchart describes how the web-based water quality monitoring system works regarding sensor data emerging from initiation on a device through transmission, storage, and then display within a web system. Lastly, it has details about how SMS module operates with regards to SMS alerts triggering towards authorities appointed. the use of a web-based system for data visualization means that the information can be accessed by various

stakeholders, including those who may not be directly involved in field monitoring, but who need to be informed about the water quality status regularly. The system ensures a seamless flow of information, from sensors in the field to actionable insights in real-time, enabling both proactive and reactive management of aquatic ecosystems.

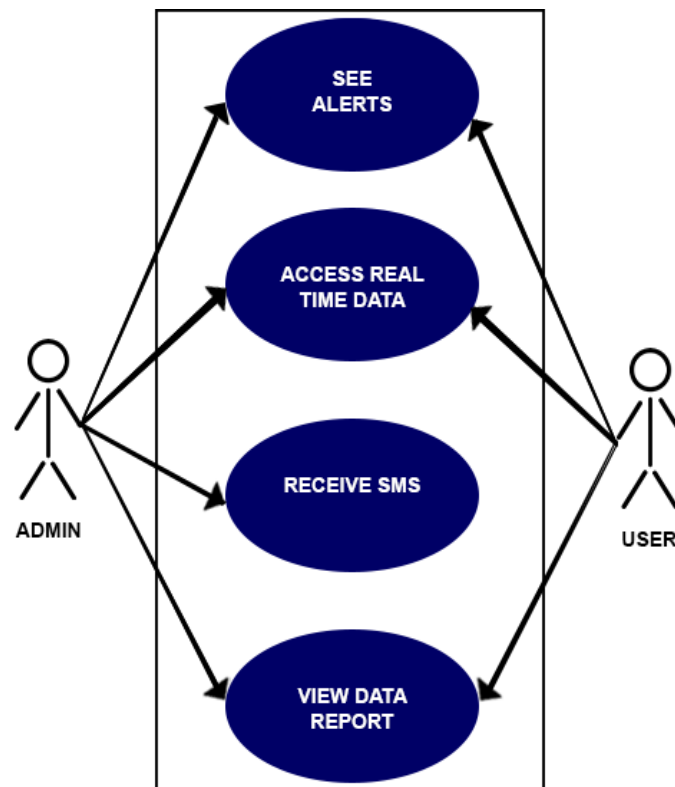


Figure 6. Use Case Diagram

Figure 6 shows the roles of Admin and Users. Admin receives text messages from the system for timely awareness if there are any breaches happening at certain thresholds or other malfunctions occurring in that particular system. Moreover, User is an important player in this case with access to real-time information which enables them to do prompt monitoring of water quality parameters. Both actors are able to benefit through alerts offered by the platform. The admin and users can see alert indicator about threshold breaches and other relevant notices. This diagram presents an activities'

overview on how a system should work in terms of effective water quality monitoring.

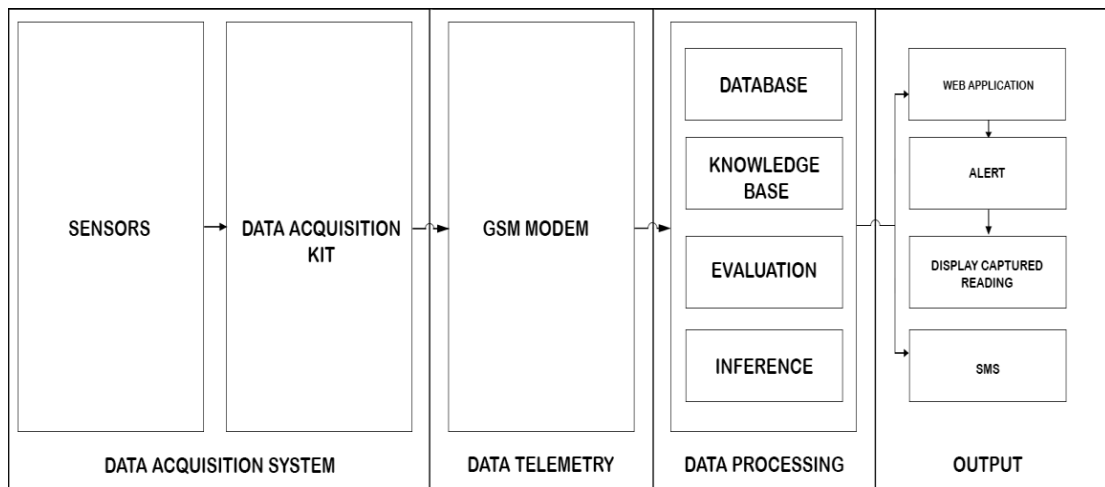


Figure 7. SMS Module Diagram

It presents the relationships between components of the system such as GSM, SMS, and designated authorities. The diagram is a way to show how information flow happens, which is from the data acquisition system, and GSM generates alerts. An SMS module processes these alerts. The process shows efficient communication by sending SMS notifications to the designated authority while the web application show an alert indicator whether the data is high or low referring for each parameters' threshold, ensuring prompt dissemination of critical water quality information. The diagram clearly shows how the system's alert mechanism operates, facilitating effective communication and response to potential water quality issues. In some remote areas, which have limited services, mobile networks are one of the most reliable ways of communication, hence the GSM and SMS modules ensure that all desired information is captured. The redundancy offered by the system is highly certain of alerting the authorities through various means by assuring backup in case one channel fails. This robustness is very important for the

issues of water quality that need quicker response in order to avoid ecological damages or threats to public health.

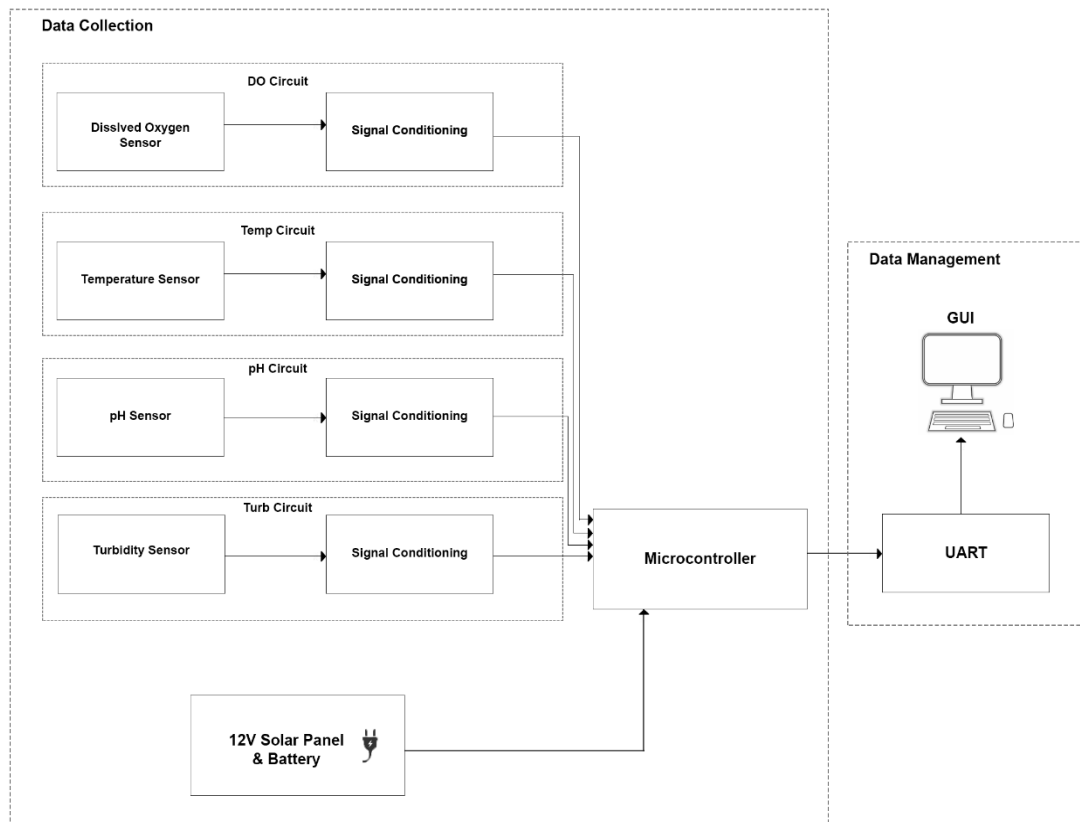


Figure 8. Sensors' Plan Diagram

The sensor plan diagram describes, in general terms, how sensors could be integrated with the Water Quality Monitoring System. It portrays the signal conditioning between sensors for efficient data flows, Arduino Uno GeekCreit as microcontroller, and Graphical User Interface (web application). Each sensor interfaces with a microcontroller for control and data transmission, as shown in the figure above. It also describes how data is obtained and how it correctly establishes and finds its source of power suitable for long periods. The schematic diagram of sensors depicts the graphical design of the sensor network and data flow in the system. This schematic enables the inputting of data sourced from sensors by showing how signal conditioning facilitated.

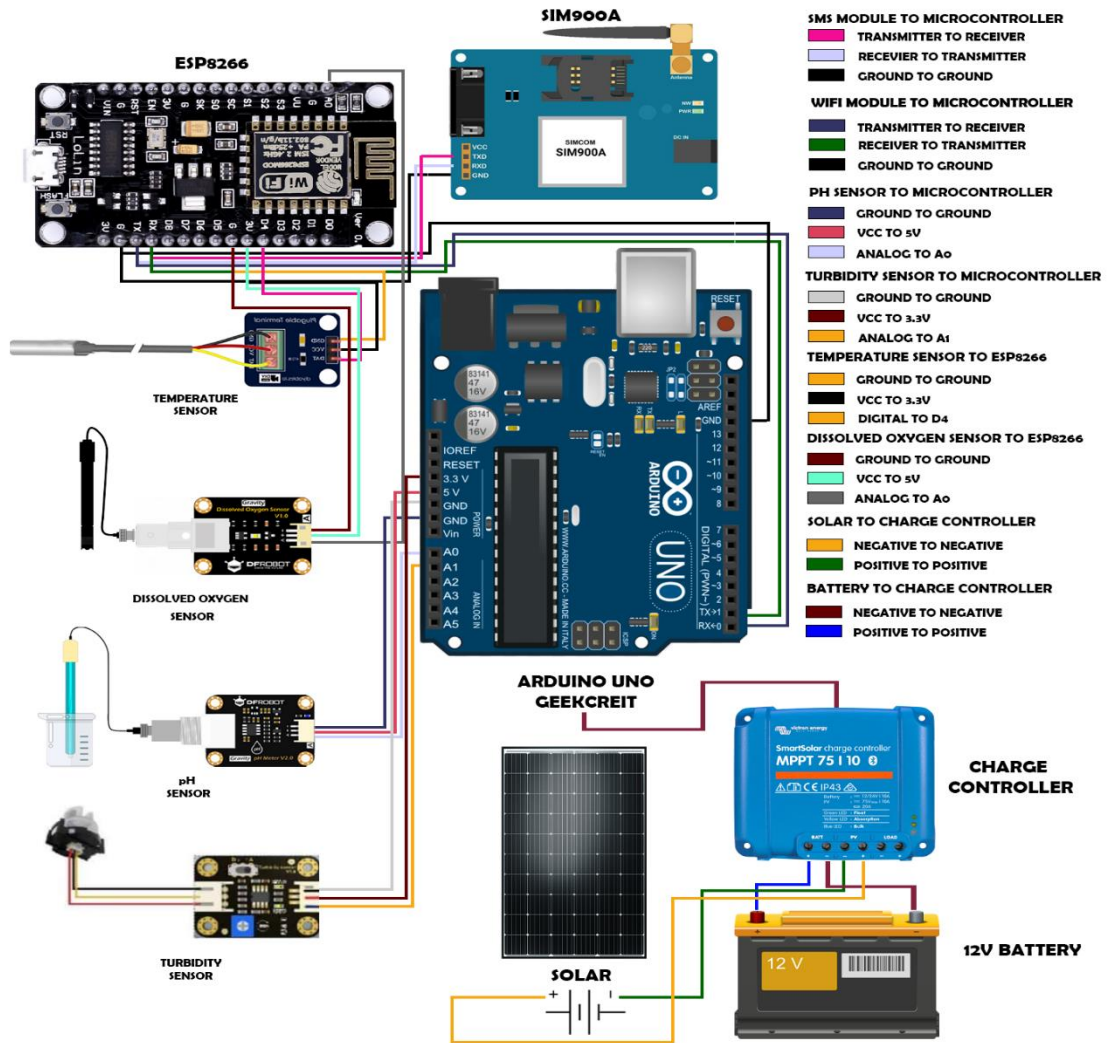


Figure 9. Arduino Circuit Diagram

Figure 9 shows the integration of hardware modules within the water quality monitoring framework. The ESP8266 NodeMCU represents as the central processing unit carrying along an inbuilt Wi-Fi scanner to ensure module to module communication. Through the documentation on cabling process, the developers executed the connection wisely, achieving the desired output for each function, such as the retrieval and transmittal, power source, and communication and interaction. The diagram also emphasizes the careful cabling and wiring process, which is critical for ensuring the reliability and stability of the system. Proper connections allow for efficient data retrieval from

sensors, consistent power supply to the components, and secure communication between modules. By meticulously documenting the wiring process, the developers ensure that each module functions correctly and that the system operates as intended, ensuring long-term durability and robustness for real-time water quality monitoring. This setup lays the foundation for reliable system performance, enabling continuous monitoring and data transmission for water quality analysis.

Table 5. Sprint Backlog (Users)

Priority	Story	Task	Est.hr./day
1	As a user, I want to view real-time water quality parameters from Tatlac Lake.	Set up sensors for pH, temperature, dissolved oxygen, and total phosphorus, and establish an Arduino-based network to connect to sensors and retrieve data.	5hrs
2	As a user, I want to know real-time chlorophyll-A Levels from Tatlac Lake.	Research and select suitable predictive classification algorithms, integrate soft sensor model for chlorophyll-A estimation tool and create frontend interface for the chlorophyll-A level value.	8hrs
3	As a user, I want to receive alerts for high chlorophyll-A levels	Define the threshold for high chlorophyll-A levels and implement an alert system in the web application.	5hrs

This initial phase involves planning the overall perspective and approach to developing the system. Setting up sensors for water quality parameters involves configuring sensors and establishing a network for data retrieval. Each module is developed according to system design specifications. For instance, a soft sensor model for a chlorophyll-A estimation tool and the frontend interface for real-time chlorophyll-A levels are developed. Developers design and develop the user interface according to user requirements. It involves the setup

for plan on calibration of sensors and how the process should done, implementing network and configuration for sensors, furthermore, having an interface for users to view real-time water quality parameters and chlorophyll-A levels.

Daily Scrum. In line with the sprint goals, developers held brief 30- minute daily meetings to discuss progress and consider any possible adjustments to the sprint or product backlog. These meetings helped developers remain focused on their goals throughout the sprint.

Sprint Execution. The developers and collaborators held regular meetings to communicate specific needs and the process of implementing the model into a web application to meet the sprint targets. This collaboration guaranteed that developers understood the deliverables and anticipated the workload ahead.

Sprint Review Meeting. Weekly gatherings were dedicated to reviewing progress and analyzing any essential changes. The developers evaluated the work so far and discussed more features or changes to make sure the functionality of the system was in line with the user requirements.

Sprint Retrospective. The developers emphasized on weekly goals, scrutinizing and analyzing interactions, tools, technologies, and relationships. Front-end development portrayed user-centered design, which made planning improvements easier to track.

Project Testing and Evaluation. To provide quality and integrity to the system, developers implemented testing techniques at several stages of development. Each module and every component was tested to determine the functionality as well as performance of the system. This rigorous testing process included unit testing for individual components, integration testing to ensure seamless

communication between modules, and system-level testing to verify the overall performance and reliability of the water quality monitoring system.

Software Testing

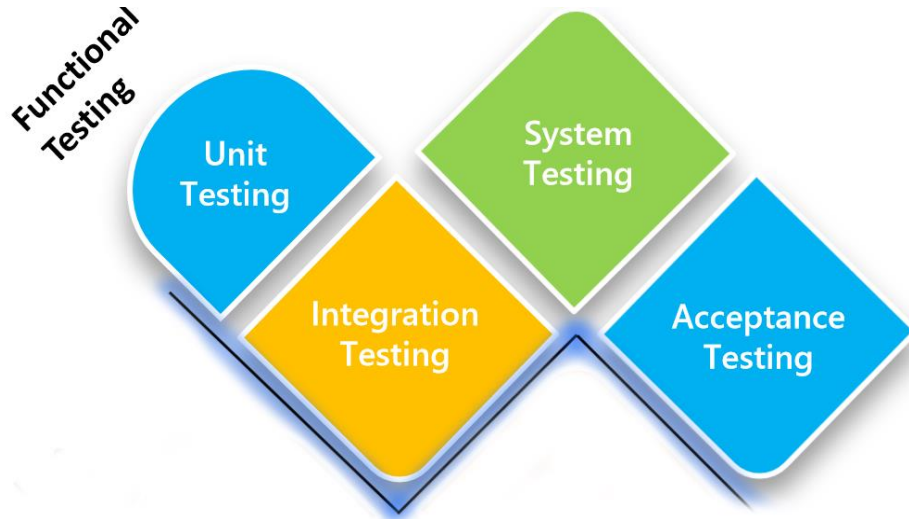


Figure 10. Levels of Testing

Figure 10 shows the various stages of software testing where individual software units or components undergo analyzation. The primary aim of this process is to ensure that each element of the software functions correctly. The figure outlines the levels of software testing, indicating the specific units and components tested at each stage. The objective is to guarantee the proper operation of every software component. The unit refers to the smallest testable part of any software, typically featuring one or more inputs and producing an output. In procedural programming, units might be individual functions, methods, or programs. Object-oriented programming allows the system to be broken down into modular, reusable components called objects. There is an object that stands for each idea in the monitoring system, and this promotes a structured approach to the system, making it scalable by adding new units or modifying existing parts of it. Effective unit testing is a must for achieving

complete knowledge about the whole system for the processes of writing and maintaining code.

Unit Testing

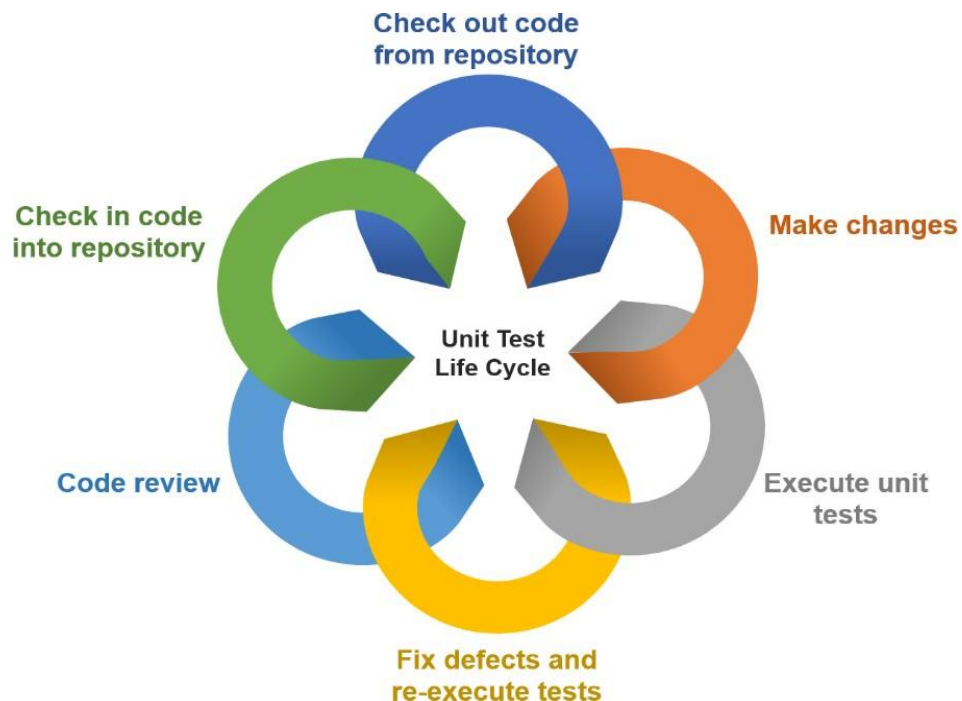


Figure 11. Unit Testing

Unit testing is the process in which particular parts of the codes developed by the developers are isolated to check how a system works. In this manner, the developers can catch and rectify the errors or defects at the right time in the development cycle. This practice apart from improving the overall quality helps in debugging and document creation as well. This therefore evaluates every aspect of the program against the original specifications, taking into consideration each component's functionality and how dependable each unit is about the features and capabilities it can offer. In assessing each unit, developers can therefore assure that each feature works just as expected, hence making the system as reliable and stable as possible. This process is not only for detecting errors but also helps in the debugging process as

developers can easily determine the exact location of any defect, thereby enhancing development and troubleshooting activities.

Integration Testing

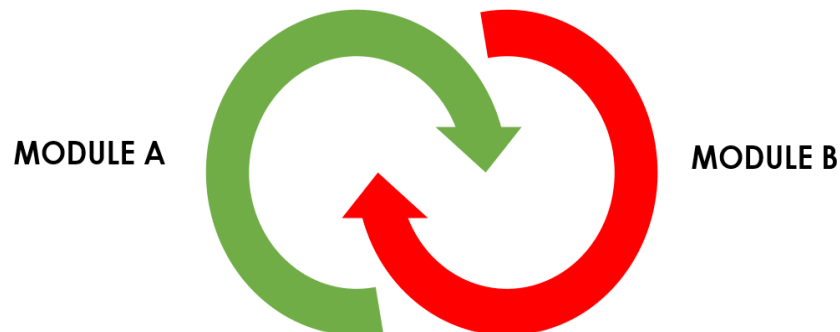


Figure 12. Integration Testing

Figure 12 depicts Integration Testing Process; the process is to test the functionality of a particular module after integrating it with another module. In the integration modules, the developments ensure that every aspect of the system is analyzed in a strict manner so that they do not break down. Developers trace how one module interacts with every component and then address with its functionality and ensure that the modules' behavior is corrected.

System Testing

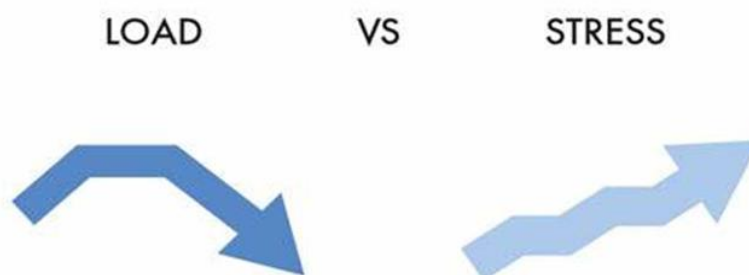


Figure 13. System Testing

In order to find out the robustness and reliability of such this web application meant for water quality metric monitoring, the developers a did

comprehensive stress and load evaluation. Load testing was essentially just running different volumes of data inputs against the system. The idea would be to test the system's performance under normal operations and high usage. Stress and load testing results had come with vital understandings on the application's performance, which resulted in necessary improvements for proper, reliable, real-time data collection, and for user interactions in the dynamic environment.

Acceptance Testing



Figure 14. User Acceptance Testing

The last step is acceptance testing, which is a final inspection of the user-validated project in terms of release-readiness. Here, it is checked whether the system satisfies all specifications according to requirements and acceptance criteria of its users. After this, various test cases are designed that focus largely on functionality and usability. These tests can be performed in any sequence. They can be performed systematically, manually, or automatically. Each test case result will be documented. The User Acceptance Testing can be ready for many purposes such as testing functionality, usability, and general experience a user might have with a system. Necessary changes will be made based on the feedback results compared with the standards set. The specified criteria are accomplished; hence, the system becomes deployment/presentation ready. Therefore, with the attainment of all stipulated criteria and the succeeding test, the final acceptance has been successfully

completed and denotes that it is fully operational, easy to use, and dependable for the intended function and requirement that the system possesses user satisfaction in its entirety.

Evaluation Procedure

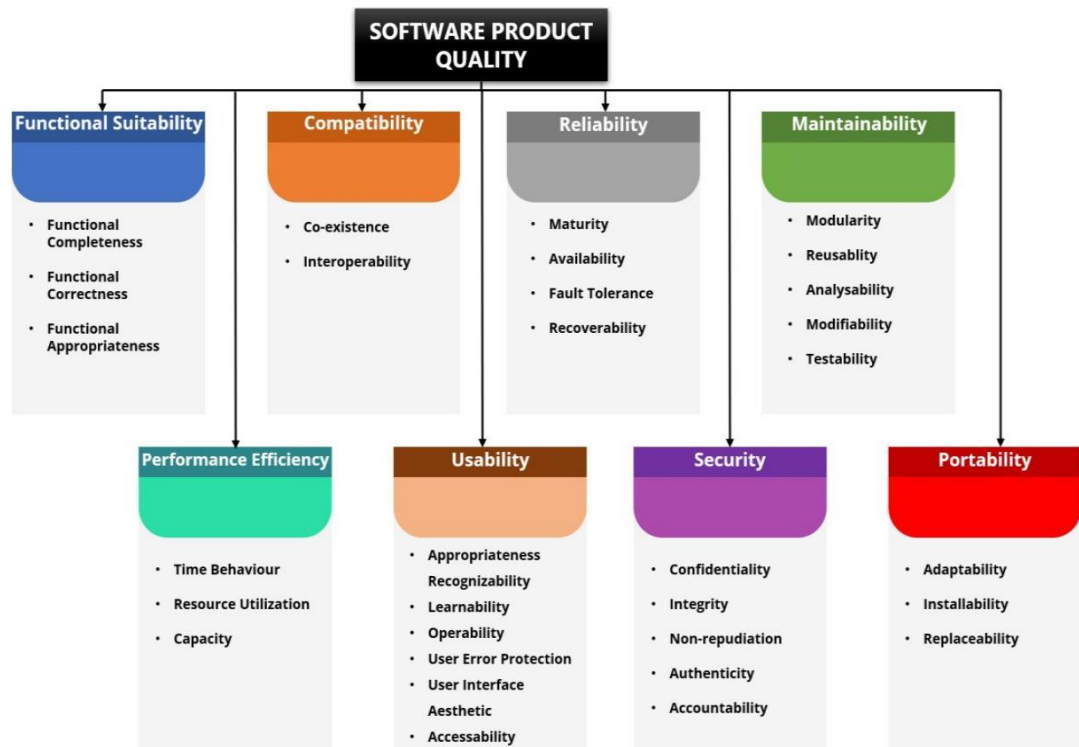


Figure 15. Quality Characteristics of ISO 25010

To ensure the quality of the developed web application, it must adhere to standards such as ISO 25010, which identifies what constitutes a good product. This standard is used as a foundation for assessing the quality of software products. The ISO 25010 model has eight quality characteristics, representing different aspects of a software product for evaluation.

It was imperative to apply ISO 25010 when evaluating the Chloro Watch's functionality and general characteristics. The project's respondents and I.T. professionals used this criterion since it is mostly utilized to appraise program consistency. Thus, assessments based on ISO 25010 include both

functional and non-functional properties such as internal design features, external structure, etc.

Users and I.T. specialists executed beta testing in the developed web-based system during the development process. Likert Scalar adopted the ISO 25010 quality model as a reference point for evaluation purposes. The Chloro Watch Web System testing focused on Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The Likert Scalar responses used in the assessment include SA (Strongly Agree), A (Agree), D (Disagree), and SD (Strongly Disagree).

Population of the Study

Researchers surveyed Brgy to build a system and have precise information. Tadalac, Los Baños, Laguna, with the help of staff, the researchers surveyed 20 users, 3 staff, 2 I.T. This procedure prioritizes empowering authorities to sustain local individuals' lives regarding their living conditions, issues, and potential for change regarding Tadalac Lake.

Table 6. 4-Point Likert Scale Technique

RATING	MEAN RANGE	INTERPRETATION
4	3.25 – 4.00	Strongly Agree
3	2.50 – 3.24	Agree
2	1.75 – 2.49	Not Agree
1	1.00 – 1.74	Strongly Not Agree

The developers utilized the Likert Scale as a survey technique, displayed in the table above, to evaluate user satisfaction with the web application. Utilizing a percentage-based approach, they assess the dimensions of appraisal expressed by respondents.

CHAPTER IV

In this chapter, the study's findings are presented and discussed in relation to the project's goals. The results aim to provide insights into the effectiveness of the implemented strategies and the impact on the outcomes.

RESULTS BY OBJECTIVES

1. Gather data for a water quality parameter such as pH, dissolved oxygen, temperature, and turbidity through Arduino-based sensors.

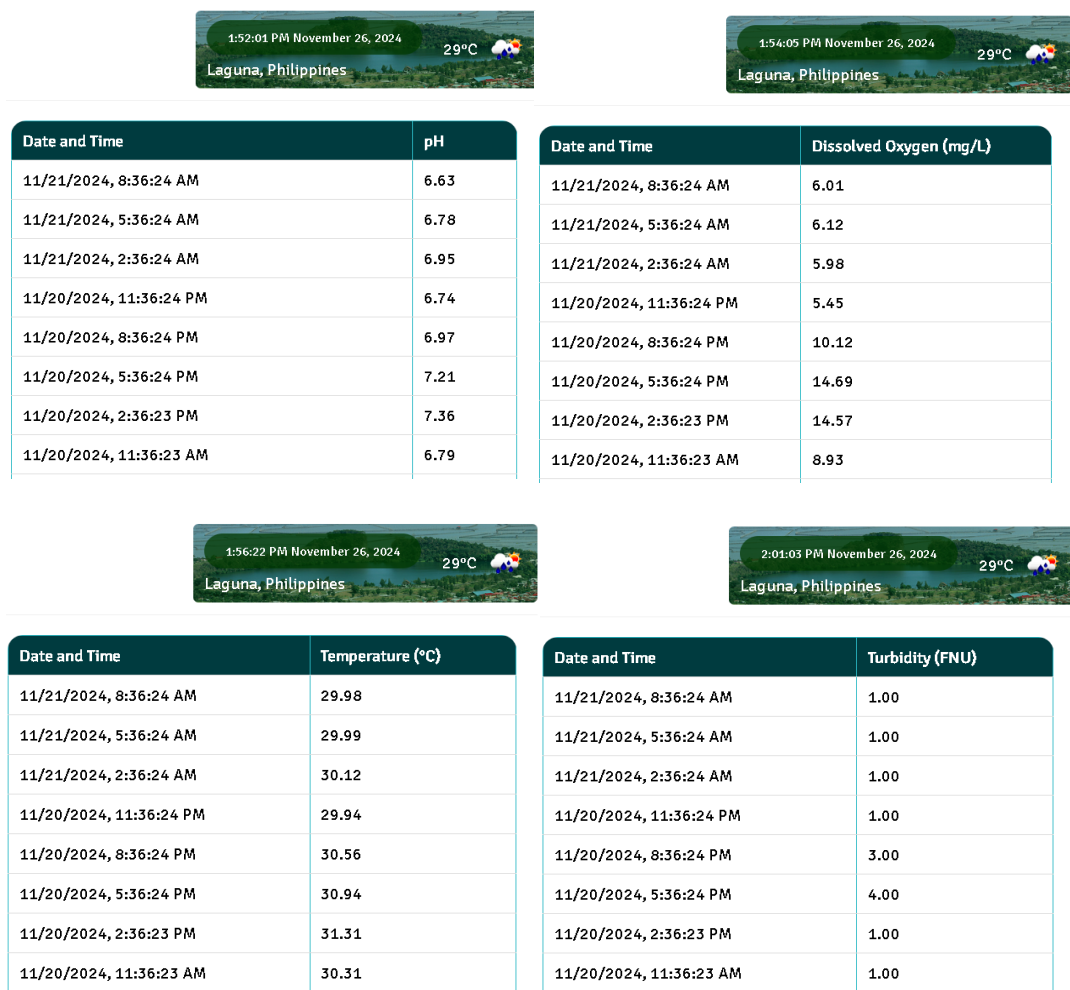


Figure 16. Data Report

These parameters significantly contributed to AquaSense, fulfilling its function with the prediction of Chlorophyll-a. To collect data, the developers used sensors compatible with Arduino. Performing accurate readings is one of

the goals; the sensors undergone in calibration by marking it with a standard scale of readings that supports AquaSense regarding the reliability of the model. The data collected from the sensors represent to train the model in the long run, achieving the desired performance aligned with efficient monitoring of water quality.

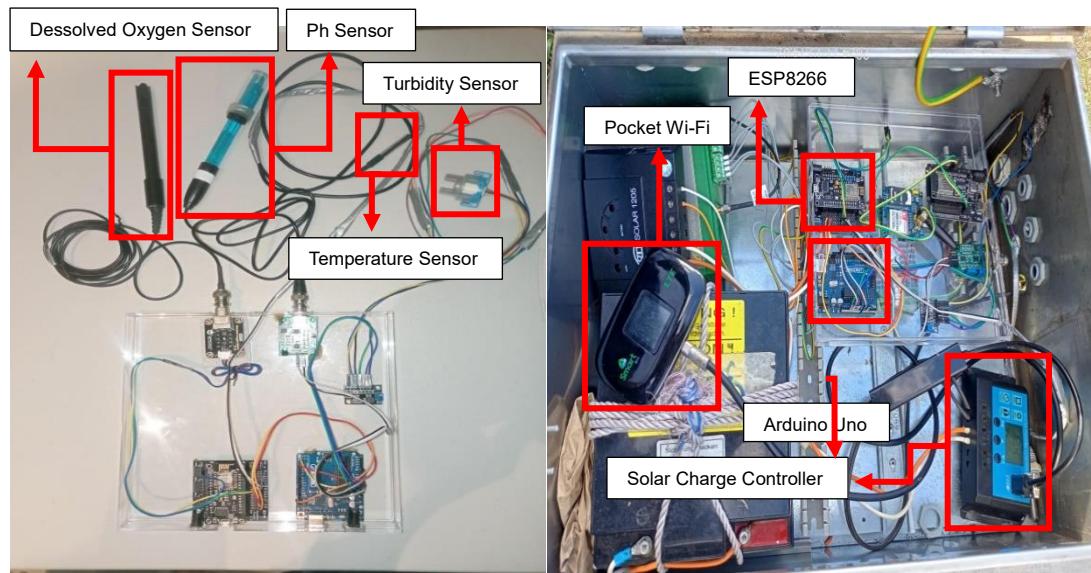


Figure 17. Arduino-based Sensors

As shown in figure 17, the setup is specifically designed to measure key water quality parameters: pH, turbidity, dissolved oxygen, and temperature. To ensure the accuracy and reliability of the readings, each sensor undergoes rigorous calibration prior to deployment. Calibration procedures usually expose the sensors to known standard solutions or controlled conditions that represent known values of each parameter involved. Sensors were connected to an Arduino Uno Geekcreit and ESP8266 microcontroller for data processing. Serial communication method was used to transfer data from Arduino to ESP. Using an ESP8266, pocket Wi-Fi, the system transmits real-time readings directly into the Firebase database, which does permit instant remote monitoring and access; uses that sensor reading with a data-informed

approach in predicting chlorophyll, making sure timely action-oriented results are provided to the water resources management.



Figure 18. Battery, Solar Panel, and Large Buoy

A reliable and renewable source of energy has been provided to ensure that the water quality monitoring system installed at Tadalac Lake will continue its operations without interruption. The system uses a 12V solar panel that is combined with a solar charge controller and a 12V Yuasa NPL24-12I battery. The solar panel is mounted on a large buoy where the battery, the solar charge controller, and other hardware devices are also located. Thus, establishing this reliable and very effective energy setup for monitoring apparatus becomes feasible. Apart from the ease of deployment, attaching the solar panel, battery, and charge controller into the structure of the buoy ensures the steadiness and ruggedness of a power source system. This will ensure that the monitoring system remains operational at all times, even in remote locations where other forms of energy cannot be used. In addition, the purpose of renewable energy allows the system to operate independently, without requiring human intervention.

2. Develop a web-based system that monitors water quality such as pH, DO, Temp, Turb, and Chl-A.

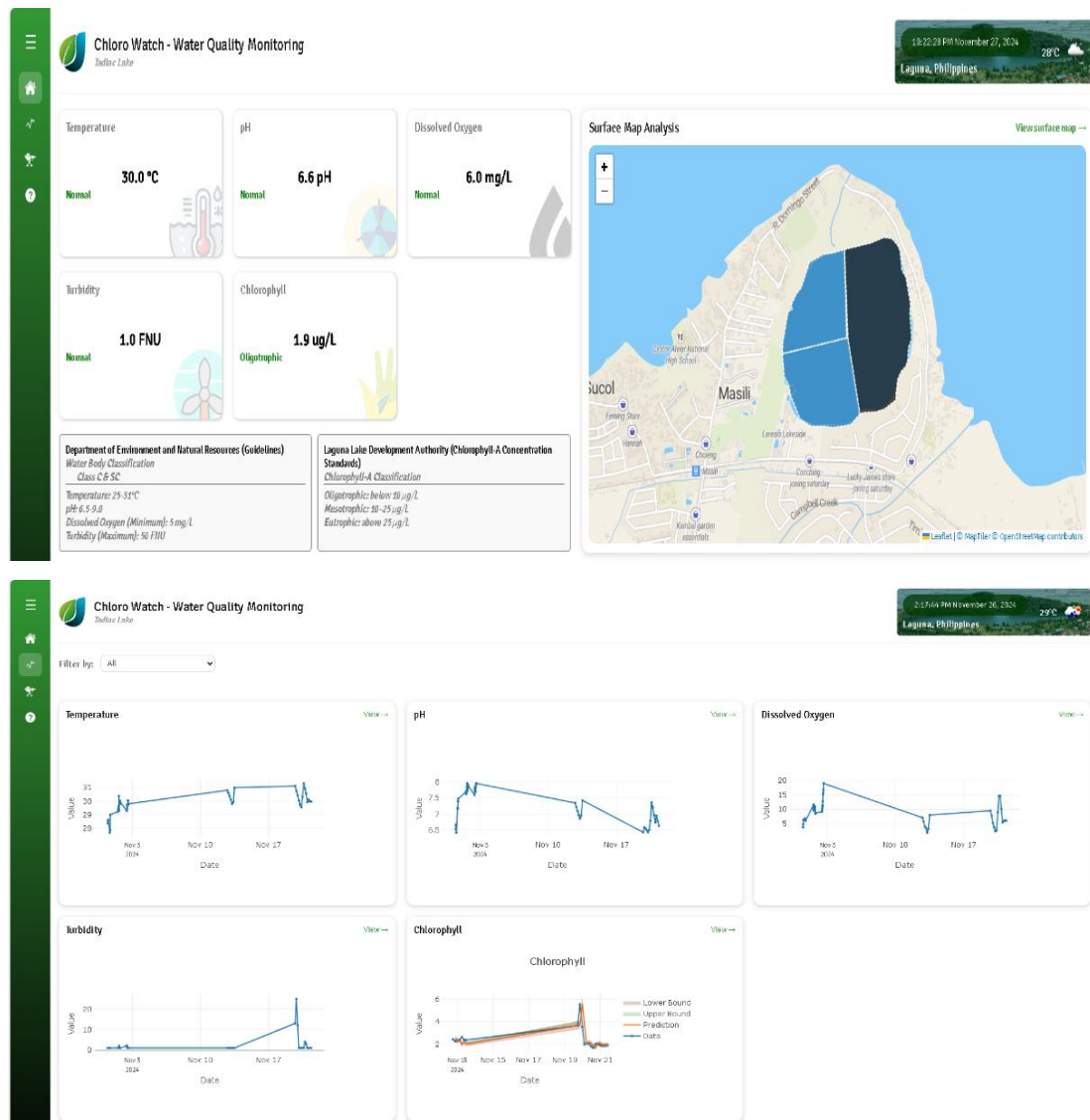


Figure 19. Data Visualization

The system was designed to monitor various parameters; as shown in the figure above, the dashboard shows parameters' values to easily know the current condition of the lake at a given time. The data showcased in the figure provide a comprehensive overview of water quality conditions. Parameter graphs can provide data records and insights regarding the behavior of the lake. The platform puts users by providing them with control over the analysis.

Overall, this comprehensive monitoring system enables effective assessment and management of the lake's health and behavior over time.

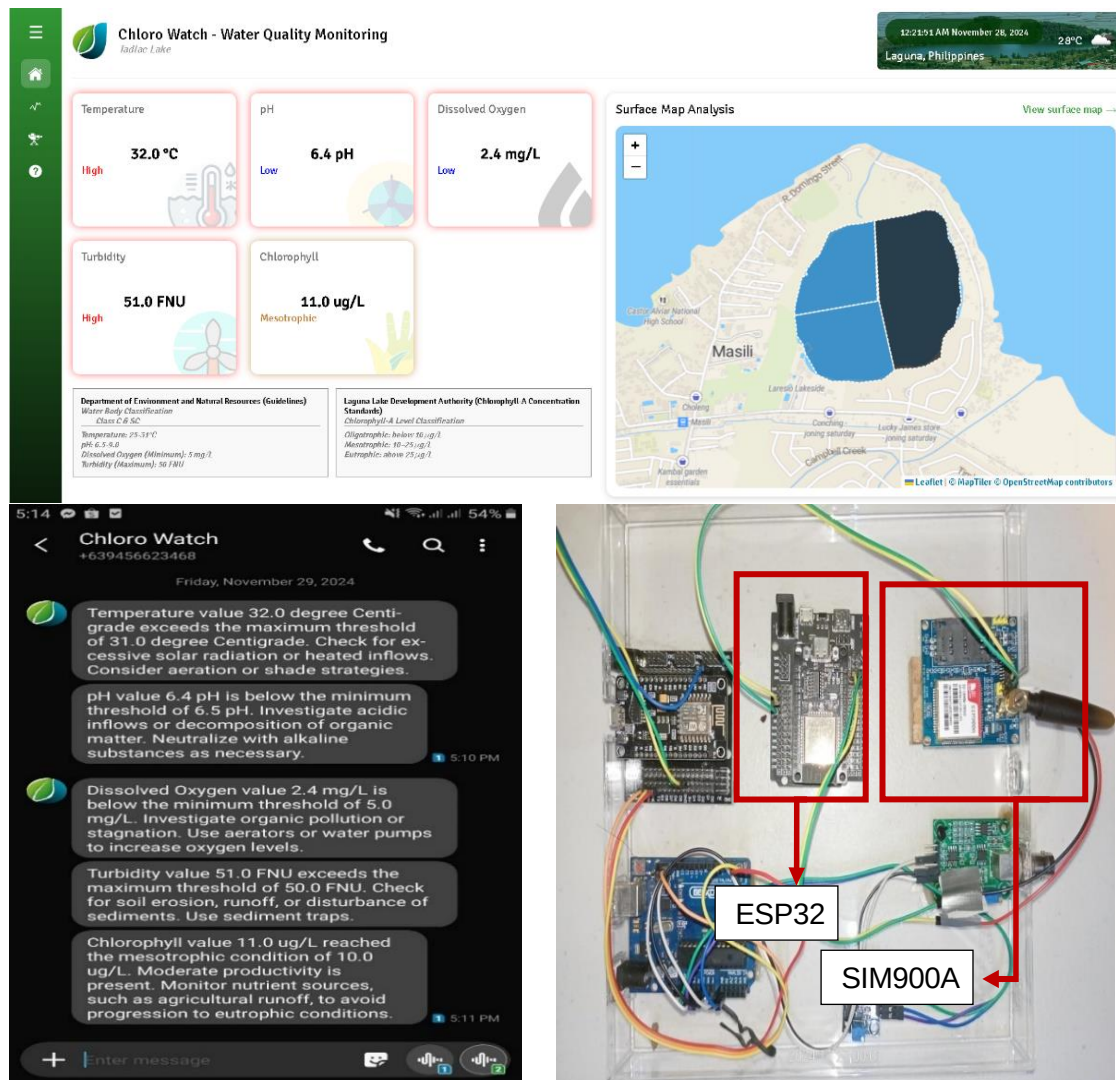


Figure 20. Alert Indicator and GSM-Based SMS

The SMS Module, as depicted in figure 20, is an integral component of the system designed to ensure notification. The developers utilize hardware serial communication between the SIM900A and the ESP32 microcontroller. Sensors' data, including pH, dissolved oxygen, temperature, and turbidity, are transmitted to Firebase via an ESP8266. The ESP32 retrieves these readings, analyzes them against predefined thresholds, and, upon detecting a breach, uses the SIM900A to send real-time SMS alerts to authorities. The system

ensures swift dissemination of critical information. Specifically in monitoring chlorophyll levels, while mesotrophic conditions represent moderate productivity. The decision to alert at mesotrophic levels reflects an evidence-based strategy prioritizing prevention over remediation.

3. Test and perform an evaluation of the web-based system through a software review using ISO 25010.

Survey Questionnaire Results

The results are collected from the respondents, who are staffs from Barangay Tadalac, potential users, and I.T.

Table 7. Post Evaluation Result.

ITEM	MEAN	SD	INTERPRETATION
Functional Completeness Functional	3.64	0.52	Strongly Agree
Correctness	2.96	0.95	Agree
Functional Appropriateness Time	3.36	0.61	Strongly Agree
Behaviour	3.12	0.66	Agree
Resource Utilization	3.20	0.63	Agree
Capacity	3.04	0.64	Agree
Co-existence	3.16	0.65	Agree
Appropriateness Recognizability	3.24	0.72	Agree
Learnability	2.96	0.86	Agree
Operability	2.96	0.86	Agree
Use Error Protect	3.36	0.60	Strongly Agree
User Interface	3.28	0.65	Strongly Agree
Aesthetic	3.36	0.63	Strongly Agree
Accessibility	3.28	0.63	Strongly Agree
Maturity	3.23	0.68	Agree
Availability	3.23	0.70	Agree
Fault Tolerance	3.62	0.52	Strongly Agree
Recoverability	3.48	0.61	Strongly Agree
Confidential	3.27	0.45	Strongly Agree
Integrity	3.40	0.57	Strongly Agree
Non-Repudiation	3.35	0.65	Strongly Agree
Accountability	3.40	0.56	Strongly Agree
Authenticity	3.40	0.56	Strongly Agree
Modularity	3.40	0.56	Strongly Agree
Reusability	3.40	0.56	Strongly Agree
Analyzability	3.40	0.56	Strongly Agree
Modifiability	2.88	0.80	Agree
Testability	3.23	0.67	Agree
Adaptability	3.27	0.64	Strongly Agree
Installability	3.40	0.65	Strongly Agree
Replaceability	3.35	0.61	Strongly Agree
Total	3.25	0.60	Strongly Agree

Table 7 shows the respondents' overall evaluations and the developed system's acceptance level. The system achieved high scores in functional completeness, correctness, and appropriateness, confirming its ability to monitor water quality and generate threshold-based alerts reliably. Performance metrics highlight its efficiency in real-time data transmission, low-latency alerting, and resource utilization. Compatibility with existing hardware and scalability for future enhancements further emphasize its robustness. Usability scores reflect an intuitive interface that facilitates ease of operation for stakeholders, while reliability assessments confirm consistent availability and fault tolerance. Respondents highlighted the system's potential to improve proactive environmental management significantly. The user-friendly interface ensures smooth navigation and minimizes errors during operation. The system's ability to integrate seamlessly with existing workflows boosts its adoption potential. Its scalability offers opportunities for broader applications in water quality monitoring initiatives. The system was also commended for its fault tolerance, ensuring continued functionality even in challenging conditions. Feedback from respondents indicates strong satisfaction with the system's security features, such as confidentiality and integrity. Moreover, the system's modular design was praised for enabling easier maintenance and future upgrades. The weighted mean of 3.25 was interpreted as strongly agreeing, indicating that the respondents confirmed the system's functionality and that the system passed the software evaluation based on ISO 25010. Thus, the results confirm that the system is, indeed, an efficient proactive water quality management system.

T-test Result

Table 8. Data Collection for Testing of Difference of Water Quality Parameter

Group	PH	TURB	TEMP	DO
1	8.38	1.01	30.94	14.77
2	8.90	3.91	31.70	11.78
1	8.34	1.02	30.83	14.09
2	8.91	4.16	31.70	12.36
1	8.53	1.01	31.27	15.17
2	8.98	4.26	31.80	12.56
1	8.62	1.03	31.69	14.49
2	8.91	4.66	31.70	13.00
1	7.52	1.04	28.38	3.70
2	7.83	4.66	29.10	2.97
1	7.64	1.02	28.56	4.79
2	7.58	4.20	28.75	3.24
1	7.67	1.05	28.63	6.33
2	7.61	4.36	28.65	3.89
1	7.42	1.01	28.63	6.33
2	7.68	4.17	28.60	4.72
1	7.65	1.07	28.69	6.62
2	7.70	4.70	28.80	4.98

Data shown in Table 8, the comparison of water quality parameters measured by two different sensor systems Group 1 refers to the reading captured using Arduino-based sensors Group 2 refers to the measurement captured from the laboratory sensors. Samples of water have been collected from Tadalac Lake. Data sets from Group 1 and Group 2 are being used to conduct a t-test, which is the statistical method of determining if there is any significant difference between the measurements taken by the two sensor systems. This analysis concluded that the Arduino sensor system could be used as reliable in the water quality assessment. This is very crucial because it indicates the relevance of this Arduino-based sensor system in real-world applications. It would serve as a cost-effective, effective alternative to the regular methods of laboratory monitoring yet retaining accuracy and reliability. This validation builds the credibility of the Arduino-based system.

Table 9. Independent Samples T-Test Result

	T	df	p
pH	-0.951	16	0.356
Turbidity	-35.984	16	< .001*
Temperature	-0.507	16	0.619
Dissolved Oxygen	0.842	16	0.412

*P Value significance level <0.05 * = weak, <0.01 ** = moderate, <0.001 *** = strong*

From the analysis above, it is observed that the sensors based on Arduino is good enough to determine pH and temperature while for dissolved oxygen, it is satisfactory but with high variation in turbidity measurements. This means that the Arduino system is partly efficient in water quality assessment; however, there is more improvement needed in recalibrating the turbidity sensor to be more accurate and comparable with laboratory standards. This may be due to the fact that the generic turbidity sensor that is being used with the Arduino is intrinsically not as approximate as the laboratory specialized equipment, which might provide a less precise reading. Specifically, enhancing the calibration of the generic turbidity sensor or exploring higher-precision alternatives could improve measurement accuracy and alignment with laboratory standards. These improvements would increase the system's overall accuracy and credibility, ensuring that it delivers valid data for water quality management, environmental protection, and regulatory compliance.

Results of Software Testing

Table 10. Unit Testing for Web Application and Devices

Module	Expected Output	Actual Output	Response Time/Duration	Remarks
Wi-Fi	The esp8266 can transfer the sensors' readings into the database.	The web application successfully transfers the readings into the database.	1.2 seconds	Passed
Data Visualization	The web application can display graphical views of the history of the data.	The web application successfully displayed the graphs for each water quality parameter.	2.5 seconds	Passed
Alert Indicator	The web application can alert the user if there is an exceeding threshold for each water quality parameter, whether high or low.	The web application successfully alerts the user if there is an exceeding threshold.	1 second	Passed
Data Report	The web application can record the data.	The web application successfully recorded the data.	2.5 seconds	Passed
SMS	The sim900a can send an SMS to designated authorities when the readings' threshold value exceeds.	The sim900a successfully sent an SMS.	3.5 seconds	Passed
Power Source	The devices should run continuously on battery, even during periods of no sunlight.	The system successfully operates the battery without interruption, even during low light.	14 hrs	Passed

The findings were obtained through testing on the system's units. The results indicate that all modules have passed the tests successfully, with response times well within acceptable limits. The system is efficient and reliable in transferring, visualizing, and alerting processes and even communication; therefore, functional requirements to adequately monitor parameters in water quality are all met. This, consequently confirms that the system fully operates and each module properly operates. The successful test outcomes also underscore the system's potential for future scalability.

Table 11. Integration Testing

Description	Expected Output	Actual Output	Response Time	Remarks
Software	The web application can show data on water quality parameters through graphs.	The web application successfully displayed data through graphs.	1.5 seconds	Passed
Hardware	The device can transmit the water quality data to a database into the web application.	The device successfully displayed the water quality data to the web application.	1.5 seconds	Passed

The integration testing confirms that both the software and hardware components are functioning correctly when combined. The software has efficiently rendered graphical representations of water quality data, and the data communication from the hardware to the web application is also efficiently done by both, hence completing the needed response times of 1.5 seconds. These results exhibit that the integration of software with hardware successfully meets the set requirements for the performance of the system. Besides, since everything seems to be running fine when tested on this system, it also matches

the designed needs very closely. However, to achieve continuous system accuracy and prevent any level of performance deterioration, maintenance of the software as well as hardware parts must be done periodically. These will help alleviate possible issues, and ensure optimal system operation.

Table 12. Stress Test and Load Test under System Testing

Description	Steps	Expected Output	Actual Output	Remarks
Stress Test: 1. High Sensor Data Input	Simulate a high quantity of sensor inputs	The system should maintain real-time monitoring for all parameters.	The system successfully maintains real-time monitoring for all parameters.	Passed
Stress Test: 2. Database performance	Assess database performance under load	The database should handle transactions without delay	No delay observed in database write operations	Passed
Load Test 3. Continuous Data Input	Simulate continuous data from the sensor with a fast time interval	Data should be logged accurately without loss	All data logged without loss	Passed

The stress and load tests provided evidence that the system is reliable concerning heavy data loads and continuous inputs. In real-time during the stress test, high sensor data input, the transactions of the database did not face delays. Another test done on the ability of the system was based on continuous sensor data logging. The test continued to increase the input data but never encounter data loss. These results show that the system scales up robustly, which allows it to meet all demands of high-performance and continuously processing data environments. However, reliability becomes highly relevant in applications where real-time monitoring and data integrity are essential. Finally,

periodic maintenance will be indispensable toward ensuring the system remains competent to deal with expanding loads of data over time.

Table 13. Acceptance Testing

CHARACTERISTICS	MEAN	SD	INTERPRETATION
Functionality	3.32	0.69	Strongly Acceptable
Performance Efficiency	3.12	0.64	Acceptable
Compatibility	3.20	0.69	Acceptable
Usability	3.20	0.70	Acceptable
Reliability	3.37	0.63	Strongly Acceptable
Security	3.36	0.56	Strongly Acceptable
Maintainability	3.26	0.64	Strongly Acceptable
Portability	3.34	0.63	Strongly Acceptable
Overall	3.27	0.65	Strongly Acceptable

The acceptance test results are indicative of the software that strongly performs on the critical dimensions of functionality, reliability, security, maintainability, and portability, rated as "Strongly Acceptable." Such ratings indicate that indeed the software meets the objectives and is reliable in terms of operation under a number of conditions. Although the software rates as "Acceptable" in terms of performance efficiency, compatibility, and usability, these areas could need further improvements to enhance the user experience. In general, with an average rating of 3.27 and a standard deviation of 0.65, the software is "Strongly Acceptable," meaning it was mostly accepted and can be regarded as appropriate for fulfilling expectations concerning water quality monitoring. These results reflect the robustness and effectiveness of the developed system in addressing core operational needs.

CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATION

This chapter shows the overview of the study's findings and improvements, and articulates conclusions drawn from the system's development. It further explains the possible optimization with emphasis on areas where the system can be expanded or refined to meet changing user requirements and technological changes.

Summary

The initiative addresses the urgent global issue of deteriorating water quality, with a particular focus on the threat of eutrophication, which significantly impacts aquatic environments as well as communities relying on these resources. This study focused on developing a web-based application that was to be used for the early detection and continuous monitoring of chlorophyll-A along with other key water quality indicators. The system uses Arduino-based sensors for the collection of data on pH, dissolved oxygen, temperature, and turbidity. T-test confirmed the system's validity with a good correlation between sensor data and laboratory measurements. Furthermore, the calibration process of the turbidity sensor also needs refinement. Improvement on the calibration process or search for a more accurate alternative will improve measurement accuracy and align well with the laboratory's standards.

The development process applied Agile Scrum methodology to deliver iterative improvements and user-centered design. The system effectively collected pH, dissolved oxygen, temperature, and turbidity for water quality parameters through calibrated sensors. The collected data demonstrated

reliability, enabling precise predictions of chlorophyll-A level using the AquaSense model. A user-friendly web interface was developed with the project, which offers a real-time dashboard in terms of visualizing water quality metrics, historical data trends, and an alert indicator and SMS mechanism for threshold breaches. The solar-powered devices and buoy ensured that the system was sustainable in remote environments.

Evaluation on ISO 25010 reveals high functional completeness, usability, and efficiency with rich feedback from stakeholders that confirms practical value. The testing and evaluation processes were quite extensive in as much as to ensure that the system is reliable and robust. Software testing entails unit, integration, system, and acceptance tests aimed at verifying functionality and smooth interaction of system components. Testing results further emphasized the strength of the system. Unit testing showed that the individual parts work correctly, while integration testing proved that the different modules function smoothly. Stress and load testing demonstrated that it can handle different conditions without negatively affecting its performance.

Chloro Watch, by aligning with the Sustainable Development Goals, supports proactive water quality management, ensuring sustainable aquatic ecosystems. From the study, it has been found that the system is scalable, reliable, and impactful in solving the problems of water quality monitoring challenges, especially in solving eutrophication and supporting sustainable water resource practices. This can be seen in its capacity to facilitate real-time monitoring and early intervention, addressing the challenges of water quality on time. This ability to monitor will even help in early detection besides facilitating

informed decision-making in the drive towards sustainable management of resources.

Conclusion

This study has successfully achieved its goals to implement the developed web-based water quality monitoring framework using Arduino-based sensors for acquiring critical metrics for water quality. The results show that the system has successfully achieved the goal of approximate data collection and also demonstrated reliability in real-time monitoring. Hence, this project was concluded based on its objectives:

1. Gather data for a water quality parameter such as pH, dissolved oxygen, temperature, and turbidity through Arduino-based sensors.

The Arduino-based sensors were applied and met the goals; providing accurate and reliable data regarding critical water quality measurements: pH, dissolved oxygen, temperature, and turbidity. Calibration protocols were followed, whereby sensor outputs were confirmed against standard set values in order to enhance the credibility of the acquired data. Data collection and transmission with a solar-powered system indicates that these systems can sustainably be cost-effective and deployed in remote aquatic environments.

2. Develop a web-based system that monitors water quality such as pH, DO, Temp, Turb, and Chl-A.

The system was designed to provide real-time visualization as well as historical analysis of water quality data. The system comprised an interactive dashboard that displayed trends in parameters as well as forecasts of chlorophyll-A, which enables the user to manage the lake's health efficiently. Equipped with the necessary tools to analyze the dynamics of parameters

over time, it includes graphical insights to strengthen the analysis. The user-friendly interface ensures that anyone without technical knowledge can utilize the system's functionalities. It encourages forward thinking on deciding how to approach problems with water quality, particularly such as oxygen depletion, eutrophication, or adverse blooms of algae.

3. Test and perform an evaluation of the web-based system through a software review using ISO 25010.

The ISO 25010 evaluation confirmed its satisfactory in all user needs. Barangay staffs, IT experts, and the potential users rated functional suitability, performance efficiency, and usability very well. The system contained good amounts of fault tolerance with the successful execution of functions since it was well-presented interfaces that made the system quite reliable and easy to observe continuously. The results have indicated the system's feasibility for real-life applications, mostly in those areas where prompt data and assessment are key requirements in solving environmental challenges.

Recommendation

Given the findings and limitations of the study, several recommendations are made for future work:

1. Expansion of Data Capture for Model Enhancement - The data from the monitoring system will have continuous analysis in order to identify trends, patterns, and possible problems that exist regarding water quality. These insights will be used to enhance predictive models of chlorophyll-A level and other indices of water quality so that the model is kept in line with reality, giving a precise predictions and forecasts.

2. Collaboration with Local Government Units (LGUs) - It will be critical to have partnerships with local government institutions, environmental organizations, and other community stakeholders in order to use mutual knowledge and assets in solving the water quality issues. Inclusion of such stakeholders in developing and implementing monitoring programs would most likely enhance the effectiveness of the system and build more support for conservation efforts.

3. Expansion of Sensor for Additional Water Quality Parameters - Define the strategy for improving the ability and future expansion of the system so that it can effectively keep pace with changing environmental conditions and incorporate more parameters in water quality that are applicable. This involves expanding sensor networks to cover larger spatial areas and monitoring emerging pollutants, ensuring high-quality sensors are used to maintain accuracy and reliability. The approach emphasizes the integration of advanced technologies, scalable systems, and strategic investments to create a comprehensive and adaptive monitoring framework.

4. Securing Buoy Infrastructure for Sensor Deployment - Resilient and secure buoy infrastructure should be developed in effective deployment and safe guarding of sensors and other monitoring tools in aquatic ecosystems. Essentially, the buoy infrastructure should be designed to extreme environmental condition like typhoons that may disturb the stability of the sensor and the reliability of data obtained.

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APPENDICES

APPENDIX

A

Input/Output/Reports Screenshot

Input

Arduino Uno Code

```
const long readInterval = 1000; // Interval to read temperature every second
unsigned long lastReadMillis = 0; // Store the last time a reading was made
//TURBIDITY
int sensorPin = A1; // A0 for Arduino / 36 for ESP
//PH
float calibration_value = 21.34 - 0.7;
int phval = 0;
unsigned long int avgval;
int buffer_arr[10],temp;
float ph_act;
float sensorValue = 0;
float voltage = 0;
float turbidity = 0;
float volt = 0;
void setup() {
  Serial.begin(115200);
}
void loop() {
  unsigned long currentMillis = millis();
  // Check if it's time to read the temperature (every second)
  if (currentMillis - lastReadMillis >= readInterval) {
    lastReadMillis = currentMillis;
    //Turbidity
    sensorValue = analogRead(sensorPin);
    voltage = sensorValue * (3.3 / 1024.0);
    // Calculate turbidity as a float, mapping from 0-202 to 100-0
    turbidity = map(sensorValue, 0, 205, 100, 0);
    turbidity = max(turbidity, 1); // Ensure turbidity does not drop below 0.1
    //ph
    for(int i=0;i<10;i++)
    {
      buffer_arr[i]=analogRead(A0);
      delay(30);
    }
    for(int i=0;i<9;i++)
    {
      for(int j=i+1;j<10;j++)
      {
        if(buffer_arr[i]>buffer_arr[j])
        {
          temp=buffer_arr[i];
          buffer_arr[i]=buffer_arr[j];
          buffer_arr[j]=temp;
        }
      }
    }
    avgval = 0;
    for(int i=2;i<8;i++)
    avgval+=buffer_arr[i];
    volt=(float)avgval*5.0/1024/6;
    ph_act = -5.70 * volt + calibration_value;
    String output = String(ph_act) + "," + String(turbidity);
    Serial.println(output); // Send data to ESP8266
  }
}
```

ESP8266 Code

```
#include <OneWire.h>
#include <DallasTemperature.h>
#include <ESP8266WiFi.h>
#include <FirebaseESP8266.h>
#define FIREBASE_HOST "sample123" // Replace with your Firebase Database URL
#define FIREBASE_AUTH "sample123" // Replace with your Firebase Database Secret
// Wi-Fi credentials
#define WIFI_SSID "sample123" // Replace with your Wi-Fi SSID
```

```

#define WIFI_PASSWORD " sample123" // Replace with your Wi-Fi password
#define DO_PIN A0
#define VREF 3300 // VREF (mv)
#define ADC_RES 1024 // ADC Resolution
// Single-point calibration Mode=0
// Two-point calibration Mode=1
#define TWO_POINT_CALIBRATION 1 // Enable two-point calibration
// Single point calibration needs to be filled CAL1_V and CAL1_T
#define CAL1_V (609) // mv
#define CAL1_T (38) // °C
// Two-point calibration needs to be filled CAL2_V and CAL2_T
#define CAL2_V (341) // mv
#define CAL2_T (11) // °C
// DS18B20 setup
#define ONE_WIRE_BUS D4 // Pin for DS18B20
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);
FirebaseData firebaseData;
FirebaseAuth auth;
FirebaseConfig config;
// DO table for calibration
const uint16_t DO_Table[41] = {
  14460, 14220, 13820, 13440, 13090, 12740, 12420, 12110, 11810, 11530,
  11260, 11010, 10770, 10530, 10300, 10080, 9860, 9660, 9460, 9270,
  9080, 8900, 8730, 8570, 8410, 8250, 8110, 7960, 7820, 7690,
  7560, 7430, 7300, 7180, 7070, 6950, 6840, 6730, 6630, 6530, 6410};
uint16_t ADC_Raw;
uint16_t ADC_Voltage;
float readDO(uint32_t voltage_mv, uint8_t temperature_c) {
  #if TWO_POINT_CALIBRATION == 0
    uint16_t V_saturation = (uint32_t)CAL1_V + (uint32_t)35 * temperature_c - (uint32_t)CAL1_T * 35;
    return (voltage_mv * DO_Table[temperature_c] / V_saturation);
  #else
    uint16_t V_saturation = (int16_t)((int8_t)temperature_c - CAL2_T) * ((uint16_t)CAL1_V - CAL2_V) /
    ((uint8_t)CAL1_T - CAL2_T) + CAL2_V;
    return (voltage_mv * DO_Table[temperature_c] / V_saturation);
  #endif
}
unsigned long previousMillis = 0; // Store the last time a reading was taken
const long readInterval = 1000; // Interval to read temperature every second
const long outputInterval = 30000; // Output interval for every 30 seconds
unsigned long lastReadMillis = 0; // Store the last time a reading was made
float lastCelsius = 0; // Store the last Celsius temperature
float lastFahrenheit = 0;
float lastDO_mg_per_L=0;
float lastADC_Raw = 0;
float lastADC_Voltage = 0;
float pH = 0;
float Turbidity = 0;
void setup() {
  Serial.begin(115200);
  Serial.println("Starting...");
  // Initialize the DS18B20 sensor
  sensors.begin();
  // Connect to Wi-Fi
  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  Serial.print("Connecting to Wi-Fi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println();
  Serial.println("Connected to Wi-Fi");
  // Configure Firebase
  config.host = FIREBASE_HOST;
  config.signer.tokens.legacy_token = FIREBASE_AUTH;
  // Initialize Firebase

```

```

    Firebase.begin(&config, &auth);
    Firebase.reconnectWiFi(true);
}
void loop() {
    unsigned long currentMillis = millis();
    // Check if it's time to read the temperature (every second)
    if (currentMillis - lastReadMillis >= readInterval) {
        lastReadMillis = currentMillis;
        // Request temperature readings from the DS18B20 sensor
        sensors.requestTemperatures();
        // Get the temperature in Celsius
        lastCelsius = sensors.getTempCByIndex(0);
        lastFahrenheit = sensors.toFahrenheit(lastCelsius);
        uint8_t Temperaturet = (uint8_t)lastCelsius; // Cast to uint8_t for DO calculation
        // Read raw voltage from the DO sensor
        lastADC_Raw = analogRead(DO_PIN);
        lastADC_Voltage = uint32_t(VREF) * lastADC_Raw / ADC_RES;
        // Calculate DO and format to 2 decimal places
        lastDO_mg_per_L = readDO(lastADC_Voltage, Temperaturet) / 1000.0; // Convert to mg/L
        if (Serial.available() > 0) {
            String data = Serial.readStringUntil('\n');
            // Parse pH and turbidity values from string
            int commaIndex = data.indexOf(',');
            if (commaIndex > 0) {
                String pHString = data.substring(0, commaIndex);
                String turbidityString = data.substring(commaIndex + 1);
                pH = pHString.toFloat();
                Turbidity = turbidityString.toFloat();
            }
        }
    }
    // Check if it's time to output the readings (every 30 seconds)
    if (currentMillis - previousMillis >= outputInterval) {
        previousMillis = currentMillis;
        Serial.print("Temperature in Celsius: " + String(lastCelsius) + "\t");
        Serial.print("Temperature in Fahrenheit: " + String(lastFahrenheit) + "\t");
        Serial.print("ADC RAW: " + String(lastADC_Raw) + "\t");
        Serial.print("ADC Voltage: " + String(lastADC_Voltage) + "\t");
        Serial.print("pH: " + String(pH) + "\t");
        Serial.print("Turbidity: " + String(Turbidity) + "\t");
        Serial.println("DO: " + String(lastDO_mg_per_L, 2) + "\t"); // Format to 2 decimal places
        // Create a unified JSON object for all parameters
        FirebaseJson dataJson;
        dataJson.set("timestamp/.sv", "timestamp"); // Common timestamp
        // Add each parameter to the unified JSON object
        dataJson.set("Temperature", lastCelsius);
        dataJson.set("Temperaturefar", lastFahrenheit);
        dataJson.set("DissolvedOxygen", lastDO_mg_per_L);
        dataJson.set("DOadc", lastADC_Raw);
        dataJson.set("DOvolt", lastADC_Voltage);
        dataJson.set("pH", pH);
        dataJson.set("Turbidity", Turbidity);
        // Send the unified data JSON to Firebase
        if (Firebase.pushJSON(firebaseData, "/SensorData", dataJson)) {
            Serial.println("SensorData sent to Firebase successfully");
        } else {
            Serial.print("Failed to send data to Firebase: ");
            Serial.println(firebaseData.errorReason());
        }
        // Optionally update the current values under individual nodes
        if (Firebase.setJSON(firebaseData, "/CurrentData", dataJson)) {
            Serial.println("CurrentData updated in Firebase successfully");
        } else {
            Serial.print("Failed to update CurrentData in Firebase: ");
            Serial.println(firebaseData.errorReason());
        }
    }
}

```

```
}
```

React code

```
import React, { useEffect, useState, useRef } from 'react';
import axios from 'axios';
import { Link } from 'react-router-dom';
import PlotlyCharts from '../charts/PlotlyChartsSmall';
import BigPlotlyCharts from '../charts/PlotlyCharts';
import { ref, onValue } from 'firebase/database';
import { database } from '../firebase';

/**
 * Water component fetches water quality and prediction data, allowing users to analyze and visualize
 * the data.
 *
 * @returns {JSX.Element} Water component.
 */
function Water() {
  const [data, setData] = useState([]);
  const [secondData, setSecondData] = useState([]);
  const [predictedData, setPredictedData] = useState([]);
  const [predictedDataTwo, setPredictedDataTwo] = useState([]);
  const [predictedQuality, setPredictedWaterQuality] = useState([]);
  const [filterOption, setFilterOption] = useState('Year');
  const [selectedAnalysis, setSelectedAnalysis] = useState('All');
  const prevFilterOption = useRef(filterOption);
  const [temperatureData, setTemperatureData] = useState([]);
  const [dissolvedoxygenData, setDissolvedOxygenData] = useState([]);
  const [phData, setpHData] = useState([]);
  const [turbidityData, setTurbidityData] = useState([]);
  const [chlorophyllData, setChlorophyllData] = useState([]);
  const [forecastedData, setForecastedData] = useState([]);
  const [upperBound, setUpperBound] = useState([]);
  const [lowerBound, setLowerBound] = useState([]);
  const [error, setError] = useState(null);
  useEffect(() => {
    const temperatureRef = ref(database, 'SensorData'); // Adjust path to your historical data
    const unsubscribe = onValue(
      temperatureRef,
      (snapshot) => {
        const data = snapshot.val();
        if (data) {
          // Map data to include timestamps and format them for the graph
          const formattedData = Object.keys(data).map((key) => {
            const entry = data[key];
            return {
              x: new Date(entry.timestamp), // Convert timestamp to Date object for x values
              y: entry.Temperature, // Celsius value as y values
            };
          });
          setTemperatureData(formattedData);
        } else {
          setError('No temperature data available.');
```

```

// Map data to include timestamps and format them for the graph
const formattedData = Object.keys(data).map((key) => {
  const entry = data[key];
  return {
    x: new Date(entry.timestamp), // Convert timestamp to Date object for x values
    y: entry.DissolvedOxygen, // Celsius value as y values
  };
});

setDissolvedOxygenData(formattedData);
} else {
  setError('No dissolvedoxygen data available.');
```

```

    },
    (error) => {
      console.error('Error fetching data:', error);
      setError('Failed to fetch turbidity data.');
```

```

    if (data) {
      const formattedData = Object.keys(data).map((key) => {
        const entry = data[key];
        return {
          x: new Date(new Date(entry.timestamp).getTime() + 3 * 60 * 60 * 1000),
          y: entry.Upper_Bound_Chlorophyll,
        };
      });
      setUpperBound(formattedData);
    }
  },
  (error) => {
    console.error('Error fetching upper bound data:', error);
    setError('Failed to fetch upper bound data.');
```

```

*/
const handleAnalysisChange = (e) => {
  setSelectedAnalysis(e.target.value);
};
return (
  <div className='analysis-header'>
    <div className='flex-header'>
      { /* Dropdown for selecting time range */ }
    <div className='header-inputs'>
      <span>Filter by: </span>
    </div>
    { /* Dropdown for selecting analysis type */ }
    <div className='header-inputs'>
      <select onChange={handleAnalysisChange} value={selectedAnalysis}>
        <option>All</option>
        <option>Temperature</option>
        <option>pH</option>
        <option>Dissolved Oxygen</option>
        <option>Turbidity</option>
        <option>Chlorophyll</option>
      </select>
    </div>
  </div>
  {selectedAnalysis === 'All' ? (
    <div className='data-analysis'>
      <div className='analysis-item'>
        <div className='custom-analysis'>
          <div className='home-map-flex'>
            <h1>Temperature</h1>
            <Link to='/temperature-details'>View →</Link>
          </div>
          <div className='graph'>
            <div className='plotly-chart'>
              <PlotlyCharts
                data={}
                parameter="
                parameterX={parameterX0}
                parameterY={parameterY0}
                predY={}
                predX={}
                upperBound={}
                lowerBound={}
                filterOption={filterOption}
              />
            </div>
          </div>
        </div>
      </div>
      <div className='analysis-item'>
        <div className='custom-analysis'>
          <div className='home-map-flex'>
            <h1>pH</h1>
            <Link to="/ph-details">View →</Link>
          </div>
          <div className='graph'>
            <div className='plotly-chart'>
              <PlotlyCharts
                data={}
                paramater="pH"
                parameterX={parameterX1}
                parameterY={parameterY1}
                predY={}
                predX={}
                upperBound={}
                lowerBound={}
                filterOption={filterOption}
              />
            </div>
          </div>
        </div>
      </div>
    </div>
  ) : (
    <div className='data-analysis'>
      <div className='analysis-item'>
        <div className='custom-analysis'>
          <div className='home-map-flex'>
            <h1>Temperature</h1>
            <Link to='/temperature-details'>View →</Link>
          </div>
          <div className='graph'>
            <div className='plotly-chart'>
              <PlotlyCharts
                data={}
                parameter="
                parameterX={parameterX0}
                parameterY={parameterY0}
                predY={}
                predX={}
                upperBound={}
                lowerBound={}
                filterOption={filterOption}
              />
            </div>
          </div>
        </div>
      </div>
      <div className='analysis-item'>
        <div className='custom-analysis'>
          <div className='home-map-flex'>
            <h1>pH</h1>
            <Link to="/ph-details">View →</Link>
          </div>
          <div className='graph'>
            <div className='plotly-chart'>
              <PlotlyCharts
                data={}
                paramater="pH"
                parameterX={parameterX1}
                parameterY={parameterY1}
                predY={}
                predX={}
                upperBound={}
                lowerBound={}
                filterOption={filterOption}
              />
            </div>
          </div>
        </div>
      </div>
    </div>
  )
);

```

```

    </div>
  </div>
</div>
</div>
<div className='analysis-item'>
  <div className='custom-analysis'>
    <div className='home-map-flex'>
      <h1>Dissolved Oxygen</h1>
      <Link to="/dissolvedoxygen-details">View →</Link>
    </div>
    <div className='graph'>
      <div className='plotly-chart'>
        <PlotlyCharts
          data={}
          paramater="Dissolved Oxygen"
          parameterX={parameterX2}
          parameterY={parameterY2}
          predY={}
          predX={}
          upperBound={}
          lowerBound={}
          filterOption={filterOption}
        />
      </div>
    </div>
  </div>
</div>
<div className='analysis-item'>
  <div className='custom-analysis'>
    <div className='home-map-flex'>
      <h1>Turbidity</h1>
      <Link to="/turbidity-details">View →</Link>
    </div>
    <div className='graph'>
      <div className='plotly-chart'>
        <PlotlyCharts
          data={}
          paramater="Turbidity"
          parameterX={parameterX3}
          parameterY={parameterY3}
          predY={}
          predX={}
          upperBound={}
          lowerBound={}
          filterOption={filterOption}
        />
      </div>
    </div>
  </div>
</div>
<div className='analysis-item'>
  <div className='custom-analysis'>
    <div className='home-map-flex'>
      <h1>Chlorophyll</h1>
      <Link to="/chlorophyll-details">View →</Link>
    </div>
    <div className='graph'>
      <div className='plotly-chart'>
        <PlotlyCharts
          parameter="Chlorophyll"
          parameterX={chlorophyllData.map(item => item.x)}
          parameterY={chlorophyllData.map(item => item.y)}
          predY={forecastedData.map(item => item.y)}
          predX={forecastedData.map(item => item.x)}
          upperBound={upperBound.map(item => item.y)}
          lowerBound={lowerBound.map(item => item.y)}
          filterOption={} // Format date and time in hover labels
        />
      </div>
    </div>
  </div>
</div>

```

```

        />
      </div>
    </div>
  </div>
</div>
): (
<div className="whole-data-analyze">
  <p>
    {selectedAnalysis}
  </p>
  {selectedAnalysis === 'Temperature' && (
    <div className='graph'>
      <div style={{ overflowX: 'auto', whiteSpace: 'nowrap' }}>
        {/* Component for rendering big Plotly charts */}
        <BigPlotlyCharts
          data={[]}
          parameter={}
          parameterX={parameterX0}
          parameterY={parameterY0}
          predY={}
          predX={}
          upperBound={}
          lowerBound={}
          filterOption={}
        />
      </div>
    </div>
  )}
  {selectedAnalysis === 'pH' && (
    <div className='graph'>
      {/* Component for rendering big Plotly charts */}
      <BigPlotlyCharts
        data={[]}
        parameter={}
        parameterX={parameterX1}
        parameterY={parameterY1}
        predY={}
        predX={}
        upperBound={}
        lowerBound={}
        filterOption={filterOption}
      />
    </div>
  )}
  {selectedAnalysis === 'Dissolved Oxygen' && (
    <div className='graph'>
      {/* Component for rendering big Plotly charts */}
      <BigPlotlyCharts
        data={[]}
        parameter={}
        parameterX={parameterX2}
        parameterY={parameterY2}
        predY={}
        predX={}
        upperBound={}
        lowerBound={}
        filterOption={filterOption}
      />
    </div>
  )}
  {selectedAnalysis === 'Turbidity' && (
    <div className='graph'>
      {/* Component for rendering big Plotly charts */}
      <BigPlotlyCharts
        data={[]}
        parameter={}

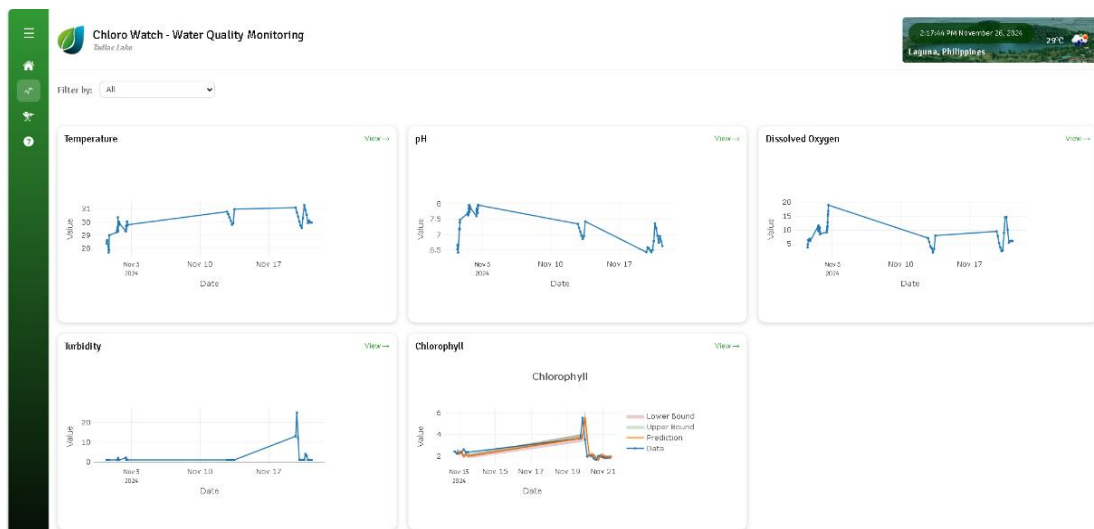
```

```

parameterX={parameterX3}
parameterY={parameterY3}
predY={[]}
predX={[]}
upperBound=[]
lowerBound=[]
filterOption={filterOption}
/>
</div>
)}
{selectedAnalysis === 'Chlorophyll' && (
  <div className='graph'>
    /* Component for rendering big Plotly charts */
    <BigPlotlyCharts
      parameter="Chlorophyll"
      parameterX={chlorophyllData.map(item => item.x)}
      parameterY={chlorophyllData.map(item => item.y)}
      predY={forecastedData.map(item => item.y)}
      predX={forecastedData.map(item => item.x)}
      upperBound={upperBound.map(item => item.y)}
      lowerBound={lowerBound.map(item => item.y)}
      filterOption={[]} // Format date and time in hover labels
    />
  </div>
)}
</div>
)}
</div>
);
};
export default Water;

```

Output



The image shows a water quality monitoring dashboard displaying real-time data for parameters such as temperature, pH, dissolved oxygen, turbidity, and chlorophyll levels. Each metric is shown with a time series graph, tracking changes over time, with predictions indicated by upper and lower bounds. The system helps assess water quality, with data from Laguna, Philippines.

Input

Model Integration Code

```
import streamlit as st
import pandas as pd
import joblib
import numpy as np
import requests
import json
import re
import pytz
from datetime import datetime
import threading
import time

# Firebase Configuration
firebase_config = {
    "apiKey": "Sample123",
    "authDomain": " Sample123",
    "databaseURL": " Sample123",
    "projectId": " Sample123",
    "storageBucket": " Sample123",
    "messagingSenderId": " Sample123",
    "appId": " Sample123"
}

# Firebase Realtime Database URL
database_url = firebase_config['databaseURL']

# Load the stacking model and scaler
stacking_model_loaded = joblib.load('stacking_ensemble_model.joblib')
scaler_loaded = joblib.load('scaler.pkl')

# Streamlit app title
st.title('Chlorophyll Prediction and Forecasting Model')

# Function to sanitize Firebase keys
def sanitize_key(key):
    """Sanitize Firebase keys to remove invalid characters."""
    return re.sub(r'[$#\[\]/.], ' _ ', key)

# Recursive function to sanitize nested dictionaries
def sanitize_dict(d):
    """Recursively sanitize keys in a nested dictionary."""
    if isinstance(d, dict):
        return {sanitize_key(k): sanitize_dict(v) for k, v in d.items()}
    else:
        return d

# Function to fetch the latest sensor data from Firebase using the REST API
def get_latest_sensor_data():
    try:
        sensor_data_ref = f'{database_url}/SensorData.json'
        response = requests.get(sensor_data_ref)

        if response.status_code == 200:
            sensor_data = response.json()
            # Sort the data by timestamp and get the latest entry
            latest_data = max(sensor_data.values(), key=lambda x: x['timestamp'])

            # Sanitize the data and extract timestamp
            sanitized_data = sanitize_dict(latest_data)
            timestamp = datetime.fromtimestamp(sanitized_data['timestamp'] / 1000.0)
            singapore_tz = pytz.timezone('Asia/Singapore')
            timestamp = timestamp.replace(tzinfo=pytz.utc).astimezone(singapore_tz)
```

```

# Prepare the sensor data as a DataFrame
sensor_df = pd.DataFrame({
    'Temp (°C)': [sanitized_data['Temperature']],
    'Turbidity (FNU)': [sanitized_data['Turbidity']],
    'pH': [sanitized_data['pH']],
    'DO (mg/L)': [sanitized_data['DissolvedOxygen']],
    'year': [timestamp.year],
    'month': [timestamp.month],
    'day': [timestamp.day],
    'day_of_week': [timestamp.weekday()],
    'day_of_year': [timestamp.timetuple().tm_yday],
    'quarter': [(timestamp.month - 1) // 3 + 1],
    'hour': [timestamp.hour]
})
return sensor_df, timestamp
else:
    st.write(f"Error fetching data from Firebase: {response.text}")
    return pd.DataFrame(), None
except Exception as e:
    st.write(f"Error fetching data from Firebase: {e}")
    return pd.DataFrame(), None

# Forecast future chlorophyll levels based on the latest data
def forecast_future_chlorophyll(df, stacking_model, scaler, features, steps=10,
uncertainty_factor=0.1):
    last_row = df[features].iloc[-1:].copy()

    forecasts = []
    upper_bounds = []
    lower_bounds = []

    for i in range(steps):
        # Simulate a "next" time step (roll over the day/month)
        last_row['day_of_year'] += 1
        last_row['day'] = (last_row['day'] % 31) + 1
        last_row['day_of_week'] = (last_row['day_of_week'] + 1) % 7
        if last_row['day'].iloc[0] == 1:
            last_row['month'] = (last_row['month'] % 12) + 1
            if last_row['month'].iloc[0] == 1:
                last_row['year'] += 1
            last_row['quarter'] = (last_row['month'] - 1) // 3 + 1

        # Scaling and prediction
        last_row_scaled = scaler.transform(last_row)
        forecast_log = stacking_model.predict(last_row_scaled)
        forecast = np.expm1(forecast_log) # Reverse log transformation

        # Calculate the upper and lower bounds
        upper_bound = forecast[0] * (1 + uncertainty_factor) # Add 10% to forecast
        lower_bound = forecast[0] * (1 - uncertainty_factor) # Subtract 10% from forecast

        forecasts.append(forecast[0])
        upper_bounds.append(upper_bound)
        lower_bounds.append(lower_bound)

    return forecasts, upper_bounds, lower_bounds

# Save predictions and forecasts to Firebase using the REST API
def save_prediction_to_firebase(predicted_chlorophyll, sensor_data, forecast_values, upper_bounds,
lower_bounds):
    predicted_chlorophyll = float(predicted_chlorophyll)

    # Use only the first value from forecast, upper bound, and lower bound arrays
    first_forecast_value = float(forecast_values[0])
    first_upper_bound = float(upper_bounds[0])
    first_lower_bound = float(lower_bounds[0])

```

```

# Sanitize sensor data keys
sensor_data_values = {
    sanitize_key(key): float(value[0]) if isinstance(value[0], np.float64) else int(value[0])
    for key, value in sensor_data.items()
}

# Push the prediction and forecast data to Firebase
prediction_ref = f'{database_url}/Predictions.json'
singapore_tz = pytz.timezone('Asia/Singapore')
timestamp = datetime.now(singapore_tz).strftime('%Y-%m-%d %H:%M:%S')
data = {
    **sensor_data_values,
    'Predicted_Chlorophyll': predicted_chlorophyll,
    'Forecasted_Chlorophyll': first_forecast_value,
    'Upper_Bound_Chlorophyll': first_upper_bound,
    'Lower_Bound_Chlorophyll': first_lower_bound,
    'timestamp': timestamp
}
response = requests.post(prediction_ref, json=data)

if response.status_code != 200:
    st.write(f"Error saving prediction: {response.text}")

# Update the current prediction separately
current_prediction_ref = f'{database_url}/CurrentPrediction.json'
current_data = {
    'Predicted_Chlorophyll': predicted_chlorophyll,
    'timestamp': timestamp
}
# Make a PUT request to update the current prediction
current_response = requests.put(current_prediction_ref, json=current_data)

if current_response.status_code != 200:
    st.write(f"Error updating current prediction: {current_response.text}")

# Background task to periodically fetch data, make predictions, and save to Firebase
def background_task():
    if 'last_processed_timestamp' not in st.session_state:
        st.session_state.last_processed_timestamp = None

    while True:
        # Fetch the latest sensor data
        fetched_sensor_data, timestamp = get_latest_sensor_data()

        if not fetched_sensor_data.empty():
            # Check if the timestamp has changed (to avoid processing the same data)
            if timestamp != st.session_state.last_processed_timestamp:
                st.session_state.last_processed_timestamp = timestamp

                # Proceed with prediction and saving logic
                features = ['Temp (°C)', 'Turbidity (FNU)', 'pH', 'DO (mg/L)', 'year', 'month', 'day',
                    'day_of_week', 'day_of_year', 'quarter', 'hour']
                X_new_scaled = scaler_loaded.transform(fetched_sensor_data[features])
                y_new_pred_log = stacking_model_loaded.predict(X_new_scaled)
                y_new_pred = np.expm1(y_new_pred_log)

                future_forecasts, upper_bounds, lower_bounds =
forecast_future_chlorophyll(fetched_sensor_data, stacking_model_loaded, scaler_loaded, features,
steps=10)
                save_prediction_to_firebase(y_new_pred[0], fetched_sensor_data, future_forecasts,
upper_bounds, lower_bounds)
            else:
                st.write("No new sensor data.")

        time.sleep(600) # Wait 10 minutes before fetching new data

```

```
# Start background task in a separate thread
if __name__ == '__main__':
    # Start the background task in a separate thread
    threading.Thread(target=background_task, daemon=True).start()

    # Main Streamlit app code (will run continuously while the background task works)
    st.write("## Continuous Chlorophyll Prediction and Forecasting")
    st.write("Prediction tasks are running in the background, check Firebase for updated predictions.")
```

React Code

```
import React, { useEffect, useState } from 'react';
import { ref, onValue } from 'firebase/database';
import { database } from '../firebase'; // Adjust the import path as needed
import PlotlyCharts from '../charts/PlotlyChartsSmall';
import './graph.css'; // Import the CSS file for styling
import Weather from './weather';
import logo from '../assets/img/favicon.png';
import { Link } from 'react-router-dom';

function ChlorophyllDetails() {
    const [chlorophyllData, setChlorophyllData] = useState([]);
    const [forecastedData, setForecastedData] = useState([]);
    const [upperBound, setUpperBound] = useState([]);
    const [lowerBound, setLowerBound] = useState([]);
    const [error, setError] = useState(null);

    useEffect(() => {
        const chlorophyllRef = ref(database, 'Predictions');
        const unsubscribe = onValue(
            chlorophyllRef,
            (snapshot) => {
                const data = snapshot.val();
                if (data) {
                    const formattedData = Object.keys(data).map((key) => {
                        const entry = data[key];
                        return {
                            x: new Date(entry.timestamp),
                            y: entry.Predicted_Chlorophyll,
                        };
                    });
                    setChlorophyllData(formattedData);
                } else {
                    setError("No chlorophyll data available.");
                }
            },
            (error) => {
                console.error("Error fetching data:", error);
                setError("Failed to fetch chlorophyll data.");
            }
        );
        return () => unsubscribe();
    }, []);

    useEffect(() => {
        const forecastedRef = ref(database, 'Predictions');
        const unsubscribe = onValue(
            forecastedRef,
            (snapshot) => {
                const data = snapshot.val();
                if (data) {
                    const formattedData = Object.keys(data).map((key) => {
                        const entry = data[key];
                        return {
                            x: new Date(new Date(entry.timestamp).getTime() + 3 * 60 * 60 * 1000), // 3 hours ahead
                            y: entry.Forecasted_Chlorophyll,
                        };
                    });
                }
            }
        );
    }, []);
```

```

    });
    setForecastedData(formattedData);
  } else {
    setError('No forecasted data available.');
```

```

  },
  (error) => {
    console.error('Error fetching forecasted data:', error);
    setError('Failed to fetch forecasted data.');
```

```

  }
);
return () => unsubscribe();
}, []);

useEffect(() => {
  const upperBoundRef = ref(database, 'Predictions');
  const unsubscribe = onValue(
    upperBoundRef,
    (snapshot) => {
      const data = snapshot.val();
      if (data) {
        const formattedData = Object.keys(data).map((key) => {
          const entry = data[key];
          return {
            x: new Date(new Date(entry.timestamp).getTime() + 3 * 60 * 60 * 1000),
            y: entry.Upper_Bound_Chlorophyll,
          };
        });
        setUpperBound(formattedData);
      }
    },
    (error) => {
      console.error('Error fetching upper bound data:', error);
      setError('Failed to fetch upper bound data.');
```

```

    }
  );
  return () => unsubscribe();
}, []);

useEffect(() => {
  const lowerBoundRef = ref(database, 'Predictions');
  const unsubscribe = onValue(
    lowerBoundRef,
    (snapshot) => {
      const data = snapshot.val();
      if (data) {
        const formattedData = Object.keys(data).map((key) => {
          const entry = data[key];
          return {
            x: new Date(new Date(entry.timestamp).getTime() + 3 * 60 * 60 * 1000),
            y: entry.Lower_Bound_Chlorophyll,
          };
        });
        setLowerBound(formattedData);
      }
    },
    (error) => {
      console.error('Error fetching lower bound data:', error);
      setError('Failed to fetch lower bound data.');
```

```

  }
);
return () => unsubscribe();
}, []);

// Combine recent and forecasted data into one array for table rendering
const combinedData = chlorophyllData.map((recentData, index) => {
  const forecastData = forecastedData[index] || {};

```

```

const upper = upperBound[index] ? upperBound[index].y : null;
const lower = lowerBound[index] ? lowerBound[index].y : null;

return {
  recent: recentData,
  forecasted: forecastData,
  upperBound: upper,
  lowerBound: lower,
};
});

return (
  <div className='analysis-page'>
    {/* Overview Dashboard */}
    <div className='overview-dash'>
      {/* Left Section of Overview */}
      <div className='over-left'>
        <div className='overview-profile'>
          {/* University Logo */}
          <Link to='/><img src={logo} alt="Ispu-logo"/></Link>
          <span>
            {/* University Name and Campus */}
            <h1>Chloro Watch - Water Quality Monitoring</h1>
            <p>Tadlac Lake</p>
          </span>
        </div>
      </div>
      {/* Right Section of Overview - Weather Component */}
      <div className='over-right'>
        <Weather />
      </div>
    </div>

    <div className="details-container">
      <div className="chart-and-table">
        <div className="chart-container">
          {error ? (
            <p>{error}</p>
          ) : (
            <PlotlyCharts
              parameter="Chlorophyll"
              parameterX={chlorophyllData.map(item => item.x)}
              parameterY={chlorophyllData.map(item => item.y)}
              predY={forecastedData.map(item => item.y)}
              predX={forecastedData.map(item => item.x)}
              upperBound={upperBound.map(item => item.y)}
              lowerBound={lowerBound.map(item => item.y)}
              filterOption={[]}
              className="water-plot"
              hovertemplate="%{x}%Y-%m-%d %H:%M}: %{y:.2f} mg/L" // Format date and time in hover
            >
              labels
            </PlotlyCharts>
          )}
        </div>

        <div className="data-table">
          <table>
            <thead>
              <tr>
                <th>Date and Time</th>
                <th>Recent</th>
                <th>Prediction</th>
                <th>Upper Bound</th>
                <th>Lower Bound</th>
              </tr>
            </thead>
            <tbody>

```

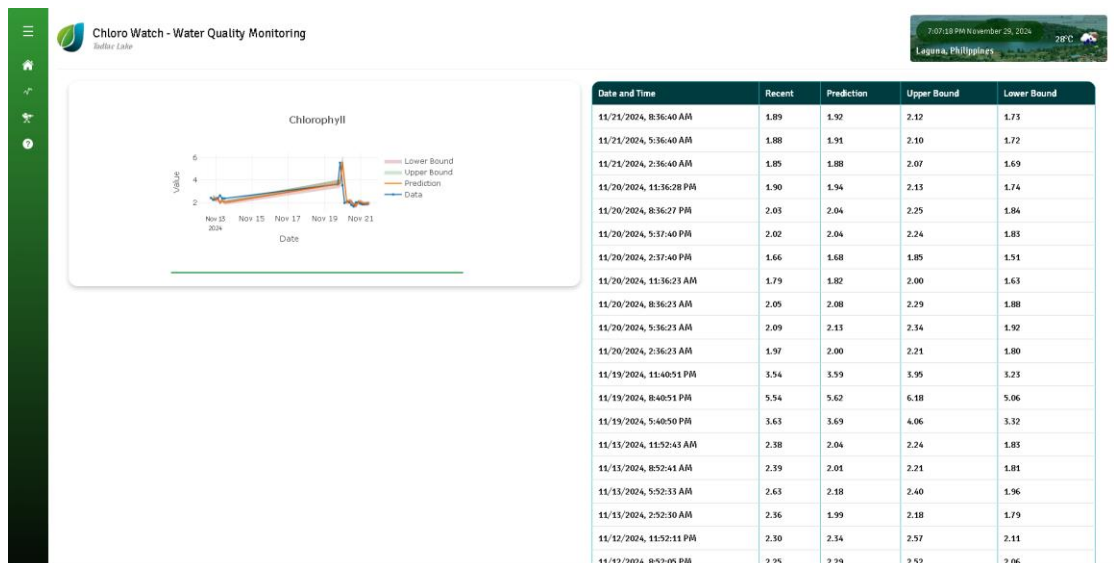
```

    {...combinedData}.reverse().map((data, index) => (
    <React.Fragment key={index}>
    { /* Render Recent Data */ }
    <tr>
    <td>{data.recent.x ? data.recent.x.toLocaleString() : 'N/A'}</td>
    <td>{data.recent.y !== null && data.recent.y !== undefined ? data.recent.y.toFixed(2) :
    'N/A'}</td>
    <td>{data.forecasted.y !== null && data.forecasted.y !== undefined ?
    data.forecasted.y.toFixed(2) : 'N/A'}</td>
    <td>{data.upperBound !== null && data.upperBound !== undefined ?
    data.upperBound.toFixed(2) : 'N/A'}</td>
    <td>{data.lowerBound !== null && data.lowerBound !== undefined ?
    data.lowerBound.toFixed(2) : 'N/A'}</td>
    </tr>
    </React.Fragment>
    ))}
  </tbody>
</table>
</div>
</div>
</div>
</div>
);
}

export default ChlorophyllDetails;

```

Ouput



This image shows a water quality dashboard focusing on chlorophyll levels, with a time series graph displaying recent measurements, predictions, and their upper and lower bounds. A table on the right lists detailed data for each timestamp, including actual and predicted values with bounds. This setup is part of the integration model to monitor and predict water quality changes.

Input

ESP32 Code

```
#include <WiFi.h>
#include <FirebaseESP32.h>

// Firebase and WiFi credentials
#define FIREBASE_HOST "Sample123"
#define FIREBASE_AUTH " Sample123"
#define WIFI_SSID " Sample123"
#define WIFI_PASSWORD " Sample123"

// SIM900A Configuration
#define RX_PIN 16 // GPIO 16 for RX (SIM900A TX)
#define TX_PIN 17 // GPIO 17 for TX (SIM900A RX)
HardwareSerial mySerial(1); // Use Serial1 for SIM900A communication
FirebaseData firebaseData;
FirebaseAuth auth;
FirebaseConfig config;

// Threshold values
float temperatureMin = 25.0, temperatureMax = 31.0;
float pHMin = 6.5, pHMax = 9.0;
float dissolvedOxygenMin = 5.0, dissolvedOxygenMax = 1000.0;
float turbidityMax = 50.0;
float chlorophyllMeso = 10.0;
float chlorophyllEutro = 25.0;

// Variables to track the last timestamps
String lastCurrentDataTimestamp = "";
String lastCurrentPredictionTimestamp = "";

// Function to send SMS using SIM900A
void sendSMS(String phoneNumber, String message) {
  mySerial.println("AT");
  delay(500);
  mySerial.println("AT+CMGF=1"); // Set SMS mode to text
  delay(500);
  mySerial.print("AT+CMGS=\"");
  mySerial.print(phoneNumber);
  mySerial.println("\");
  delay(500);
  mySerial.println(message);
  mySerial.write(26); // End-of-message character (Ctrl+Z)
  delay(1000); // Short delay to ensure SMS is sent
}

// Generalized threshold check function
void checkThreshold(String key, float value, float min, float max, String unit, String lowInstruction,
String highInstruction) {
  if (value < min || value > max) {
    String instruction = (value < min ? lowInstruction : highInstruction);
    String alertMessage = key + " value " + String(value, 1) + " " + unit +
      (value < min ? " is below the minimum threshold of " : " exceeds the maximum
threshold of ") +
      String(value < min ? min : max) + " " + unit + ". " + instruction;
    sendSMS("+639770710102", alertMessage);
  }
}

// Specific chlorophyll threshold check
void checkChlorophyllThreshold(String key, float value, float meso, float eutro, String unit) {
  if (value >= meso || value > eutro) {
    String condition = ((value >= meso && value < eutro) ? "reached the mesotrophic condition of " :
"exceeds the maximum mesotrophic condition of ");
    String instruction = ((value >= meso && value < eutro)
```

```

        ? "Moderate productivity is present. Monitor nutrient sources, such as agricultural
runoff, to avoid progression to eutrophic conditions."
        : "High nutrient levels may cause algal blooms. Implement nutrient reduction
strategies and consider aeration to manage oxygen depletion risks.");
    String alertMessage = key + " value " + String(value, 1) + " " + unit + " " + condition +
        String((value >= meso && value < eutro) ? meso : eutro) + " " + unit + ". " +
instruction;
    sendSMS("+639770710102", alertMessage);
}
}
void setup() {
    Serial.begin(115200);
    // Initialize SIM900A
    mySerial.begin(9600, SERIAL_8N1, RX_PIN, TX_PIN);
    delay(3000); // Allow time for SIM900A to initialize
    // Connect to WiFi
    WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
    Serial.print("Connecting to Wi-Fi");
    while (WiFi.status() != WL_CONNECTED) {
        delay(500);
        Serial.print(".");
    }
    Serial.println("\nConnected to Wi-Fi");
    // Configure Firebase
    config.host = FIREBASE_HOST;
    config.signer.tokens.legacy_token = FIREBASE_AUTH;
    Firebase.begin(&config, &auth);
    Firebase.reconnectWiFi(true);
    Serial.println("Ready for Firebase monitoring...");
}
void loop() {
    // Monitor CurrentData for changes
    if (Firebase.getJSON(firebaseData, "/CurrentData")) {
        FirebaseJson jsonData = firebaseData.jsonObject();
        FirebaseJsonData jsonDataValue;

        if (jsonData.get(jsonDataValue, "timestamp")) {
            String currentDataTimestamp = jsonDataValue.stringValue;
            if (currentDataTimestamp != lastCurrentDataTimestamp) {
                lastCurrentDataTimestamp = currentDataTimestamp;
                if (jsonData.get(jsonDataValue, "Temperature")) {
                    float temperatureValue = atof(jsonDataValue.stringValue.c_str());
                    checkThreshold("Temperature", temperatureValue, temperatureMin, temperatureMax, "degree
Centigrade",
                        "Investigate cooling sources and consider maintaining optimal temperature.",
                        "Check for excessive solar radiation or heated inflows. Consider aeration or shade
strategies.");
                }
                if (jsonData.get(jsonDataValue, "pH")) {
                    float pHValue = atof(jsonDataValue.stringValue.c_str());
                    checkThreshold("pH", pHValue, pHMin, pHMax, "pH",
                        "Investigate acidic inflows or decomposition of organic matter. Neutralize with alkaline
substances as necessary.",
                        "Check for algal blooms or excessive alkaline inflows. Consider adding acidic
substances.");
                }
                if (jsonData.get(jsonDataValue, "DissolvedOxygen")) {
                    float doValue = atof(jsonDataValue.stringValue.c_str());
                    checkThreshold("Dissolved Oxygen", doValue, dissolvedOxygenMin, dissolvedOxygenMax,
"mg/L",
                        "Investigate organic pollution or stagnation. Use aerators or water pumps to increase
oxygen levels.",
                        "Check for equipment calibration errors or unusual conditions. No immediate action
may be necessary.");
                }
                if (jsonData.get(jsonDataValue, "Turbidity")) {
                    float turbidityValue = atof(jsonDataValue.stringValue.c_str());

```

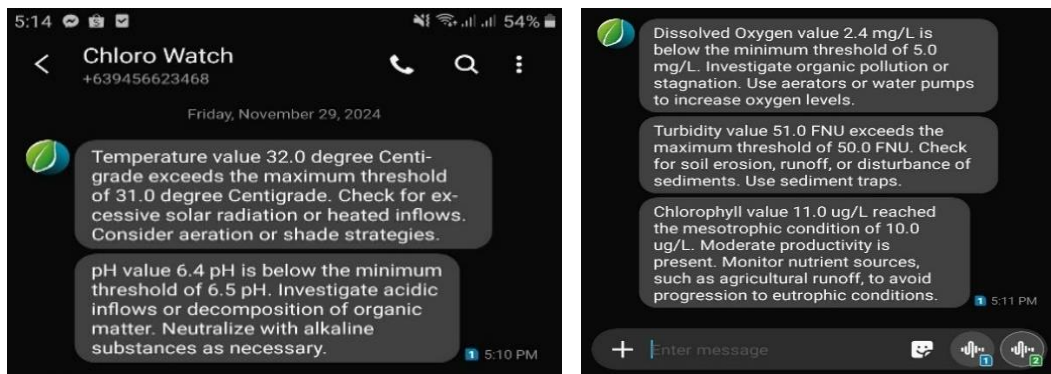
```

        checkThreshold("Turbidity", turbidityValue, 0, turbidityMax, "FNU",
            "Check for equipment calibration errors or unusual conditions. No immediate action
            may be necessary.",
            "Check for soil erosion, runoff, or disturbance of sediments. Use sediment traps.");
    }
}
} else {
    Serial.println("Failed to read CurrentData: " + firebaseData.errorReason());
}
// Monitor CurrentPrediction for changes
if (Firebase.getJSON(firebaseData, "/CurrentPrediction")) {
    FirebaseJson jsonPrediction = firebaseData.jsonObject();
    FirebaseJsonData jsonPredictionValue;
    if (jsonPrediction.get(jsonPredictionValue, "timestamp")) {
        String currentPredictionTimestamp = jsonPredictionValue.stringValue;
        if (currentPredictionTimestamp != lastCurrentPredictionTimestamp) {
            lastCurrentPredictionTimestamp = currentPredictionTimestamp;

            if (jsonPrediction.get(jsonPredictionValue, "Predicted_Chlorophyll")) {
                float chlorophyllValue = atof(jsonPredictionValue.stringValue.c_str());
                checkChlorophyllThreshold("Chlorophyll", chlorophyllValue, chlorophyllMeso, chlorophyllEutro,
                "ug/L");
            }
        }
    }
} else {
    Serial.println("Failed to read CurrentPrediction: " + firebaseData.errorReason());
}
delay(5000); // Check every 5 seconds
}

```

Output



This image shows SMS alerts from an ESP32 device, providing real-time water quality data and recommendations. The messages include values for temperature, pH, dissolved oxygen, turbidity, and chlorophyll, with suggested actions for improvement, such as aeration or adjusting pH levels. These notifications help manage water conditions effectively.

APPENDIX

B

Publishable Paper

CHLORO WATCH: WEB SYSTEM FOR WATER QUALITY MONITORING OF CHLOROPHYL LEVELS

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Keywords: Eutrophication, Chlorophyll, Water Quality Monitoring, Arduino

ABSTRACT

This paper introduces the real-time online platform called Chloro Watch. Monitoring of chlorophyll-A and critical water quality parameters, including pH dissolved oxygen, temperature, and turbidity within freshwater ecosystems. Chloro Watch, designed to address the serious threat of eutrophication. Arduino-based sensors with a data visualization platform thus enable continuous, precise monitoring and timely alerts. Critical review confirmed the significant levels of validity, responsiveness, and functionality of the system. All essential metrics comply with the ISO 25010 software quality standards. In particular, the system showed great robustness in functional completeness, the effectiveness of performance, alongside usability, as evidenced by favorable ISO assessments from the targeted users and technical experts. Further testing affirmed the reliability of sensor readings, indicating the system's efficacy in supporting water quality assessments. The overall assessment of water quality suggests that Chloro Watch has significant promise as a reliable method for water quality management, pushing both ecological sustainability and social consciousness forward. The system's design further underscores opportunities for scalability and potential partnerships with local agencies, enhancing its impact on sustainable water resource practices.

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1. INTRODUCTION

Chlorophyll-A is a great indicator of eutrophication in water bodies (Zhang, 2023). Chlorophyll-A concentration represents as a water quality parameter used to evaluate the condition of eutrophication; due to its visibility, algae growth may be noticed (Yang X., 2020). Seasonal events such as rainfalls and storms, and agricultural activities are the key factors contributing to an increasing amount of nutrients in water bodies (Akinawo, 2023). Indeed, aquaculture can also contribute to eutrophication in the lake (Villaruel, 2022). Eutrophication threatens water quality worldwide; reducing aquatic food supplies and fish production are concerns (Wurtsbaugh, 2019). Eutrophication can be a cause for the increasing number of algae. Harmful Algal Blooms can affect the ecosystem in aquatic life because of toxin production. Additionally, they cover large areas of freshwater bodies, indicating that sunlight is not reaching lower depths, which can inhibit the growth of other plants. When the algae die, it sinks to the bottom, and the decomposition of bacteria occurs; it produces toxic substances that can harm aquatic organisms, and these effects lead to a decrease in oxygen levels in the water. Concerning oxygen depletion, algae do not process photosynthesis at night; it is their respiration period. During the mentioned period, algae consume oxygen quickly, so oxygen levels drop rapidly due to their massive number. With these factors, some species may die due to suffocation, which may lead to ecosystem imbalances.

Inland municipal fisheries production in the Philippines was estimated at 205.54 metric tons in the year 2021; it was reduced by 14.7% in the year 2022, estimated at 175.35 thousand metric tons; suddenly, another reduction occurred in the year 2023, 174.58 thousand metric tons was recorded (PSA, 2024). There must be a cause for the considerable reduction. As Santiago (2022) stated, tons of bangus, tilapia, and other fish have died due to fishkill in Lake Tawiran, located at Obanda, Bulacan, in 2022, causing losses for fishers. Within the same year, another fish kill occurred at fishponds in Taal Lake in part of Agoncillo, Batangas, affecting at least 29 tons of tilapia, amounting to P3.2 million (BFAR, 2022). With the ongoing terrible incidents, a massive wave of fish kill hit South Cotabato's Lake Sebu, which began on Wednesday, January 25, 2023, and has affected millions of pesos in tilapia over three days. The continuing fish kill, known locally as "kamahong," is the third to attack the lake since

early January, bringing the total number of afflicted fish cages to 2,023 as of Friday, January 27 (Rebollido, 2023). Fish kills can occur due to several factors, including excessive nutrients leading to algae growth, low levels of dissolved oxygen, toxic compounds, and sustained increases in water temperature. Therefore, it is vital to maintain water quality, manage nutrient levels, promote sustainable practices, and prevent toxic releases to mitigate the occurrence of fish kills in freshwater ecosystems. With the said issues, people should come up with a solution, providing environmental insights to locals and an efficient way of monitoring water quality through modern technologies.

Remote sensing gives a huge perspective when visualizing an entire lake and tracking how it changes over time. The gathered data can be used to understand how lakes work and determine their water quality (Dörnhöfer, 2016). Sentinel-2 high-resolution satellite images are one of the most efficient ways to detect harmful algal blooms, particularly in small areas (Khalili, 2019). Accessing quality remote sensing data is crucial, but it can be limited or expensive, and the accuracy and applicability of existing tools and algorithms can vary depending on the specific condition of the lake being monitored. Factors such as the depth and size of the lake and the type of water pollutants can influence how accurately these tools perform (Dörnhöfer, 2016). Additionally, there can be an issue regarding using Satellite-based sensors due to climate change; cloud contamination would be the factor, and the low transparency of processed images will be a concern regarding their performance and accuracy (Zhang, 2023).

The researchers came up with the analysis, as mentioned, that rainfalls and storms are causes of eutrophication due to floods or runoffs; it was clearly important that during rainy seasons, monitoring of chlorophyll levels in lakes is a must. However, the remote sensing method has a low accuracy during rainy seasons. To come up with the solution, the researchers executed an on-site deployment of sensors, employing Arduino technologies for data retrieval and transmission. As low accuracy was not a concern, the obtained values of the water quality parameters could be utilized to determine and forecast chlorophyll levels in the lakes. A web-based system, facilitated by access to a database, is implemented to display transmitted data that offers real-time monitoring capabilities.

This study is conducted to address the alarming issue of eutrophication in Lakes, recognizing its impact on water quality and ecosystem health. Through monitoring the chlorophyll-A level, the study aims to develop a practical method to assess the eutrophication state, mainly focusing on the lakes' vulnerability to nutrient enrichment and algal blooms.

METHODOLOGY

Project Design

The system architecture efficiently connects user requirements with a technical plan for water quality monitoring. It ensures efficient data collection, processing, and presentation, aligning with monitoring and management objectives.

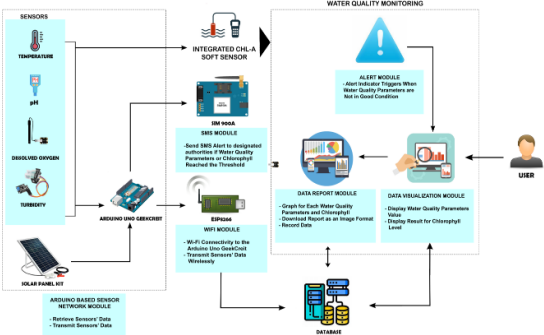


Figure 2. System Architecture

As shown in Figure 2, the web application acts as the interface facilitating user interaction with the monitoring and data retrieval process. When users interact with the web application, an established connection to sensors is implemented, and they are tasked with measuring water quality parameters, such as pH, dissolved oxygen, temperature, and turbidity in the designated environment. These parameters form the basis for fulfilling the Chlorophyll-A soft sensor model requirements. The parameters' sensors are instrumental in gathering real-time data, subsequently used to evaluate, determine, and predict the chlorophyll levels within the monitored area. The data transmission occurs from the microcontroller device, powered by Solar Panel Kit. Data is transmitted wirelessly to the database, enabling users to access timely and accurate information regarding water quality parameters and chlorophyll concentration values, especially data reports to determine the changes and behaviour of the data over time and when the threshold value of water quality parameters, and chlorophyll levels exceed, there will be an alert indicator features for users to be notified due to alarming critical levels

occurred, and an SMS will be sent to the designated authorities responsible for monitoring the area.

Materials

Software - Listed below is the software developers use to develop the web-based system, including the reasons and features.

Database set up - Firebase Real-time Database is a NoSQL cloud-hosted database that stores data as JSON and synchronizes it across all connected clients in real-time. Any changes made to the database are immediately reflected on all connected devices without the need for manual updates or refreshments. This feature is handy for collaborative applications where multiple users may access and modify the same data simultaneously.

React - React is a JavaScript framework aligned with creating creative and responsive user interfaces. React is component-based. An example of a component could be a form, form field, or button on a website.

JavaScript - It is mainly utilized to make websites more dynamic and interactive. It can enable the developers to create drop-down menus, image slideshows, forms among other features that promote user's engagement in the website.

HTML - It acts as a backbone of web pages by providing a framework and organization of content. By using its markup language, developers can create headlines, paragraphs, lists, and tables, among other variety of elements that are necessary for displaying information on the site. Apart from better accessibility and search engine rankings due to clear structural cues, the use of semantic HTML tags such as <header>, <nav>, <main>, <section>, et cetera is important in a website.

Hardware

Table 2. Minimum Requirements for PC or Laptop

MINIMUM REQUIREMENTS FOR PC OR LAPTOP	SPECIFICATION
OS	WINDOWS 10 6GB
RAM	At least 236GB Intel
HARD DISK	Celeron
PROCESSOR	

Table 2 briefly shows all the important requirements necessary for smooth and error-free work on a computer or laptop. The Windows 10 operating system is recommended because some components may not be compatible with lower versions of the OS. Additionally, this system needs an internet browser and connection to fully exploit capabilities provided to it by online service providers. If these basic requirements are met, then efficiency in performance will be guaranteed,

and users will be able to maximize system resources according to their individual needs.

Data Analysis

The developer's collaborator conducted descriptive analysis on each variable, which involves calculating key descriptive statistics, including mean and standard deviation. This process provides a detailed understanding of the dispersion of all variables.

Table 3: Descriptive Statistics of Water Quality Parameters in Lake Tikub from September 1, 2022 to September 5, 2023

	coun t	mean	std	min	max
pH	390	7.89	0.6544	6.68	8.98
DO (mg/L)	390	4.72	3.44	0.31	11.83
TDS (mg/L)	390	114.94	27.92	21.5	179.9
Salinity (ppt)	390	0.168	0.147	0.04	0.67
Cond (uS/cm)	390	233.75	45.3	131.1	372
Temp (°C)	390	28.71	2.31	26.4	34.0
BOD (mg/L)	390	1.93	1.02	0.04	5.66
TSS (mg/L)	390	0.018	0.027	0.00	0.30
NO2 (ppm)	390	0.045	0.025	0.022	0.37
NO3 (ppm)	390	0.057	0.047	0.015	0.40
PO4 (ppm)	390	0.541	0.590	0.033	2.35
NH4 (ppm)	390	0.897	0.924	0.010	3.93
TN (ppm)	390	0.996	0.933	0.032	4.03
TP (ppm)	390	0.561	0.590	0.053	2.37
BGA-PC (ug/L)	390	0.178	0.210	-0.020	1.03
Chlorophyll (ug/L)	390	3.36	5.79	0.120	57.86
Turbidity (FNU)	390	1.09	1.21	-0.10	7.65
Coliform (CFU/100ml)	390	1041.01	485.27	43.0	1400.
Cu (ppm)	390	13.63	2.08	8.9	22.4
Fe (ppm)	390	45.42	7.71	25.2	65.9
Mn (ppm)	390	63.28	7.13	46.6	82.7
Zn (ppm)	390	8.65	3.55	4.60	51.9
Cr (ppm)	390	6.53	1.66	2.7	11.8
Cd (ppm)	390	12.42	1.46	9.10	17.7
Hg (ppm)	390	1.01	0.033	1.0	1.2
As (ppm)	390	3.31	0.405	2.0	4.7
Pb (ppm)	390	4.51	0.826	1.90	7.10

The developers' collaborator performs an analysis that shows readings within the normal ranges for a freshwater lake ecosystem. A few sexamples of these key parameters are pH, dissolved oxygen, conductivity, and temperature, which are the usual values in healthy aquatic environments. Nutrient levels such as nitrogen and phosphorous fall within acceptable limits, indicating a balanced nutrient composition. Also, there is no significant contamination as its concentration remains within the typical ranges of heavy metals like copper, iron and zinc. Afterwards, by examining each parameter deeply and correlating them with one another, the developers' collaborator sought to find out the parameters that had high correlations among themselves for use in predicting chlorophyll-a levels by developing a model based on such parameters. Acidity (pH), Dissolved Oxygen (DO), Temperature (Temp) and Total Phosphorus (TP) are chosen. Due to the high cost of total phosphorus sensors, the developers' collaborators

identified an alternative parameter. Turbidity, was selected as a suitable replacement.

Model Development

As the developers' collaborator role in developing the AquaSense (Integrated Chlorophyll Soft Sensor), they recognize that each parameter employed possesses distinct characteristics that influence the model.

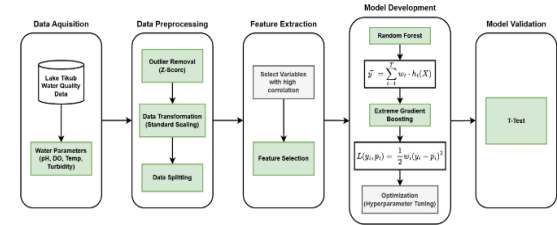


Figure 8. Model Development

To ensure the integrity of the data groups and preserve the original data trends, the developers' collaborator prioritizes the normalization of the parameters. Subsequently, a symmetrical transformation is applied to these normalized parameters, priming them to apply advanced machine-learning techniques. This method marks the initiation of the development phase of the AquaSense model.

Data

The highly correlated parameters necessitate the provision of sensors. The sensors will be responsible for gathering data originating from the lake. To validate the sensors' readings, a laboratory examination was done, which involved the collection of water samples from the same location where the sensors are deployed and comparing the results obtained from laboratory analysis with the readings recorded by the sensors at the time of sample collection. The process facilitates precise comprehension and visualization for users, enhancing the accessibility and usability of the information provided.

Project Development

The developers skillfully crafted and implemented the web-based solution using the agile scrum software development life cycle approach. Utilizing the agile scrum method, the project's progression was effectively overseen by breaking it down into distinct phases. This is necessary to continuously improve in every phase and proceed with end user participation through the evolving needs-driven development journey.

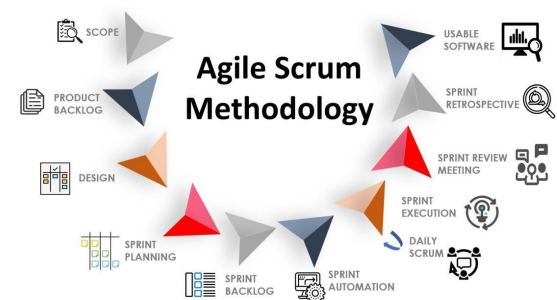


Figure 4. Agile Scrum Methodology

Figure 4 presents a project management methodology focusing on incremental development. The approach calls for dividing a project into pieces that can be accomplished in sprints of two to four weeks. These sprints are meant to prioritize and complete features quickly, which could produce a working system in stages. Projects may be continuously refined and adapted through this iterative process to move along with changing requirements and feedback. With Agile Scrum, developers use a flexible approach, enabling them to deliver projects wisely, and satisfy more users with their work.

Evaluation Procedure

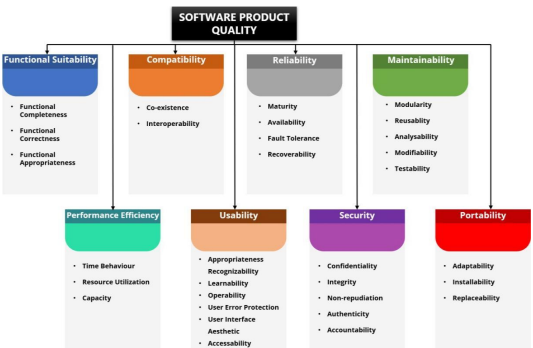


Figure 15. Quality Characteristics of ISO 25010

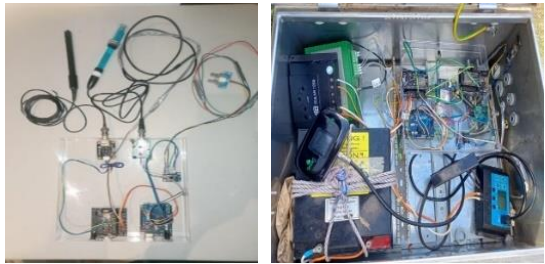
To ensure the quality of the developed web application, it must adhere to standards such as ISO 25010, which identifies what constitutes a good product. This standard is used as a foundation for assessing the quality of software products. The ISO 25010 model has eight quality characteristics, representing different aspects of a software product for evaluation.

It was imperative to apply ISO 25010 in evaluating the Chloro Watch's functionality and general characteristics. The project's respondents and IT professionals used this criterion since it is mostly utilized to appraise program consistency. Thus, assessments based on ISO 25010 include both functional and non-functional properties such as internal design features, external structure, etc.

Users and IT specialists executed beta testing in the developed web- based system during the development process. Likert Scalar adopted the ISO 25010 quality model as a reference point for evaluation purposes. The Chloro Watch Web System testing focused on Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The Likert Scalar responses used in the assessment include SA (Strongly Agree), A (Agree), D (Disagree), and SD (Strongly Disagree). By employing ISO 25010 as a guiding standard, the developers aimed to ensure that the system

met the highest standards in terms of its various quality aspects.

Population of the Study



Researchers conducted a survey at Brgy to build a system and have precise information. Tadlac, Los Baños, Laguna, with the help of staff, the researchers surveyed 20 users, 3 staff, 2 I.T. This procedure prioritizes empowering authorities to sustain local individuals' lives regarding their living conditions, issues, and potential for change regarding Tadlac Lake.

Table 6. 4-Point Likert Scale Technique

RATING	MEAN RANGE	INTERPRETATION
4	3.25 – 4.00	Strongly Agree
3	2.50 – 3.24	Agree
2	1.75 – 2.49	Not Agree
1	1.00 – 1.74	Strongly Not Agree

The developers utilized the Likert Scale as a survey technique, displayed in the table above, to evaluate user satisfaction with the web application. Utilizing a percentage-based approach, they assess the dimensions of appraisal expressed by respondents.

RESULTS BY OBJECTIVES

1. Gather data for a water quality parameter such as pH, dissolved oxygen, temperature, and turbidity through Arduino-based sensors.

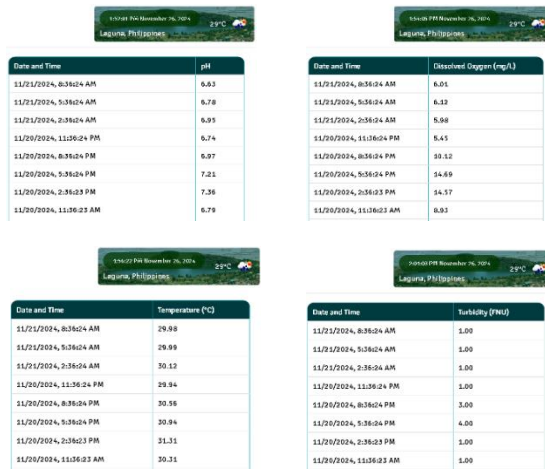


Figure 16. Data Report

These parameters significantly contributed to AquaSense, fulfilling its function with the prediction of Chlorophyll-a, to provide a data collection, the developers used sensors compatible with arduino. Performing accurate readings is one of the goals, the sensors undergone in calibration by marking it with a standard scale of readings that supports to AquaSense regarding with reliability of the model. The data collected from the sensors represent to train the model in the long run, achieving the desired performance aligned with efficient monitoring of water quality.



Figure 17. Arduino-based Sensors

As shown in figure 17, the setup specifically designed to measure key water quality parameters: pH, turbidity, dissolved oxygen, and temperature. To ensure the accuracy and reliability of the readings, each sensor undergoes rigorous calibration prior to deployment. Calibration procedures involve adjusting the sensors against standard solutions or controlled conditions that reflect known values for each parameter. These sensors are connected to an Arduino Uno Geekcreit, and ESP8266 microcontroller, which processes the collected data. The developers used serial communication approach for the transmission of data from Arduino to ESP. Using an ESP8266 and pocket wifi, the system transmits real-time readings to a Firebase database, ensuring seamless remote monitoring and data accessibility. By leveraging these sensor readings, the system enables a data-driven approach to chlorophyll prediction, providing timely and actionable insights for managing water resources.



Figure 18. Battery, Solar Panel, and Large Buoy

To ensure the continuous operation of the water quality monitoring system deployed in Tadlac Lake, a reliable and sustainable power source was implemented. The system utilizes a 12V solar panel integrated with a solar charge controller and a 12V Yuasa NPL24-12I battery. The solar panel is mounted on a large buoy, which also houses the battery, solar charge controller, and other hardware devices, providing a stable and efficient energy setup for the monitoring equipment. The integration of the solar panel, battery, and charge controller within the buoy setup not only simplifies deployment but also ensures the stability and longevity of the system's power supply. Furthermore, the renewable energy source aligns with sustainability objectives, reducing reliance on non-renewable power sources and ensuring that the system can operate autonomously in remote locations without direct human intervention.

2. Develop a web-based system that monitor water quality such as pH, DO, Temp, Turb, and Chl-A.

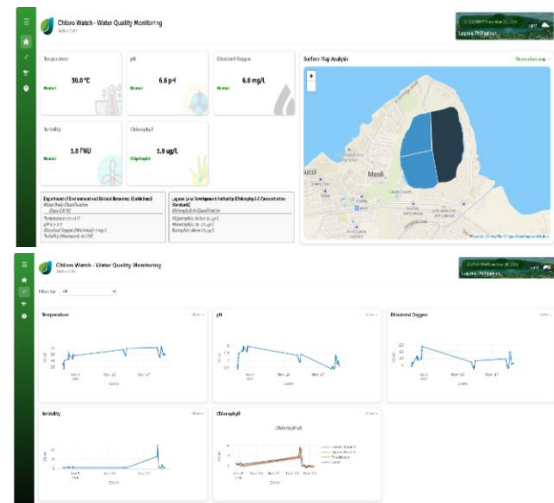


Figure 19. Data Visualization

The system was designed to monitor various parameters, as shown in figures above, the dashboard was showing parameters' values to easily know the current condition of the lake at a given time. The data showcased in figures provide a comprehensive overview of water quality conditions. Parameters' graphs can provide the data records, providing insights regarding with the behavior of the lake. The platform puts users by providing them with control over the analysis. Overall, this comprehensive monitoring system enables effective assessment and management of the lake's health and behavior over time.

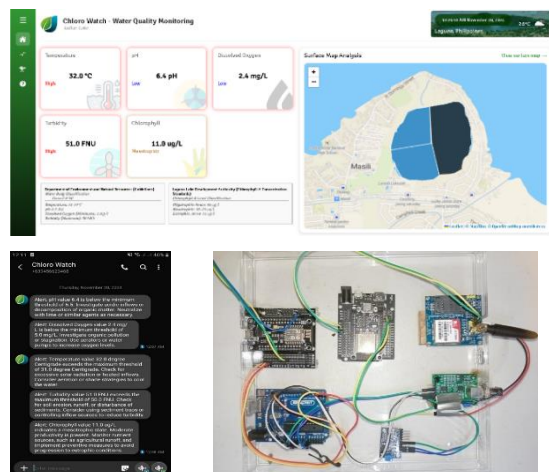


Figure 20. Alert Indicator and GSM-Based SMS

The SMS Module, as depicted in Figure 20, is an integral component of the system, designed to ensure timely notification of critical water quality threshold breaches. The developers utilize hardware serial communication between the SIM900A and the ESP32 microcontroller. Sensors' data, including pH, dissolved oxygen, temperature, turbidity, and chlorophyll-A level, are transmitted to Firebase via an ESP8266. The ESP32 retrieves these readings, analyzes them against predefined thresholds, and, upon detecting a breach, uses

the SIM900A to send real-time SMS alerts to authorities. The system ensures swift dissemination of critical information. For instance, breaches in parameters are flagged with specific details, including the measured value and recommended actions, such as investigating nutrient sources, deploying aeration solutions, etc. Specifically in monitoring chlorophyll levels, while mesotrophic conditions represent moderate productivity, this state marks a transitional phase where nutrient levels begin to escalate, increasing the risk of tipping into eutrophic conditions. Proactive alerts during this phase allow authorities to implement management measures before algal blooms manifest, potentially averting ecological degradation. The decision to alert at mesotrophic levels reflects an evidence-based strategy prioritizing prevention over remediation.

3. Test and perform an evaluation of the web-based system through a software review using ISO 25010.

Survey Questionnaire Results

The results are collected from the respondents, who are staffs from Barangay Tadalac, potential users, and I.T.

Table 7. Post Evaluation Result.

ITEM	MEAN	SD	INTERPRETATION
Functional Completeness	3.64	0.52	Strongly Agree
Functional			
Correctness	2.96	0.95	Agree
Functional	3.36	0.61	Strongly Agree
Appropriateness Time			
Behavior	3.12	0.66	Agree
Resource Utilization	3.20	0.63	Agree
Capacity	3.04	0.64	Agree
Co-existence	3.16	0.65	Agree
Appropriateness	3.24	0.72	Agree
Recognizability	2.96	0.86	Agree
Learnability			
Operability	2.96	0.86	Agree
Use Error Protect	3.36	0.60	Strongly Agree
User Interface	3.28	0.65	Strongly Agree
Aesthetic	3.36	0.63	Strongly Agree
Accessibility	3.28	0.63	Strongly Agree
Maturity	3.23	0.68	Agree
Availability	3.23	0.70	Agree
Fault Tolerance	3.62	0.52	Strongly Agree
Recoverability	3.48	0.61	Strongly Agree
Confidential	3.27	0.45	Strongly Agree
Integrity	3.40	0.57	Strongly Agree
Non-Repudiation	3.35	0.65	Strongly Agree
Accountability	3.40	0.56	Strongly Agree
Authenticity	3.40	0.56	Strongly Agree
Modularity	3.40	0.56	Strongly Agree
Reusability	3.40	0.56	Strongly Agree
Analyzability	3.40	0.56	Strongly Agree
Modifiability	2.88	0.80	Agree
Testability	3.23	0.67	Agree
Adaptability	3.27	0.64	Strongly Agree
Installability	3.40	0.65	Strongly Agree
Replaceability	3.35	0.61	Strongly Agree
Total	3.25	0.60	Strongly Agree

Table 7 shows the respondents' overall evaluations and the developed system's acceptance level. The system achieved high scores in functional completeness, correctness, and appropriateness, confirming its ability to reliably monitor water quality and generate threshold-based alerts. Performance metrics highlight its efficiency in real-time data transmission, low-latency alerting, and resource utilization. Compatibility with existing hardware and scalability for future enhancements further emphasize its robustness. Usability scores reflect an intuitive interface that facilitates ease of operation for stakeholders, while

reliability assessments confirm consistent availability and fault tolerance. Additionally, the system's modular design ensures maintainability and portability, making it adaptable for deployment in diverse environments. The weighted mean of 3.25 was interpreted as strongly agreeing, indicating that the respondents confirmed to system's functionality and that the system passed the software evaluation based on ISO 25010. These findings affirm the system's capability to support proactive and effective water quality management.

T-test Result

Table 8. Data Collection for Testing of Difference of Water Quality Parameter

Group	PH	TURB	TEMP	DO
1	8.38	1.01	30.94	14.77
2	8.90	3.91	31.70	11.78
1	8.34	1.02	30.83	14.09
2	8.91	4.16	31.70	12.36
1	8.53	1.01	31.27	15.17
2	8.98	4.26	31.80	12.56
1	8.62	1.03	31.69	14.49
2	8.91	4.66	31.70	13.00
1	7.52	1.04	28.38	3.70
2	7.83	4.66	29.10	2.97
1	7.64	1.02	28.56	4.79
2	7.58	4.20	28.75	3.24
1	7.67	1.05	28.63	6.33
2	7.61	4.36	28.65	3.89
1	7.42	1.01	28.63	6.33
2	7.68	4.17	28.60	4.72
1	7.65	1.07	28.69	6.62
2	7.70	4.70	28.80	4.98

The data presented in Table 8 compares the water quality measurements collected using two different sensor systems: Group 1 represents the Arduino-based sensor readings, and Group 2 represents the laboratory sensor readings. The water samples were taken from Tadalac Lake, dataset from Group 1 and Group 2 is being used to conduct a t-test, which is a statistical method to determine if there is no significant difference between the readings of the two sensor systems, indicating that the Arduino sensor system is reliable for monitoring water quality.

Table 9. Independent Samples T-Test Result

	t	df	p
pH	-0.951	16	0.356
Turbidity	-0.951	16	< .001*
Temperature	-35.984	16	0.619
Dissolved	-0.507	16	0.412
Oxygen	0.842	16	

The comparison shows that while the Arduino-based system demonstrates adequate accuracy in measuring pH, temperature and dissolved oxygen, it shows a notable discrepancy in turbidity readings. These findings indicate that, while the Arduino system is partially effective for water quality monitoring, further refinement is necessary, particularly in recalibrating the turbidity sensor, to achieve better accuracy and consistency with laboratory standards. The discrepancy in turbidity may be attributed to limitations inherent in the generic turbidity sensor used with the Arduino, which may not provide the same level of precision as specialized laboratory equipment. Specifically, enhancing the calibration of the generic turbidity sensor or exploring higher-precision alternatives could improve measurement accuracy and alignment with laboratory

standards. With these refinements, the Arduino-based system could become a more reliable tool for practical water quality monitoring applications.

Results of Software Testing

Table 10. Unit Testing for Web Application and Devices

Module	Expected Output	Actual Output	Response Time/Duration	Remarks
Wifi	The esp8266 can transfer the sensors' readings in database.	The web application successfully transfer the readings in database.	1.2 seconds	Passed
Data Visualization	The web application can show the history of data through graph.	The web application successfully displayed the graphs for each water quality parameter.	2.5 seconds	Passed
Alert Indicator	The web application can alert the user if there is an exceeding threshold for each water quality parameter whether high or low.	The web application successfully alerts the user if there is an exceeding threshold.	1 second	Passed
Data Report	The web application can record the data.	The web application successfully recorded the data.	2.5 seconds	Passed
SMS	The sim900a can send an sms to designated authorities when readings' threshold value exceed.	The sim900a successfully sent an sms.	3.5 seconds	Passed
Power Source	The devices should run continuously on battery, even during periods of no sunlight	The system successfully operates battery without interruption even during low light.	14 hrs	Passed

The findings were obtained through testing on the system's units. The results indicate that all modules have passed the tests successfully, with response times well within acceptable limits. The system demonstrates reliability and efficiency in data transfer, visualization, alerting, and communication, ensuring it meets the functional requirements for monitoring water quality parameters effectively.

Table 11. Integration Testing

Description	Expected Output	Actual Output	Response Time	Remarks
Software	The web application can show the water quality parameters data through graphs.	The web application successfully displayed data through graphs.	1.5 seconds	Passed

Hardware	The device can transmit the water quality data to database into web application.	The device successfully displayed the water quality data to web application.	1.5 seconds	Passed
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The integration testing results indicate that both the software and hardware components of the system are functioning as intended when combined. The results show that the integration between the software and hardware components is successful, with both achieving the expected functionality and meeting the required response times. Additionally, periodic maintenance of both the hardware and software components should be implemented to ensure long-term functionality and accuracy.

Table 12. Stress Test and Load Test under System Testing

Description	Steps	Expected Output	Actual Output	Remarks
Stress Test: 1. High Sensor Data Input	Simulate a high quantity of sensor data inputs	The system should maintain real time monitoring for all parameters	The system successfully maintains real time monitoring for all parameters	Passed
Stress Test: 2. Database performance	Assess database performance under load	Database should handle transactions without delay	No delay observed in database write operations	Passed
Load Test 3. Continuous data Input	Simulate a continuous data from sensor with a fast time interval	Data should be logged accurately without loss	All data logged without loss	Passed

The results of the stress and load tests demonstrate that the system is capable of sustaining high performance, even under heavy data loads and continuous input conditions. The system successfully passed all tests, ensuring both reliable real-time monitoring and accurate data logging under stress scenarios. This underscores the system's robustness and scalability, making it well-suited for environments that demand continuous and intensive data processing.

Table 13. Acceptance Testing

CHARACTERISTICS	MEAN	SD	INTERPRETATION
Functionality	3.32	0.69	Strongly Acceptable
Performance Efficiency	3.12	0.64	Acceptable
Compatibility	3.20	0.69	Acceptable
Usability	3.20	0.70	Acceptable
Reliability	3.37	0.63	Strongly Acceptable
Security	3.36	0.56	Strongly Acceptable
Maintainability	3.26	0.64	Strongly Acceptable
Portability	3.34	0.63	Strongly Acceptable
Overall	3.27	0.65	Strongly Acceptable

The acceptance testing results indicate that the software exhibits strong acceptance in several key areas, notably in functionality, reliability, security, maintainability, and portability, all of which are rated as "Strongly Acceptable." This suggests that the software effectively meets its intended purposes and performs reliably under various conditions. Areas like performance efficiency, compatibility, and usability are rated as "Acceptable," indicating that while these aspects are satisfactory, there may still be room for improvement to enhance the overall user experience and compatibility with other systems. The overall rating of 3.27 with a standard deviation of 0.65 reflects a generally positive reception, deeming the software "Strongly Acceptable" in meeting the expectations for water quality monitoring tasks.

SUMMARY, CONCLUSIONS, AND RECOMMENDATION

This chapter provides an overview of the findings of the study, summarizes the progress achieved, and presents conclusions from system development. Furthermore, it outlines recommendations for system improvements, and future implementation strategies.

Summary

The project addresses the critical global issue of water quality degradation, particularly the threat of eutrophication, which significantly impacts aquatic ecosystems and communities reliant on these resources. The study focused on developing a web-based application designed to facilitate early detection and continuous monitoring of chlorophyll-A and other essential water quality parameters. The system integrates Arduino-based sensors to collect data on pH, dissolved oxygen, temperature, and turbidity. T-test results confirmed the system's validity, achieving a strong correlation between sensor data and laboratory measurements. Further refinement for turbidity sensor is needed. Enhancing the calibration process or exploring more precise alternatives could improve measurement accuracy and better align with laboratory standards.

The development process employed the Agile Scrum methodology to ensure iterative improvements and user-centric design. The system effectively collected water quality parameters, including pH, dissolved oxygen, temperature, and turbidity, through calibrated sensors. The gathered data demonstrated reliability, enabling precise predictions of chlorophyll-A level using the AquaSense model. A user-friendly web interface was developed as part of the project, featuring real-time dashboard for visualizing water quality metrics, historical data trends, and an alert indicator and SMS mechanism for threshold breaches. The system's sustainable operation was ensured through the integration of a solar-powered devices and buoy, enabling autonomous functionality in remote environments.

The ISO 25010 evaluation highlighted high functional completeness, usability, and efficiency, with strong feedback from stakeholders affirming its practical

value. Extensive testing and evaluation processes confirmed the system's reliability and robustness. Software testing, encompassing unit, integration, system, and acceptance tests, validated the functionality and seamless interaction of system components. Software testing results further underscored the system's resilience. Unit testing ensured that individual components performed correctly, while integration testing verified seamless functionality across modules. Stress and load testing demonstrated the system's capacity to handle dynamic conditions without compromising performance.

By aligning with Sustainable Development Goals, Chloro Watch promotes proactive water quality management, ensuring sustainable aquatic ecosystems. The study concludes that the system is a scalable, reliable, and impactful tool for addressing water quality monitoring challenges, particularly in addressing eutrophication and supporting sustainable water resource practices.

1. Gather data for a water quality parameter such as pH, dissolved oxygen, temperature, and turbidity through Arduino-based sensors.

The system achieved precise and consistent data collection for critical water quality parameters, including pH, dissolved oxygen, temperature, and turbidity, using Arduino-based sensors. Calibration processes ensured that sensor readings were aligned with standards, enhancing the reliability of the collected data. Moreover, the use of a solar-powered system ensured uninterrupted data collection and transmission, demonstrating the feasibility of deploying sustainable, cost-effective monitoring solutions in remote aquatic environments.

2. Develop a web-based system that monitors water quality such as pH, DO, Temp, Turb, and Chl-A.

The system was designed to provide real-time visualization and historical analysis of water quality data. It featured an intuitive dashboard that displayed parameter trends and chlorophyll-A predictions, allowing users to monitor the lake's health efficiently. The system empowered users with tools to assess water quality changes over time, supported by graphical insights into parameter behavior. Its accessible design ensures that even non-technical users can benefit from its functionalities, fostering proactive decision-making to address water quality issues, particularly those related to oxygen depletion, eutrophication and harmful algal blooms.

3. Test and perform an evaluation of the web-based system through a software review using ISO 25010.

The evaluation of the web system under ISO 25010 standards confirmed its effectiveness in meeting user requirements. Functional suitability, performance efficiency, and usability received high ratings from respondents, including staff, IT professionals, and potential end-users. The system demonstrated robust fault tolerance, operability, and user interface design,

making it reliable and easy to use for continuous monitoring. These results validate the system's potential for real-world application, particularly in scenarios where timely data and analysis are crucial for mitigating environmental challenges.

Recommendation

1. Expansion of Data Capture for Model Enhancement - Implement ongoing analysis of the data collected from the monitoring system to identify trends, patterns, and potential areas of concern regarding water quality. Use these insights to refine predictive models for chlorophyll-A levels and other water quality parameters, ensuring that they accurately represent real-world conditions and provide reliable forecasts.

2. Collaboration with Local Government Units (LGUs) - Foster partnerships with local government agencies, environmental organizations, and community stakeholders to leverage shared expertise and resources in managing water quality issues. Engaging these stakeholders in the development and implementation of monitoring initiatives can enhance the effectiveness of the system and garner broader support for conservation efforts.

3. Expansion of Sensor for Additional Water Quality Parameters - Plan for the scalability and future expansion of the water quality monitoring system to accommodate evolving environmental conditions and additional water quality parameters of interest. This includes investing in flexible technologies and infrastructure that can adapt to changing needs, as well as expanding the sensor network to cover a larger geographic area or monitor new pollutants.

4. Securing Buoy Infrastructure for Sensor Deployment - Establishing robust and secure buoy infrastructure for the deployment and protection of water quality sensors and other monitoring devices in aquatic environments, particularly in lakes. This infrastructure will be designed to withstand harsh environmental conditions, including extreme weather events such as typhoons, and other natural disasters that may impact sensor stability and data reliability.

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APPENDIX

C

ISO 25010 QUESTIONNAIRE

ISO 25010 QUESTIONNAIRE

Dear respondents,

We are students of the Laguna State Polytechnic University – Los Baños Campus under the College of Computer Studies conducting a research on the project entitled Chloro Watch: **Water Quality Monitoring of Chlorophyll level**. We would like to seek your opinion on the acceptability of the said project by scoring the statements below by checking the appropriate numbers, wherein 1 = not acceptable; 2 = slightly acceptable; 3 = moderately acceptable; and 4 = highly acceptable. Rest assured that any information that you provide will be treated with utmost confidentiality (RA 10173. Data Privacy Act) and will be used for research purposes only. Thank you for your cooperation.

<Names of Proponents>

NAME OF RESPONDENT: _____

Categories			1	2	3	4
Functional Suitability						
1	Functionality Completeness	The software includes all necessary features and functions required for water quality monitoring. <i>(Ang software ay may lahat ng mahalagang tampok at kakayahan na kailangan para sa pagsubaybay ng kalidad ng tubig.)</i>				
2	Functionality Accuracy	The software accurately performs its intended functions, providing reliable water quality monitoring, predictions based on sensor data inputs. <i>(Ang software ay tumpak na gumaganap ng mga naka-takdang tungkulin nito, nagbibigay ng maaasahang pagsubaybay sa kalidad ng tubig at mga prediksiyon batay sa mga input mula sa sensor data.)</i>				
3	Functionality Appropriateness	The software's features and functions are suitable and relevant for water quality monitoring. <i>(Ang mga tampok at kakayahan ng software ay lubos na angkop at naaayon para sa masusing pagsubaybay at pagtatasa ng kalidad ng tubig.)</i>				
Performance Efficiency			1	2	3	4
4	Time Behaviour	The software performance perform tasks within acceptable time limits, providing timely predictions to users. <i>(Ang pagganap ng software ay mahusay sa pagtupad ng mga gawain sa loob ng tinatanggap na oras, na nagbibigay ng napapanahong prediksiyon sa mga gumagamit.)</i>				

5	Resource Utilication	The software optimizes the use of the system resource to ensure efficient water quality monitoring. <i>(Ang software ay nag-aayos at nagpapabuti ng paggamit ng mga yaman ng sistema upang matiyak ang mas epektibong pagsubaybay sa kalidad ng tubig.)</i>				
6	Capacity	The software can handle the expected data related to water quality monitoring without performance degradation. <i>(Ang software ay kayang hawakan ang inaasahang datos ukol sa pagsubaybay ng kalidad ng tubig nang walang pagbaba sa pagganap.)</i>				
Compatibility			1	2	3	4
7	Co-existence	The software can coexist with other water quality monitoring systems and software applications without conflicts. <i>(Ang software ay maaaring makipag-ugnayan sa ibang mga sistema at aplikasyon para sa pagsubaybay ng kalidad ng tubig nang walang anumang alitan o kaguluhan.)</i>				
8	Interoperability	The software integrates and interoperates with external sensors, data sources, and platforms for comprehensive water quality analysis. <i>(Ang software ay nagtataglay ng kakayahang magsanib at makipag-ugnayan sa mga panlabas na sensor, pinagkukunan ng datos, at mga plataporma upang magkaroon ng komprehensibong pagsusuri ng kalidad ng tubig.)</i>				
Usability			1	2	3	4
9	Appropriateness Recognicability	The software's interface is intuitive and easily recpgnizable, facilitating effient user navigation and interaction. <i>(Ang interface ng software ay madaling maunawaan at kilala, na nagpapadali sa maayos at mabilis na pag-navigate at pakikipag-ugnayan ng mga gumagamit.)</i>				
10	Learnability	Users can quickly learn how to use the software for water quality monitoring. <i>(Madali at mabilis matutunan ng mga gumagamit kung paano gamitin ang software para sa pagsubaybay ng kalidad ng tubig.)</i>				

11	Operability	The software is easy to operate and control, enabling users to perform water quality monitoring tasks effectively. <i>(Madali ang paggamit at kontrol ng software, na nagbibigay-daan sa mga gumagamit na maisagawa ang mga gawain sa pagsubaybay ng kalidad ng tubig nang epektibo.)</i>				
12	User Error Protection	The software includes features to prevent user errors and mitigate their impact on water quality monitoring. <i>(Ang software ay may mga tampok na naglalayong pigilan ang mga pagkakamali ng gumagamit at bawasan ang epekto ng mga ito sa pagsubaybay ng kalidad ng tubig.)</i>				
13	User Interface Aesthetics	The software's user interface is visually appealing and enhances user experience. <i>(Ang user interface ng software ay kaakit-akit sa mata at nagpapahusay sa karanasan ng mga gumagamit.)</i>				
14	Acceptability	The software is accessible to any user, providing inclusive access to water quality monitoring. <i>(Ang software ay madaling ma-access ng sinumang gumagamit, na nagbibigay ng inklusibong pag-access sa pagsubaybay ng kalidad ng tubig.)</i>				
Reliability			1	2	3	4
15	Maturity	The software has achieved a high level of maturity in its development and deployment, ensuring stable and reliable water quality monitoring. <i>(Ang software ay nakamit ang mataas na antas ng pag-unlad at pag-deploy, na tinitiyak ang matatag at maaasahang pagsubaybay ng kalidad ng tubig..)</i>				
16	Availability	The software is consistently available for water quality monitoring tasks, minimizing downtime and service interruptions. <i>(Ang software ay palaging available para sa mga gawain sa pagsubaybay ng kalidad ng tubig, na nagpapababa ng oras ng pagka-down at mga pagka-abala sa serbisyo.)</i>				

17	Fault tolerance	The software can continue operating reliably in the event of minor faults or errors during water quality monitoring. <i>(Ang software ay patuloy na makakapag-operate ng maaasahan kahit na may mga maliliit na pagkakamali o depekto habang isinasagawa ang pagsubaybay ng kalidad ng tubig.)</i>				
18	Recoverability	The software can recover gracefully from failures or disruptions, ensuring continuity in water quality monitoring. <i>(Ang software ay kayang makabawi nang maayos mula sa mga pagkabigo o pagkagambala, na tinitiyak ang tuloy-tuloy na pagsubaybay sa kalidad ng tubig.)</i>				
Security			1	2	3	4
19	Confidentiality	The software protects sensitive water quality data and user information from unauthorized access or disclosure. <i>(Ang software ay nagpoprotekta sa sensitibong datos ng kalidad ng tubig at impormasyon ng gumagamit mula sa hindi awtorisadong pag-access o pagbubunyag.)</i>				
20	Integrity	The software ensures the integrity of water quality monitoring, preventing unauthorized tampering or modification. <i>(Ang software ay tinitiyak ang integridad ng pagsubaybay ng kalidad ng tubig, na pumipigil sa hindi awtorisadong pagmamaniipula o pagbabago.)</i>				
21	Non-reputation	The software provides mechanism to prevent users from denying their actions or transactions related to water quality monitoring. <i>(Ang software ay nagbibigay ng mekanismo upang pigilan ang mga gumagamit na tanggihan ang kanilang mga aksyon o transaksyon kaugnay ng pagsubaybay ng kalidad ng tubig.)</i>				
22	Accountability	The software tracks and logs user actions and transactions related to water quality monitoring. <i>(Ang software ay nagmamamati at nagtatala ng mga aksyon at transaksyon ng gumagamit na may kaugnayan sa pagsubaybay ng kalidad ng tubig.)</i>				

23	Authenticity	The software verifies the authenticity of users and data sources involved in water quality monitoring. <i>(Ang software ay nire-verify ang pagiging tunay ng mga gumagamit at pinagkukunan ng datos na kasangkot sa pagsubaybay ng kalidad ng tubig.)</i>				
Maintainability			1	2	3	4
24	Modularity	The software is modularity structured, allowing for easy maintenance and updates to individual components. <i>(Ang software ay nagpoprotekta sa sensitibong datos ng kalidad ng tubig at impormasyon ng gumagamit mula sa hindi awtorisadong pag-access o pagbubunyag.)</i>				
25	Reusability	The software components related to water quality monitoring can be used in other applications or environment. <i>(Ang mga bahagi ng software na may kaugnayan sa pagsubaybay ng kalidad ng tubig ay maaaring magamit sa ibang aplikasyon o kapaligiran.)</i>				
26	Analyzability	The software includes features for analyzing and diagnosing issues related to water quality monitoring, facilitating maintenance and troubleshooting. <i>(Ang software ay may mga tampok para sa pagsusuri at pag-diagnose ng mga isyu kaugnay ng pagsubaybay ng kalidad ng tubig, na nagpapadali sa pagpapanatili at paglutas ng problema.)</i>				
27	Modifiability	The software is easily modified to accommodate changes or updates. <i>(Ang software ay madaling baguhin upang tumugon sa mga pagbabago o update.)</i>				

28	Testability	The software components related to water quality monitoring are designed for effective testing, ensuring reliability and accuracy in monitoring. <i>(Ang mga bahagi ng software na may kaugnayan sa pagsubaybay ng kalidad ng tubig ay dinisenyo para sa epektibong pagsusuri, na tinitiyak ang pagiging maaasahan at katumpakan sa pagsubaybay.)</i>				
Portability			1	2	3	4
29	Adaptability	The software can adapt to different environments and configurations for water quality monitoring, ensuring consistent performance across various settings. <i>(Ang software ay kayang umangkop sa iba't ibang kapaligiran at mga configuration para sa pagsubaybay ng kalidad ng tubig, na tinitiyak ang pare-parehong pagganap sa iba't ibang setting..)</i>				
30	Installability	The software can be easily installed and configured for water quality monitoring task without significant effort or technical expertise. <i>(Ang software ay madaling i-install at i-configure para sa mga gawain sa pagsubaybay ng kalidad ng tubig nang walang malaking pagsisikap o teknikal na kaalaman.)</i>				
31	Replaceability	The software can be replaced or upgrade with minimal disruption to water quality monitoring operations and data integrity. <i>(Ang software ay maaaring palitan o i-upgrade nang may kaunting pagka-abala sa mga operasyon ng pagsubaybay ng kalidad ng tubig at integridad ng datos.)</i>				

Signature of the Respondent

APPENDIX

D

Grammar Rate

THESIS PART I

by DO NOT DELETE OTHER FILES OR ELSE WARRANTY WILL BE VOID

General metrics

55,967	7,850	552	31 min 23 sec	1 hr 0 min
characters	words	sentences	reading time	speaking time

Score



This text scores better than 95% of all texts checked by Grammarly

Writing Issues

143	54	89
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

THESIS PART II

by DO NOT DELETE OTHER FILES OR ELSE WARRANTY WILL BE VOID

General metrics

51,907	7,434	934	29 min 44 sec	57 min 11 sec
characters	words	sentences	reading time	speaking time

Score



This text scores better than 97% of all texts checked by Grammarly

Writing Issues

72	50	22
Issues left	Critical	Advanced

Plagiarism

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APPENDIX

E

Similarity Report

13% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Crossref database
- Crossref posted content database

Match Groups

-  **145 Not Cited or Quoted 12%**
Matches with neither in-text citation nor quotation marks
-  **12 Missing Quotations 1%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 9%  Internet sources
- 2%  Publications
- 9%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

-  **145 Not Cited or Quoted 12%**
Matches with neither in-text citation nor quotation marks
-  **12 Missing Quotations 1%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 9%  Internet sources
- 2%  Publications
- 9%  Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	
www.researchgate.net		1%
2	Submitted works	
Laguna State Polytechnic University on 2022-04-19		1%
3	Internet	
www.mdpi.com		1%
4	Internet	
www.scilit.net		1%
5	Internet	
academic-accelerator.com		0%
6	Submitted works	
Laguna State Polytechnic University on 2021-07-22		0%
7	Internet	
www.rappler.com		0%

APPENDIX

F

Curriculum Vitae



Joshua Bryan N. Valencia

CONTACT

09396517141

Joshuabryan050@gmail.com

Links

Facebook

<https://m.me/Joshua.val05>

Messenger

<https://m.me/Joshua.val05>

Skills

- CSS/HTML
- Python(Programmer Language)
- Microsoft Office
- SQL
- PHP
- Data Analysis
- Customer Check-in/Check-out
- Adobe Photoshop Editor
- Figma(Design Software)

LANGUAGES

English, Tagalog

About Me

As an IT programmer with a Bachelor of Science in Information Technology, skilled in web design and dedicated to delivering exceptional customer service. Known for creating efficient, user-centered solutions and providing technical support that enhances user experience. Committed to continuous learning and bringing both technical expertise and customer-focused insights to innovative projects.

EDUCATION

2016-2019

Jr. High School, EDUARDO BARRETTO SR. NATIONAL HIGHSCHOOL

With Honors

2019-2021

Sr. High School Diploma, AMA COMPUTER COLLEGE - CALAMBA CAMPUS

2022-2026

Bachelor of Science in Information Technology, LAGUNA STATE POLYTECHNIC UNIVERSITY - LOSBAÑOS CAMPUS

Achievements

- Programming (Oracle Database) NC III Part1
- Programming (Oracle Database) NC III Part2
- Programming (Java) NC III Part2I have a proven track record of delivering high-quality care while maintaining patient safety and satisfaction

WORK EXPERIENCE

June – September 2021

Direct Picking/J-SOX, EPPI, EPSON PRECISION PHILIPPINES INCORPORATED

- Transactions Audit Findings.
- Delivering and picking printer and projector parts in production line.
- Recording receipt code, dates and company name on receipt.



Tom D. Deocales

CONTACT

09456623468

Deocales8@gmail.com

Links

Facebook

facebook.com/profile.php?id=100009355344474

Github

github.com/tomdeocales

Skills

- CSS/HTML
 - Python
 - SQL
 - React (Javascript)
 - PHP
 - Firebase
 - Adobe Photoshop
 - Figma (Design Software)
-

LANGUAGES

English, Tagalog

About Me

As an IT student pursuing a Bachelor of Science in Information Technology, I am honing my skills in web design and development with a focus on creating efficient, user-centered solutions. I am passionate about enhancing user experiences and providing technical support through hands-on learning. Eager to continuously expand my knowledge, I strive to integrate both technical expertise and customer-focused insights into innovative projects, with a commitment to professional growth and problem-solving in the IT field.

EDUCATION

2016-2019

Jr. High School, ASIAN COMPUTER COLLEGE INC.

2019-2021

Sr. High School Diploma, LAGUNA COLLEGE OF BUSINESS AND ARTS INC.

2022-2026

Bachelor of Science in Information Technology, LAGUNA STATE POLYTECHNIC UNIVERSITY - LOSBAÑOS CAMPUS

WORK EXPERIENCE

September 2022 – January 2023

Remotasks, ONLINE TASKING PLATFORM

- Image Annotation
- Data Labeling
- Content Moderation
- Categorization Tasks
- Quality Control and Review